The Geometric Engine of Time: The Fifteen Lemmas of the Geometric Engine

G. Hoard — BioChain AI, SHD-CCP Architecture Research

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Abstract

This monograph gives complete proofs of the fifteen lemmas on which the Symbolic High Dimensional Context Compression Protocol (SHD-CCP), in its 64-bit cubic form, and the Geometric Engine of Time rest, and derives from them how many engines behave together. Lemmas 1–3 are Zhao's three lemmas in geometry (incircle–excircle homothety, centres of spiral similarity and the symmedian harmonic bundle), proved in full as the synthetic and projective foundation. Lemmas 4, 5, 10 and 11 fix the engine geometry: the tier laws $R_n = 5n^2 + 15n + 15$, $r_n = 2n + 4$ and $W_n = 16 - 2n$ with $W_\Sigma = 60$, the order-6 Hurwitz spinor unit $h = (1 - i + j + k)/2$, the harmonic cubit lengths, five disjoint spherical shells separated from the voxel cube by $(11 - 2\sqrt{3})\mathcal{C}_0 \approx 4.05$ m, and the dichotomy between rational clock closure and golden ergodic subdivision. Lemmas 6, 7, 8, 12 and 14 govern the dynamics: five-moment Vandermonde conservation across the gap, the 9 + 3 = 12 affine frame and the implied fourth $H_4 = (H_1H_2H_3)^{-1}$ as a parity element, the eight-step cycle with its Step 6 projection into hyperbolic space through the {4,3,5} honeycomb, exact holonomy telescoping, and the Fokker–Planck circularity boundary. Lemmas 9, 13 and 15 settle arithmetic and integrity: closed-form Garner inverses for the prime triplet (881, 883, 887), Niven's finite-order bound with linear bit growth, and recursive Merkle integrity as against the South Halo pre-filter. Section 6 specifies the 64-bit Einstein Tile, whose Form ID selects exact, lossy or codex readings of the payload. Section 7 derives, with no new lemma, the 3:1 face-centred-cubic interleave, the handshake windows at Steps 2 and 6, interaction without contact, twelve-engine parity memory, the collapse of three ghost tetrahedra into the central cube, and the four-structure system with its 64 logged streams. Ten scripts reproduce every computation, and all pass.

Editorial Note. This document is formatted as a continuation of the Geometric Engine of Time & SHD-CCP Technical Series (Hoard 2026a–e; BioChain AI / Geometric Labs, Papers 1–18) and follows the layout of *The Geometric Engine of Time: A Proposal for a Standardization of the Cubit* (Hoard 2026d). It is a unified projective, quaternionic and metrological monograph for the SHD-CCP 64-bit cubic protocol, expanding Zhao's *Three Lemmas in Geometry* (2010) to fifteen lemmas on the plan of the audit Hoard (2026f), and drawing on Garvan (2019), Kaihnsa et al. (2017) and OpenAI (2026). The questions that lie outside the defined system, and the record of how this text was reached, are in two companion papers, *Open Problems of the Geometric Engine* and *Audit and Revision Record of the Fifteen Lemmas*. All claims are annotated with status tags: **[PROVEN]**, **[ESTABLISHED]**, **[CORRESPONDENCE]**, **[DEFINITION]**, **[COMPUTED]**, **[EMPIRICAL]**, **[HEURISTIC]**, **[CONJECTURAL]**, **[OPEN]**, defined under Epistemic Conventions and Scope.

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Epistemic Conventions and Scope

Every mathematical statement, physical model and architectural rule in this monograph carries one of nine status tags, the series tags established in Hoard (2026d–e). Six are applied to claims here. **[EMPIRICAL]** is reserved for a field realization of the cubit, which this monograph does not undertake, and **[CONJECTURAL]** and **[OPEN]** are used only in the companion paper *Open Problems of the Geometric Engine*.

Table 1: Status tags used in this monograph.

Tag	Meaning	Representative Example in This Monograph
[PROVEN]	A deductive mathematical theorem or lemma with a complete, self-contained proof in this document.	Lemmas 1–15; (0,2;4,4/3) = −1; $h^6 = +1$; closed-form c_{13}; $H = (Q^{(12)})^{-1}$.
[ESTABLISHED]	Standard mathematics or physics cited from peer-reviewed literature and used without re-proof.	Legendre's 3-square theorem; Hurwitz's norm theorem; Kronecker-Weyl; Niven's theorem.
[CORRESPONDENCE]	Two finite sets of equal cardinality or two equal numbers; no structure-preserving map is claimed.	64 voxels ↔ 64 sextic isotopy classes; 12 frame coordinates ↔ degree-12 Hessian.
[DEFINITION]	A formal architectural, metrological, or bit-layout convention adopted by the protocol.	$\mathcal{C}_0 := v_{local}/(16\, f_{cal})$; centered cubes $K = [-1,1]^3$, $V = [-2,2]^3$; Einstein Tile layout.
[COMPUTED]	An exact numerical or symbolic evaluation of a stated formula, verified by script.	$\mathcal{C}_0(20°C, 50\%\ RH) = 53.748$ cm; gap $11 - 2\sqrt{3} \approx 7.536\mathcal{C}_0$.
[EMPIRICAL]	A comparison against physical or experimental measurements with stated uncertainty.	Field realization of $\mathcal{C}_0$ in a physical room (the GUM budget U = 0.22 mm, k = 2, is **[COMPUTED]**).

Tag	Meaning	Representative Example in This Monograph
[HEURISTIC]	An engineering or physical modeling choice motivated by analogy rather than deductive necessity.	Δt_{geo} = 1 s anchoring; TQF scores and Fokker-Planck sieve as Helix A rankers.
[CONJECTURAL] (companion paper)	A mathematical or physical hypothesis requiring new proof or empirical validation.	Golden Stability Conjecture ($\overline{E}_{1/\varphi} < \overline{E}_{1/2}$); Oracle Bias Hypothesis ($\beta > 1$).
[OPEN] (companion paper)	A stated mathematical problem awaiting resolution.	Structure-preserving 64-voxel/sextic bijection; Bunyakovsky prime-norm lattice lines.

Rule for mixed claims. Where two tags are joined by a slash, the statement has both characters. **[ESTABLISHED / PROVEN]** marks a standard result that is nonetheless proved in full here (Lemmas 1–3 and 11); **[PROVEN / DEFINITION]** marks a lemma that adopts a convention alongside its theorems (Lemmas 7 and 8); **[PROVEN / CORRESPONDENCE]** marks a lemma one of whose parts is a match of counts only (Lemmas 4 and 5). Inside a lemma of mixed status, each part carries its own tag. A computed value is tagged **[COMPUTED]** even when the formula it evaluates is proven.

What this monograph does not claim.

- It does not claim a structure-preserving map between the 64 voxels and the 64 rigid isotopy classes of real plane sextics, or between any other pair of equal counts; such matches are tagged **[CORRESPONDENCE]** (Lemma 5 Part 6, §7.7).
- It does not claim that any map from an elliptic-curve group into the 64-bit tile inverts the discrete logarithm in O(1); Lemma 9 Part 4 proves three obstructions.
- It does not treat the 8-bit South Halo as an integrity check. The South Halo is a cache pre-filter, and integrity rests on the recursive Merkle root (Lemma 15).
- It does not claim that the Fokker–Planck sieve computes primality. The sieve samples a prime set it is given (Lemma 14 Part 4).
- It does not rely on the blowup claim of OpenAI (2026), whose overall correctness is publicly disputed; only the results re-proved here are used (Lemma 4 Part 6′, Lemma 6).
- It does not claim that the hierarchies of Section 7 store more than the bits written. Their gains are erasure tolerance, phase throughput and a conserved invariant (§7.4, §7.6).
- It does not settle the open problems P1–P16, which are collected in the companion paper.

Global coordinate and centering convention. **[DEFINITION]** Of the two natural placements of the voxel cube, corner-anchored $[0, 4\mathcal{C}_0]^3$ and origin-centered $[-2\mathcal{C}_0, 2\mathcal{C}_0]^3$, this monograph adopts the **Origin-Centered Convention** globally:

1. **Standardized Local Acoustic Cubit:** $\mathcal{C}_0(f_{cal}; t, p, x_w, x_c) := \lambda/16 = v_{local}(t, p, x_w, x_c)/(16\, f_{cal})$ (**[DEFINITION]**). Unless otherwise noted, all spatial coordinates are expressed in dimensionless units of $\mathcal{C}_0$.
2. **Centered Inner Core (Cube 1):** $K = [-1, 1]^3$ (half-edge $R_1 = 1\mathcal{C}_0$, full edge length $L_{core} = 2\mathcal{C}_0 = \lambda/8$, containing the $2 \times 2 \times 2 = 8$ core voxel centers at $(\pm 1/2, \pm 1/2, \pm 1/2)$).
3. **Centered Voxel Cube (Cube 2):** $V = [-2, 2]^3$ (half-edge $R_2 = 2\mathcal{C}_0$, full edge length $L_{cube} = 4\mathcal{C}_0 = \lambda/4$, containing all $4 \times 4 \times 4 = 64$ voxel centers).
4. **Symmetry Advantage of the Centered Convention:** The rotational symmetry group of the centered cube $V = [-2, 2]^3$ is the full 24-element **octahedral rotation group O ⊂ SO(3)**, whose spinor double-cover preimage in S^3 is the 48-element **binary octahedral group 2O**, which contains all **24 Hurwitz units** (including the inter-tier spinor unit $h = (1 - i + j + k)/2$). By contrast, a corner-anchored cube $[0, 4]^3$ is preserved by only 3 rotations in SO(3) (the cyclic permutations about (1,1,1)) and is *not* preserved by h.
5. **Edge-Length vs. Half-Edge Harmonic Convention:** In **Lemma 5 (Part 3)**, the projective harmonic bundle $(0, 2\mathcal{C}_0; 4\mathcal{C}_0, (4/3)\mathcal{C}_0) = -1$ is a statement about **full edge lengths** (or corner-anchored intervals $[0, 2\mathcal{C}_0] \subset [0, 4\mathcal{C}_0]$), yielding the Klein-wave merge length $F = (4/3)\mathcal{C}_0 = \lambda/12$ (phase $\pi/6$). In radial half-edges from the center O ($B = 0$, $C = 1\mathcal{C}_0$, $E = 2\mathcal{C}_0$), the harmonic conjugate is $F_{rad} = (2/3)\mathcal{C}_0 = \lambda/24$ (phase $\pi/12$), whose diameter/full-span is $2F_{rad} = (4/3)\mathcal{C}_0 = \lambda/12$.

Relation to the series. This monograph imports results from the Geometric Engine of Time and SHD-CCP technical series (Hoard 2026a–e) and from the SHD-CCP protocol notes (Hoard 2026g). Appendix C lists each source and the results that rely on it. No proof depends on a paper that is not restated here, and every imported result is checked by script (Appendix A).

Notation. Inline mathematics is set in plain text where it reads unambiguously (for example $R_n = 5n^2 + 15n + 15$); fractions, matrices and display equations are typeset. Spatial coordinates are in units of the cubit $\mathcal{C}_0$ unless stated. Parts of a lemma are cited as "Lemma 4 Part 5"; Propositions 6.1–6.5 belong to Step 6 of Lemma 8, and Proposition 11.4 to Lemma 11.

1 Introduction

1.1 The Four-Layer Deductive System

The 15 lemmas of the Geometric Engine are structured into four deductive layers, plus a system layer of propositions derived from them (Table 2).

Table 2: The deductive layers and their governing lemmas.

Layer	Governing Lemmas	Mathematical & Architectural Scope
Layer I: Synthetic & Projective Foundation	**Lemmas 1, 2, 3**	Classical Euclidean homothety, isotomic tangency conservation, Miquel spiral similarity centers, symmedian quadrance ratios, Apollonius circles, and projective harmonic bundles.
Layer II: Engine Geometry, Nesting, & Winding Dichotomy	**Lemmas 4, 5, 10, 11**	Upper Manifold ($\mathcal{M}_{\text{upper}}$) tier laws & Hurwitz unit h; Lower Manifold ($\mathcal{M}_{\text{lower}}$) harmonic cubit lengths & 4D Hurwitz prime completeness; strict spatial separation $\mathcal{M}_{\text{upper}} \cap \mathcal{M}_{\text{lower}} = \varnothing$; and rational clock closure vs. golden ergodic subdivision.
Layer III: Dynamics, Coupling, Holonomy, & Prime Sieve	**Lemmas 6, 7, 8, 12, 14**	Cross-gap 5-moment Vandermonde conservation; the 9 + 3 = 12 affine frame bundle $\mathfrak{aff}(3)$ & Markov current J; the 8-step "π-cycle" & C_9 non-degenerate K_5 4-simplex; exact holonomy telescoping $H = (Q^{(12)})^{-1}$; and the Fokker-Planck Gibbs/harmonic measure & circularity boundary.
Layer IV: Modular-Cubic Arithmetic, Bit Growth, & Security	**Lemmas 9, 13, 15**	Prime-Triplet mod-30 CRT cubic indexing with closed-form inverses (c_{12}, c_{23}, c_{13}) & ECDLP obstructions; unbounded denominator growth in $S^3 \cap \mathbb{Q}^4$ vs. Niven's finite-order bound & projective integer arithmetic; and recursive Merkle integrity vs. South Halo cache filtering.
Layer V: System Level	**§7 (derived from Lemmas 1–15)**	Ghost projection into $\mathbb{H}^3$; 3:1 FCC interleave and implied fourth; scale-free and {4,3,5} hyperbolic stacking; handshake windows k = 2, 6; non-contact channels; twelve-engine parity memory and the third phase.

1.2 Summary of the Fifteen Lemmas

This monograph establishes the mathematical foundation for the **64-bit cubic geometric form of the Symbolic High Dimensional Context Compression Protocol (SHD-CCP)** and **The Geometric Engine of Time**. Taking Yufei Zhao's classical *Three Lemmas in Geometry* (**Lemmas 1, 2, and 3**: incircle/excircle homothety, centers of spiral similarity, and symmedian projective harmonic bundles) as the synthetic and projective point of departure, it proves twelve domain-specific lemmas (**Lemmas 4 through 15**) organized into the hierarchy of Table 2:

- **Layer I — Classical Synthetic & Projective Foundation (Lemmas 1–3):** Establishes incircle/excircle homothety and isotomic equality BD = CX = s − b, BX = CD = s − c (**Lemma 1**), Miquel circle localization and paired spiral similarities (**Lemma 2**), and symmedian quadrance division BF/(FC) = Q_3/Q_2, Apollonius circle realization $EA^2 = EB \cdot EC$, and projective harmonic bundles (B, C; E, F) = −1 (**Lemma 3**).
- **Layer II — Geometry, Nesting, and Winding Dichotomy of the Engine (Lemmas 4, 5, 10, 11):**
  - **Lemma 4 (Upper Manifold $\mathcal{M}_{\text{upper}}$ Standardization):** Proves the tier-winding invariants $\varrho_n + \mathcal{W}_n = 20\mathcal{C}_0 = (5/4)\lambda$ and $W_\Sigma = 60$, inter-tier signed-permutation spiral similarities $\mathcal{S}_n(\mathbf{x}) = \alpha_n \mathbf{U}_n \mathbf{x}$ about the axis $(1/\sqrt{3})(-1,1,1)^T$, actuated temperature-invariant mode scaling, the Order-6 Hurwitz spinor unit $h = (1 - i + j + k)/2$ ($h^2 = -\overline{h}$, $h^3 = -1$, $h^4 = -h$, $h^6 = +1$), the Legendre symbol $((\cdot)/5)$ partition of the 30-node wheel matching the level sets of χ_{60}^2 for Garvan's mod-60 mock-theta character χ_{60}, and torus-rectangle separation for multiplexing, proven by reproducing OpenAI (2026, Lemma 6.1).
  - **Lemma 5 (Lower Manifold $\mathcal{M}_{\text{lower}}$ Standardization & Hurwitz Completeness):** Proves symmedian quadrance division, binary octahedral group (2O) core-to-cube scaling $\mathbf{x} \mapsto 2\rho_g(\mathbf{x})$ on centered cubes $K = [-\mathcal{C}_0, \mathcal{C}_0]^3 \to V = [-2\mathcal{C}_0, 2\mathcal{C}_0]^3$, the edge-length projective harmonic bundle $(0, 2\mathcal{C}_0; 4\mathcal{C}_0, (4/3)\mathcal{C}_0) = -1$ (and centered bundle $(0, \mathcal{C}_0; 2\mathcal{C}_0, (2/3)\mathcal{C}_0) = -1$) realized by the Apollonius circle of radius $2\sqrt{2}\,\mathcal{C}_0$, the 4D Hurwitz lift resolving Legendre's $p \equiv 7 \pmod 8$ gap (with the bounded box $[-2,2]^4$ realizing exactly the primes 2, 3, 5, 7, 13), non-collinear prime triangles, two-product 128-bit Gram-TQF evaluation, and the $48 + 2 \cdot 8 = 64$ real plane sextic and degree-2 K3 correspondence.
  - **Lemma 10 (Global Nesting and Separation Theorem):** Proves that the 5 toroidal tiers occupy disjoint spherical shells [11, 19], [29, 41], [57, 73], [95, 115], [143, 167] $\mathcal{C}_0$ separated by gaps {10, 16, 22, 28}$\mathcal{C}_0$, while

the centered voxel cube $V = [-2\mathcal{C}_0, 2\mathcal{C}_0]^3$ lies inside $\|\mathbf{x}\| \le 2\sqrt{3}\,\mathcal{C}_0 \approx 3.464\mathcal{C}_0$, establishing a strict radial separation of $11 - 2\sqrt{3} \approx 7.536\mathcal{C}_0$ (4.05 m) between $\mathcal{M}_{\text{upper}}$ and $\mathcal{M}_{\text{lower}}$ ($\mathcal{M}_{\text{upper}} \cap \mathcal{M}_{\text{lower}} = \emptyset$).

- **Lemma 11 (The Winding Dichotomy Theorem — Rational Closure vs. Golden Ergodicity):** Proves that linear torus flows close iff the frequency ratio is rational (governing the integer-winding stator clock $W_n \in \mathbb{Z}$; the 0.5 s Caesium closure is proven separately as pulse arithmetic of $W_\Sigma = 60$ in Proposition 11.4) and are injective, dense, and equidistributed iff the ratio is irrational (governing the Golden Clifford subdivision/offload layer), and proves via the Steinhaus–Sós Three-Distance Theorem that golden subdivision $k/\varphi \bmod 1$ produces **exactly two gap lengths** (in ratio φ) whenever $N = F_m$ is a Fibonacci number.

- **Layer III — Cross-Manifold Dynamics, Holonomy, and the Fokker-Planck Sieve (Lemmas 6, 7, 8, 12, 14):**
  - **Lemma 6 (Cross-Gap 5-Moment Vandermonde Conservation):** Proves generalized Vandermonde invertibility of the five cumulative radial moments $\mathfrak{m} = (M, I, J, S, C_p)$ across five ordered radial bands bridging the separation gap, together with Banach fixed-point solvability of the quadratic moment-matching system $Bc + \mathcal{Q}(c,c) = d$, the sharpened 2 + 3 block structure of the moment Jacobian (invertible iff $\alpha \notin \{-1/2, 1/2, 3/2\}$), and exterior stress agreement proven from explicit integrated stress formulas.
  - **Lemma 7 (The 9 + 3 = 12 Affine Frame Bundle $\mathfrak{aff}(3)$, Central Tier T_2, & Group Inverse H_4):** Proves that the 9 linear micro-axis entries $L \in \mathfrak{gl}(3,\mathbb{R})$ and the 3 components of the single combined barycentric translation $\mathbf{t} \in \mathbb{R}^3$ form the 12-dimensional affine Lie algebra $\mathfrak{aff}(3) = \mathfrak{gl}(3,\mathbb{R}) \ltimes \mathbb{R}^3$, identifies T_2 ($r_2 = 8$, $W_2 = 12$) as the unique arithmetic mean/median tier, and proves the Markov tree-theorem current $J = (\Pi_+ - \Pi_-)/Z$ and BCH cross-product expansion of the unique group inverse $H_4 = (H_1\ H_2\ H_3)^{-1} \in S^3$; it proves $J > 0$ unconditionally for the rational valve kernel ($\Pi_- = \rho_{ba}\rho_{cb}\rho_{ac}\Pi_+$), and that H_4 is a single-erasure parity element for its three parents; for valves that are the same at every cell, it proves a positive stationary current on the whole 64-cell lattice $(\mathbb{Z}/4)^3$.
  - **Lemma 8 (The 8-Step "π-Cycle" & C_9 Non-Degenerate 4-Simplex Theorem):** Formalizes the 8 steps $\theta_k = k\pi/4$, proves the exact conformal null-cone identity $-2X(\mathbf{x}) \cdot X(\mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|^2$ for $X(\mathbf{x}) = (\mathbf{x}, (1/2)\|\mathbf{x}\|^2, 1) \in \mathbb{R}^{4,1}$, uses the **$C_9$ offsets {0°, 40°, 80°}** (a 120° rotation alone maps the tetrahedron to itself) to generate **9 distinct peripheral rays** at 40° spacing on the circumsphere of Cube 1 (**Lemma 8.2**), proves that no construction can end 9 distinct rays at vertices of Cube 1 (**Proposition 8.2′**), proves that lifting the 4 tetrahedron vertices + centroid O to $(\mathbf{x}, \|\mathbf{x}\|^2) \in \mathbb{R}^4$ yields 5 affinely independent points forming a non-degenerate K_5 4-simplex (**Lemma 8.3**), and proves the 16 congruent half-cube subdivision, $E_{\max} = 4$ quadrance scaling, Golden Binomial Cube $n^3 = (m + r)^3$, D_3 trefoil symmetry with 1/6 fundamental arcs and integer folding matrices, the two visible apex peaks at k = 2, 6 (the handshake windows), and the rigorous **Step 6 projection**: the ghost map $G = \Phi \circ \mathcal{C}_{\text{rat}}$, an exact, rational, rotation-equivariant isometry from the Poincaré ball onto $\mathbb{H}^3$, with the binary dihedral spin lift $2D_3$ (order 12), the equivalent golden-Clifford (2,3) torus knot in SU(2), and **ghost boosts derived from the {4,3,5} hyperbolic honeycomb**: the six face-neighbour translations of the cube cell, each with cosh(rapidity) = φ, which carry the trefoil into six disjoint neighbouring cells.
  - **Lemma 12 (Holonomy Telescoping & Right-Acting Ordering Theorem):** Proves that the middle-to-inner blocks $A_l = \{m: \lfloor 10m/72 \rfloor = l\}$ have exact sizes (8, 7, 7, 7, 7, 8, 7, 7, 7, 7), that under right-acting middle multiplication $Q^{(10)} = (Q^{(12)})^{-1}$ and $Q^{(8)} = Q^{(10)}$, and that the cycle holonomy reduces identically to $H = Q^{(8)}Q^{(10)}Q^{(12)} = (Q^{(12)})^{-1} = q_{-1}\,q_{719}^{-1}$, which under the adopted **open convention** ($\Delta q_0 = 1$) equals the start-to-end discrepancy $q_0\ q_{719}^{-1}$ (whereas the cyclic convention makes $H \equiv 1$ vacuously, and left-acting middle multiplication fails to cancel).
  - **Lemma 14 (Fokker-Planck Stationary Gibbs Measure & Circularity Boundary):** Proves that the lattice master/Metropolis chain has the unique Gibbs stationary measure $\pi(x) \propto e^{-\beta V(x)}$, that zero-diffusion traps ($D_2 \to 0$) yield the harmonic hitting measure, that the infinite Hurwitz lattice normalizes (for every trap with $V(n) \ge \alpha \ln n$, $\beta\alpha > 2$) iff $\Sigma_{n=1}^{\infty} 24\sigma_{\text{odd}}(n)e^{-\beta V(n)} < \infty$, and proves the Circularity Boundary isolating the sieve to Helix A.

- **Layer IV — Modular-Cubic Arithmetic, Bit-Growth Limits, and Cryptographic Security (Lemmas 9, 13, 15):**
  - **Lemma 9 (Prime-Triplet Mod-30 CRT Cubic Indexing, Closed-Form c_{13}, & ECDLP Obstruction):** Proves that prime triplets (p, p + 2, p + 6) for p > 5 lie exclusively in (11, 13, 17) or (17, 19, 23) (mod 30), supplies closed-form modular inverses for all three Garner constants c_{12}, c_{23}, c_{13}, identifies (881, 883, 887) as the maximal 10-bit prime triplet ($M = 690{,}017{,}701 \approx 2^{29.4}$), and proves the Pigeonhole, Torsion-Free Homomorphism, and faithful torus-character ("gauge") obstructions on O(1) ECDLP claims.
  - **Lemma 13 (Exact Quaternionic Arithmetic, Niven's Finite-Order Bound, & Bit-Growth Theorem):** Proves that $q = (3 + 4i)/5 \in S^3 \cap \mathbb{Q}^4$ has q^n with exact denominator 5^n (unbounded denominator growth),

proves via Niven's theorem that every finite-order element in $S^3 \cap \mathbb{Q}^4$ has order $\in \{1, 2, 3, 4, 6\}$, proves that projective integer conjugation $\mathbf{x} \mapsto q\mathbf{x}\overline{q}/N(q)$ is exact with $O(k(b + 2))$ bit growth over k steps, and proves that $q_k = (1 + i)^k$ realizes the exact 8-step spinor double-cover cycle (6 signed bits unreduced, 2 bits after gcd reduction).

- **Lemma 15 (Recursive Merkle Integrity & South Halo Pre-Filter Bound):** Proves that a 64-bit label on a 12-layer 64-voxel state (2^{72} states) has ≥ 256 preimages, that the 8-bit South Halo has 2^{48} preimages on average (making it an O(1) cache filter, not an integrity check), and that full recursive Merkle hashing $\tau_d = H(G_d, \tau_{d+1})$ reduces structural integrity to the collision resistance of H (linear checksums such as CRC-32 give no integrity).

- **Layer V — System Level (§7; no new lemma):** Propositions derived from Lemmas 1–15 describing many engines together: the 3:1 face-centred-cubic interleave whose fourth lattice is forced by closure, scale-free and {4,3,5} hyperbolic stacking, the two handshake windows k = 2, 6, non-contact interaction through the five-moment channel and the projected kernel $Q_h = \sinh^2 d_{\mathbb{H}}$, the twelve-engine memory hierarchy (implied fourths as parity engines, exact inverse-tier offload, and the third 120° phase), the collapse of the three ghost tetrahedra into the central cube at Step 6, the four-structure system (input, two balancers, emergent fourth) and its 64 logged streams. **Protocol (§6):** the 64-bit Einstein Tile, whose Form ID (adopted from the SHD-CCP Tensor Transfer Protocol, Hoard 2026g) selects among exact, lossy and codex readings of the payload.

2 Layer I — Classical Synthetic and Projective Foundations (Lemmas 1, 2, and 3)

2.1 Lemma 1: Incircle Diameter, Excircle Homothety, and Isotomic Conservation

Lemma 1 (Incircle/Excircle Dilation and Isotomic Tangency Conservation). **[ESTABLISHED / PROVEN]** Zhao (2010), §1.

*Let the incircle ω of $\triangle ABC$ touch side BC at D, and let DT be the diameter of ω perpendicular to BC. If line AT meets BC at X, then X is the point of tangency of the A-excircle ω_A with BC, and the segments along BC satisfy the **isotomic conservation law**:*

$$BD = CX = s - b \iff BX = CD = s - c,$$

where s = (a + b + c)/2 is the semiperimeter, b = CA and c = AB. Equivalently, D and X are reflections of each other in the midpoint of BC.

Proof.

1. ***Homothety Mapping Incircle to Excircle:*** *The homothety $\mathcal{H}_A$ centered at A with dilation ratio r_A / r maps the incircle ω (inscribed in ∠BAC) to the A-excircle ω_A (also tangent to rays AB and AC). At $T \in \omega$, the tangent to ω is parallel to BC with ω lying on the BC side of that tangent; hence $\mathcal{H}_A(T)$ is the point of ω_A where the tangent is parallel to BC on the far side from A, which is line BC itself. Since $\mathcal{H}_A(T)$ also lies on ray AT, we have $\mathcal{H}_A(T) = AT \cap BC = X$, proving ω_A is tangent to BC at X.*
2. ***Equal-Tangent Chase:*** *Let ω touch AB, AC at F, E and let ω_A touch rays AB, AC at Z, Y. By equality of tangents from a point to a circle, BD = BF and CD = CE, so $2BD = (AB - AF) + (BC - CE) = AB + BC - AC = 2(s - b) \implies BD = s - b$. On the excircle side, $AZ = AY = s$, so $CX = CY = AY - AC = s - b$. Thus $BD = CX = s - b$. Finally, $BX = BC - CX = a - (s - b) = s - c$ and $CD = BC - BD = a - (s - b) = s - c$, so BX = CD.* ■

2.2 Lemma 2: Center of Spiral Similarity and Paired Similitudes

Lemma 2 (Center of Spiral Similarity and Miquel Intersection). **[ESTABLISHED / PROVEN]** Zhao (2010), §2.

Let A, B, C, D be four distinct points in the plane such that ABCD is not a parallelogram and lines AC and BD are not parallel. Let $X = AC \cap BD$, and let the circumcircles (ABX) and (CDX) intersect again at $O \neq X$. Then:

1. ***Algebraic Uniqueness:*** *There exists a unique spiral similarity $T(z) = z_0 + \alpha(z - z_0)$ ($\alpha \in \mathbb{C}^\times$) carrying A ↦ C and B ↦ D, given by:*

$$\alpha = \frac{c - d}{a - b}, \qquad z_0 = \frac{ad - bc}{a - b - c + d}.$$

2. ***Miquel Circle Localization:*** *The center z_0 coincides with the second intersection O of (ABX) and (CDX), so $\triangle OAC \sim \triangle OBD$ (directly similar).*
3. ***Dual Paired Similarity:*** *The same point O is also the center of the unique spiral similarity carrying A ↦ B and C ↦ D, so $\triangle OAB \sim \triangle OCD$ (directly similar).*

Proof.

1. *Subtracting $T(a) = (1-\alpha)z_0 + \alpha a = c$ and $T(b) = (1-\alpha)z_0 + \alpha b = d$ gives $\alpha(a-b) = c-d$. Since ABCD is not a parallelogram, $a-b \neq c-d \Longrightarrow \alpha \neq 1$, and solving $z_0 = (c-\alpha a)/(1-\alpha)$ yields $z_0 = (ad-bc)/(a-b-c+d)$.*
2. *Using directed angles modulo π, concyclicity on (ABX) and collinearity of A, X, C and B, X, D give $\angle(OA, AC) = \angle(OA, AX) = \angle(OB, BX) = \angle(OB, BD)$, and concyclicity on (CDX) gives $\angle(OC, CA) = \angle(OC, CX) = \angle(OD, DX) = \angle(OD, DB)$. Hence $\triangle OAC \sim \triangle OBD$ directly, so $O = z_0$.*
3. *From $\triangle OAC \sim \triangle OBD$, we have $\angle AOC = \angle BOD \Longrightarrow \angle AOB = \angle COD$ and $OC/(OA) = OD/(OB) \Longrightarrow OB/(OA) = OD/(OC)$, so $\triangle OAB \sim \triangle OCD$ directly.* ■

2.3 Lemma 3: Symmedian, Apollonius Circle, and Projective Harmonic Bundle

Lemma 3 (Symmedian Quadrance Ratio, Apollonius Circle and Projective Harmonic Bundle). **[ESTABLISHED / PROVEN]** Zhao (2010), §3.

*Let $\triangle ABC$ be inscribed in circumcircle Γ, let M be the midpoint of BC, let the tangents to Γ at B and C meet at D, let line AD meet BC at F, and, **assuming AB ≠ AC** (required for Parts 2–3), let the tangent to Γ at A meet line BC at E. Then:*

1. ***Symmedian Quadrance Ratio:*** *Line AD is the **symmedian** of $\triangle ABC$ (the reflection of median AM across the internal angle bisector of ∠BAC), and its foot F divides BC in the exact ratio of the adjacent squared sides (quadrances $Q_3 = AB^2$, $Q_2 = AC^2$):*

$$\frac{BF}{FC} = \frac{AB^2}{AC^2} = \frac{Q_3}{Q_2}.$$

2. ***External Tangent Ratio & Apollonius Circle:*** *The tangent intersection E divides BC externally in the same quadrance ratio, and satisfies the power-of-a-point / Apollonius identity:*

$$\frac{EB}{EC} = \frac{AB^2}{AC^2} = \frac{Q_3}{Q_2}, \qquad EA^2 = EB \cdot EC.$$

3. ***Projective Harmonic Bundle & Newton's Relation:*** *The four collinear points B, C, E, F form an exact **projective harmonic bundle** satisfying Newton's midpoint relation:*

$$(B, C; E, F) = \frac{EB}{EC} \div \frac{FB}{FC} = -1, \qquad MB^2 = MC^2 = MF \cdot ME.$$

Proof.

1. ***Proof of Part (1) (Cleaned Area & Sine-Rule Chase):*** *By the ratio of areas of triangles $\triangle ABF$ and $\triangle ACF$ sharing altitude from A, $BF/(FC) = AB \sin\angle BAF/(AC \sin\angle CAF)$. Applying the Law of Sines to $\triangle ABD$ and $\triangle ACD$ and using equal tangent lengths BD = CD gives:*

$$\frac{\sin\angle BAD}{\sin\angle CAD} = \frac{\frac{BD}{AD}\sin\angle ABD}{\frac{CD}{AD}\sin\angle ACD} = \frac{\sin\angle ABD}{\sin\angle ACD}.$$

By the tangent-chord theorem on Γ, ∠ABD = ∠C (or 180° − ∠C, having the same sine) and ∠ACD = ∠B (or 180° − ∠B). Hence $\sin\angle BAD/(\sin\angle CAD) = \sin C/(\sin B) = AB/(AC)$ by the Law of Sines on $\triangle ABC$. Substituting this into the area ratio yields:

$$\frac{BF}{FC} = \frac{AB}{AC} \cdot \frac{\sin\angle BAD}{\sin\angle CAD} = \left(\frac{AB}{AC}\right)^2 = \frac{Q_3}{Q_2}.$$

For the median AM, $BM/(MC) = 1 \Longrightarrow \sin\angle BAM/(\sin\angle CAM) = AC/(AB) = \sin\angle CAD/(\sin\angle BAD)$. Since ∠BAD + ∠CAD = ∠A = ∠BAM + ∠CAM and $t \mapsto \sin t/(\sin(\angle A - t))$ is strictly increasing on (0, ∠A), we have ∠BAD = ∠CAM, so AD is the isogonal conjugate of AM (the symmedian).

2. ***Proof of Part (2) (Similar Triangles & Apollonius Circle):*** *Triangles $\triangle EAB$ and $\triangle ECA$ share ∠E and have ∠EAB = ∠ECA by the tangent-chord theorem. Thus $\triangle EAB \sim \triangle ECA$, which gives $EA/(EC) = EB/(EA) = AB/(CA)$. Multiplying the first two ratios gives $EB/(EC) = AB^2/(AC^2) = Q_3/Q_2$ and cross-multiplying gives $EA^2 = EB \cdot EC$.*

3. ***Proof of Part (3) (Harmonic Bundle & Newton's Identity):*** *Because F divides BC internally and E divides BC externally in the identical ratio Q_3/Q_2, their signed cross-ratio is (B, C; E, F) = −1. Placing the origin at the midpoint M of BC so $x_B = -c$, $x_C = +c$, the harmonic relation $(x_F + c)/(c - x_F) = (x_E + c)/(x_E - c)$ cross-multiplies to $2x_F x_E = 2c^2 \Longrightarrow MF \cdot ME = c^2 = MB^2 = MC^2$.*
4. ***Degenerate case AB = AC:*** *the tangent at A is then parallel to BC, F = M, and the harmonic conjugate of M with respect to B, C is the point at infinity of line BC, so (B, C; ∞, M) = −1 holds projectively while the finite statements of Parts 2–3 do not apply.* ■

3 Layer II — Geometry, Nesting, and Winding Dichotomy of the Engine (Lemmas 4, 5, 10, and 11)

3.1 Lemma 4: Upper Manifold Standardization and Spinor Structure

Lemma 4 (Upper Manifold Tier-Winding Invariants, Signed-Permutation Spiral Similarity, Hurwitz Spinor Unit and Mod-60 Mock-Theta Partition). **[PROVEN / CORRESPONDENCE]** Tier data from Hoard (2026a); Part 3 after Hoard (2026d), Lemma 3.2; Part 5 after Garvan (2019), Theorem 5.3; Part 6′ after OpenAI (2026), Lemma 6.1.

Let the 5-tier toroidal Upper Manifold $\mathcal{M}_{upper} = \bigcup_{n=0}^{4} T_n$ be centered at the origin O and metricized by $\mathcal{C}_0 = \lambda/16 = v_{local}/(16\, f_{cal})$, with major radius $\mathcal{R}_n = R_n\, \mathcal{C}_0 = (5n^2 + 15n + 15)\mathcal{C}_0 \in \{15, 35, 65, 105, 155\}\mathcal{C}_0$, minor tube radius $\varrho_n = r_n\, \mathcal{C}_0 = (2n + 4)\mathcal{C}_0 \in \{4, 6, 8, 10, 12\}\mathcal{C}_0$, winding count $W_n = 16 - 2n \in \{16, 14, 12, 10, 8\}$, and alternating orientation $\mathrm{Rot}(T_n) = \mathrm{Rot}_x(\pi/2)$ (even n) and $\mathrm{Rot}_y(\pi/2)$ (odd n). Then:

1. ***Tier-Winding & Diameter Invariants ([PROVEN]):*** *Across all five tiers $n \in \{0, 1, 2, 3, 4\}$, the minor radius ϱ_n and winding segment $\mathcal{W}_n = W_n\, \mathcal{C}_0$ satisfy the exact linear conservation laws:*

$$\varrho_n + \mathcal{W}_n = 20\mathcal{C}_0 = \frac{5}{4}\lambda, \qquad \mathcal{D}_n + 2\mathcal{W}_n = 40\mathcal{C}_0 = \frac{5}{2}\lambda, \qquad \sum_{n=0}^{4} W_n = W_\Sigma = 60.$$

   On the minor-radius interval $[4\mathcal{C}_0, 12\mathcal{C}_0]$, ϱ_n and $\varrho_{4-n} = (12 - 2n)\mathcal{C}_0$ are mirror-symmetric (isotomic) about the central tier radius $\varrho_2 = 8\mathcal{C}_0$ ([CORRESPONDENCE]).

2. ***Inter-Tier Signed-Permutation Spiral Similarity & Weitzenböck Maximum ([PROVEN]):*** *For each $n \in \{0, 1, 2, 3\}$, the inter-tier transition matrix $\mathbf{U}_n = \mathrm{Rot}(T_{n+1})\, \mathrm{Rot}(T_n)^{-1} \in SO(3)$ is given by:*

$$\mathbf{U}_{\text{even}} = \mathrm{Rot}_y\left(\frac{\pi}{2}\right)\mathrm{Rot}_x\left(-\frac{\pi}{2}\right) = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{pmatrix}, \qquad \mathbf{U}_{\text{odd}} = \mathbf{U}_{\text{even}}^{-1} = \begin{pmatrix} 0 & 0 & -1 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

   *which is a **120° (±2π/3) signed-permutation rotation about the cube body diagonal $\hat{u} = (1/\sqrt{3})(-1, 1, 1)^T$** (satisfying $\mathbf{U}_n^3 = I_3$). With fixed anchor phases, the 3D spiral similarity $\mathcal{S}_n(\mathbf{x}) = \alpha_n\, \mathbf{U}_n\, \mathbf{x}$, where $\alpha_n = R_{n+1}/R_n = (n^2 + 5n + 7)/(n^2 + 3n + 3) \in \{7/3, 13/7, 21/13, 31/21\}$, maps the core circle and equilateral C_3 anchor triangle of T_n onto those of T_{n+1}, scaling anchor quadrances by $Q_k^{(n+1)} = \alpha_n^2\, Q_k^{(n)} = 3\mathcal{R}_{n+1}^2$ and maintaining the global Weitzenböck maximum $\chi(T_n) = \sqrt{A_R^{(n)}/S^{(n)}} = 1/\sqrt{3}$.*

3. ***Actuated Temperature-Invariant Waveguide Mode Scaling ([PROVEN / DEFINITION]):*** *In the thin-tube approximation ($r_n/R_n \in [0.08, 0.27]$), the m-th azimuthal standing-wave frequency $f_{n,m}^{maj} = 8m\, f_{cal}/(\pi R_n)$ and (p,q)-th meridian Bessel frequency $f_{n,p,q}^{min} = 8\, j'_{p,q}\, f_{cal}/(\pi r_n)$ are independent of $v_{local}(t, p, x_w, x_c)$ whenever the physical waveguide dimensions track $\mathcal{C}_0(T)$.* Exact companion (**[PROVEN]**, after Hoard 2026d Lemma 3.2): *a rigid rectangular room of $N_x \times N_y \times N_z$ cubits has modes $f_{lmn} = 8\, f_{cal}\sqrt{l^2/N_x^2 + m^2/N_y^2 + n^2/N_z^2}$, independent of v_{local} with no thin-tube approximation; the 4-cubit cube's lowest modes are 80 Hz (threefold) and $80\sqrt{2} \approx 113.1$ Hz.*

4. ***Order-6 Hurwitz Spinor Double-Cover Unit ([PROVEN]):*** *In the spinor cover $\pi : S^3 \xrightarrow{2:1} SO(3)$, $\mathrm{Rot}_x(-\pi/2)$ lifts to $q_x = (1 - i)/\sqrt{2}$ and $\mathrm{Rot}_y(\pi/2)$ lifts to $q_y = (1 + j)/\sqrt{2}$, whose product is the Hurwitz unit:*

$$h = q_y q_x = \frac{1 - i + j + k}{2} = \cos\left(\frac{\pi}{3}\right) + \left(\frac{-i + j + k}{\sqrt{3}}\right)\sin\left(\frac{\pi}{3}\right) \in \mathcal{H}_1 \subset S^3,$$

   whose rotation axis $(1/\sqrt{3})(-1, 1, 1)^T$ matches $\mathbf{U}_{even}$ via $q \mapsto h\, q\, h^{-1}$. Its powers in S^3 are:

$$h^1 = h, \quad h^2 = -\bar{h} = \frac{-1 - i + j + k}{2}, \quad h^3 = -1, \quad h^4 = -h = \frac{-1 + i - j - k}{2}, \quad h^5 = \bar{h}, \quad h^6 = +1.$$

   Left- and right-multiplication by h is a norm-preserving bijection of the Hurwitz ring $\mathcal{H}$ ($N(hq) = N(q)$).

5. ***Legendre Mod-5 & Garvan Mod-60 Partition of the 30-Node Wheel ([PROVEN / CORRESPONDENCE]):*** *The $\phi(30) = 8$ prime-admissible unit residue classes $(\mathbb{Z}/30\mathbb{Z})^\times = \{1, 7, 11, 13, 17, 19, 23, 29\}$ split into the index-2 subgroup of **quadratic residues modulo 5** and its non-residue coset:*

$$\mathcal{B}_+ = \ker\left(\frac{\cdot}{5}\right) = \{1, 11, 19, 29\} \pmod{30}, \qquad \mathcal{B}_- = \{7, 13, 17, 23\} \pmod{30}.$$

This partition coincides identically with the level sets of χ_{60}^2, where χ_{60} is the order-4 Dirichlet character mod 60 in Garvan (2019, Thm. 5.3). Explicitly, $\chi_{60} = 1$ on $\{1, 11, 19, 29\}$, i on $\{7, 13, 17, 23\}$, -1 on $\{31, 41, 49, 59\}$ and $-i$ on $\{37, 43, 47, 53\}$, so $\chi_{60}^2(b) = (b/5)$; χ_{60}^2 is constant on residue classes mod 30, while χ_{60} itself is not ($\chi_{60}(31) = -1 \neq \chi_{60}(1)$). The identification of Garvan's series with the engine's 30 nodes is [CORRESPONDENCE]; the equality $60 = W_\Sigma$ is numerical.

6. ***Auxiliary Torus Covering & Greedy Graph Coloring ([PROVEN]):*** *The integer matrix $J_g = \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}$ has $\det J_g = 14$ and eigenvalues $4 \pm \sqrt{2}$, defining a 14:1 torus covering map $Y \mapsto J_g\, Y \pmod{\mathbb{Z}^2}$ ([PROVEN]). Any finite graph with maximum degree Δ (in the application, the intersection graph of the slow supports of OpenAI 2026, §6) admits a proper vertex coloring with at most $\Delta + 1$ colors ([PROVEN]).*
Part 6′ — Torus-Rectangle Separation ([PROVEN]; OpenAI 2026, Lemma 6.1, proof reproduced): *Let $\mathcal{G}$ be a graph of maximum degree $\Delta_{\mathcal{G}}$ on a set of labels, let $\boldsymbol{v}_r$, $\boldsymbol{v}_t$ be unit eigenvectors of J_g for the eigenvalues $\Lambda_g = 4 - \sqrt{2}$ and $T_g = 4 + \sqrt{2}$, say that a label at level i has its rectangle pulled back by J_g^i, and fix $\Delta_{max} \geq 0$. There are colours $1, \ldots, \Delta_{\mathcal{G}} + 1$, rational centres $c_\nu \in \mathbb{T}^2$ and a radius $r_0 > 0$ such that whenever two adjacent labels sit at levels i and $i + \Delta$ ($0 \leq \Delta \leq \Delta_{max}$), their rectangles $c + \{\xi \boldsymbol{v}_r + \eta \boldsymbol{v}_t : |\xi|, |\eta| < 2r_0\}$, pulled back by J_g^i and $J_g^{i+\Delta}$, are disjoint. This is the separated multiplexing the engine uses. The preprint's further claim that the separated pulses span its stress cone is not used anywhere in this monograph (companion paper, Problem P13).*

Proof.

1. *$(2n + 4) + (16 - 2n) = 20$ for all n, and $\Sigma_{n=0}^{4} (16 - 2n) = 80 - 20 = 60$.*
2. *Direct matrix multiplication gives:*

$$\mathbf{U}_{\text{even}} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{pmatrix}.$$

Here $\det \boldsymbol{U}_{even} = +1$, $Tr(\boldsymbol{U}_{even}) = 0 = 1 + 2\cos\theta \Longrightarrow \theta = 2\pi/3$, and $\boldsymbol{U}_{even}\boldsymbol{v} = \boldsymbol{v} \Longleftrightarrow -v_2 = v_1$, $v_3 = v_2$, $-v_1 = v_3 \Longleftrightarrow \boldsymbol{v} \propto (-1, 1, 1)^T$. With $S = Q_1 + Q_2 + Q_3$ and $A_R = S^2 - 2\Sigma Q_i^2$, we have $S^2 - 3A_R = 6\Sigma Q_i^2 - 2S^2 = 2(3\Sigma Q_i^2 - S^2) = 2[(Q_1 - Q_2)^2 + (Q_2 - Q_3)^2 + (Q_3 - Q_1)^2] \geq 0$ (using $3\Sigma Q_i^2 - S^2 = \Sigma_{i<j}(Q_i - Q_j)^2$). Hence $\chi = \sqrt{A_R}/S \leq 1/\sqrt{3}$ (the quadrance form of Weitzenböck's inequality), with equality iff $Q_1 = Q_2 = Q_3$; for the inscribed anchor triangle the common value is $3\mathcal{R}_n^2$.
3. *Substituting $\mathcal{C}_0 = v_{local}/(16\, f_{cal})$ into $f_{n,m}^{maj} = m\, v_{local}/(2\pi R_n\, \mathcal{C}_0)$ and $f_{n,p,q}^{min} = j'_{p,q}\, v_{local}/(2\pi r_n\, \mathcal{C}_0)$ cancels v_{local}. For the room, the rigid-wall modes are $f = (v_{local}/2)\sqrt{((l/L_x)^2 + (m/L_y)^2 + (n/L_z)^2)}$ with $L_x = N_x\mathcal{C}_0$, and the same substitution gives $8\, f_{cal}\sqrt{l^2/N_x^2 + m^2/N_y^2 + n^2/N_z^2}$.*
4. *Expanding $h = ((1 + j)/\sqrt{2})((1 - i)/\sqrt{2}) = (1 - i + j - ji)/2 = (1 - i + j + k)/2$ gives $N(h) = 1$ and $Re(h) = 1/2$. Since $h^2 - h + 1 = 0$ (by $q^2 - 2\,Re(q)q + N(q) = 0$) and $\overline{h} = 1 - h$, we have $h^2 = h - 1 = -\overline{h}$, $h^3 = -1$, $h^4 = -h$, $h^5 = \overline{h}$, and $h^6 = +1$. Conjugating the basis vectors i, j, k by h gives $h\, i\, \overline{h} = -k$, $h\, j\, \overline{h} = -i$, $h\, k\, \overline{h} = j$, which is the exact column action of $\boldsymbol{U}_{even}$.*
5. *Modulo 5, the elements of $\mathcal{B}_+ = \{1, 11, 19, 29\}$ reduce to $\{1, 1, 4, 4\} = \{\pm 1\}^2$ (quadratic residues mod 5, where $(b/5) = +1$), while $\mathcal{B}_- = \{7, 13, 17, 23\}$ reduce to $\{2, 3, 2, 3\}$ (quadratic non-residues mod 5, where $(b/5) = -1$). Squaring the listed values of χ_{60} gives $+1$ exactly on the residues $\equiv 1, 4 \pmod 5$, so $\chi_{60}^2 = ((\cdot)/5)$; since b and $b + 30$ have the same residue mod 5, χ_{60}^2 descends to $\mathbb{Z}/30\mathbb{Z}$.*
6. *$\det J_g = 3(5) - 1(1) = 14$ and $Tr(J_g) = 8 \Longrightarrow \lambda^2 - 8\lambda + 14 = 0 \Longrightarrow \lambda = 4 \pm \sqrt{2}$. Ordering vertices and assigning each the smallest available color in $\{1, \ldots, \Delta + 1\}$ proves the $(\Delta + 1)$-coloring bound. **Part 6′:** greedy colouring gives $\Delta_{\mathcal{G}} + 1$ colours. For $\Delta \geq 1$, $\det(J_g^\Delta - I) = (\Lambda_g^\Delta - 1)(T_g^\Delta - 1) \neq 0$, where $\Lambda_g = 4 - \sqrt{2} > 1$ and $T_g = 4 + \sqrt{2}$. So each condition $c_\mu \equiv J_g^\Delta c_\nu \pmod{\mathbb{Z}^2}$ (and, for $\mu \neq \nu$, $c_\mu \equiv c_\nu$) cuts out a proper closed subset of the finite tuple of centres. The complement is open and dense, so it contains a rational tuple, and the finitely many differences $c_\mu - J_g^\Delta c_\nu$ stay at torus distance at least some $\delta > 0$ from $\mathbb{Z}^2$. A common point of two lifted rectangles would force $c_\mu - J_g^\Delta c_\nu \equiv J_g^\Delta e_\nu - e_\mu$ with $|e| \leq 2r_0(|\boldsymbol{v}_r| + |\boldsymbol{v}_t|)$, which is impossible once $r_0 < \delta/(2(\|J_g^{\Delta_{max}}\| + 1)(|\boldsymbol{v}_r| + |\boldsymbol{v}_t|))$.* Instance ([COMPUTED]): *with 8 colours, $\Delta_{max} = 4$ and centres of denominator 997, $\delta = 0.0058$, $\|J_g^4\| = 859.3$, and any $r_0 < 1.57 \times 10^{-6}$ works; $\det(J_g^k - I) = 7, 161, 2569, 37513, 533057$ for $k = 1, \ldots, 5$.* ■

3.2 Lemma 5: Lower Manifold Standardization, Apollonius Realization, and Hurwitz Completeness

Lemma 5 (Lower Manifold Symmedian Partition, Binary Octahedral Scaling, Harmonic Bundle, 4D Hurwitz Completeness and 128-Bit Gram-TQF Arithmetic). **[PROVEN / CORRESPONDENCE]** Part 4 (corner-box comparison) and Part 5 (Theorem 4.1) after Hoard (2026e); Part 6(b) uses the sextic census of Kaihnsa et al. (2017).

Let the Lower Manifold $\mathcal{M}_{lower}$ consist of the centered 4 × 4 × 4 voxel cube $V = [-2\mathcal{C}_0, 2\mathcal{C}_0]^3$ (edge $L_{cube} = 4\mathcal{C}_0 = \lambda/4$) and the concentric 2 × 2 × 2 Quaternion Core $K = [-\mathcal{C}_0, \mathcal{C}_0]^3$ (edge $L_{core} = 2\mathcal{C}_0 = \lambda/8$). Then:

1. ***Symmedian-Quadrance Partition ([PROVEN]):*** *On a core edge baseline BC = [0, 2$\mathcal{C}_0$] inscribed in a circular cross-section Γ, the symmedian ray AD from a non-collinear state A ∈ Γ meets BC at $x_F = (Q_3/(Q_2 + Q_3))2\mathcal{C}_0$, and its isogonal conjugate (the median AM) meets BC at the single-cubit midpoint $x_M = \mathcal{C}_0 = \lambda/16$, with $x_F = x_M \iff Q_2 = Q_3$.*
2. ***Binary Octahedral Group Core-to-Cube Scaling ([PROVEN]):*** *For a unit quaternion $g \in S^3$ with 3D rotation $\rho_g(\mathbf{x}) = g\mathbf{x}g^{-1}$, the similarity map $\mathbf{x} \mapsto 2\rho_g(\mathbf{x})$ maps the centered core $K = [-\mathcal{C}_0, \mathcal{C}_0]^3$ onto the centered voxel cube $V = [-2\mathcal{C}_0, 2\mathcal{C}_0]^3$ if and only if $\rho_g \in O$ (the 24-element octahedral rotation group), i.e., if and only if g lies in the **48-element binary octahedral group $2O \subset S^3$**, which contains all 24 Hurwitz units $\mathcal{H}_1$ (including h).*
3. ***Projective Harmonic Bundle & Apollonius Circle Realization ([PROVEN]):***
   - ***(a) Edge-Length Harmonic Bundle:*** *Along the edge-length baseline where B = 0, C = $L_{core} = 2\mathcal{C}_0 = \lambda/8$, and E = $L_{cube} = 4\mathcal{C}_0 = \lambda/4$, the projective harmonic conjugate of E with respect to B and C is the **Klein-Wave Merge Boundary**:*

     $$x_F = \frac{4}{3}\mathcal{C}_0 = \frac{\lambda}{12} \quad \left(\text{phase } kx_F = \frac{\pi}{6},\ \frac{Q_3}{Q_2} = 2\right), \qquad (B, C; E, F) = \left(0, 2\mathcal{C}_0; 4\mathcal{C}_0, \frac{4}{3}\mathcal{C}_0\right) = -1,$$

     satisfying Newton's identity $MB^2 = MC^2 = MF \cdot ME = ((1/3)\mathcal{C}_0)(3\mathcal{C}_0) = \mathcal{C}_0^2$ about M = $\mathcal{C}_0$. (In radial half-edges B = 0, C = $\mathcal{C}_0$, E = $2\mathcal{C}_0$ from the cube center, the conjugate is $F_{rad} = (2/3)\mathcal{C}_0 = \lambda/24$ with full diameter $2F_{rad} = (4/3)\mathcal{C}_0$).
   - ***(b) Apollonius Circle Realization:*** *Every point A (off line BC) on the **Apollonius circle $\mathcal{A}$ centered at E = $4\mathcal{C}_0$ of radius $R_{\mathcal{A}} = \sqrt{(EB \cdot EC)} = 2\sqrt{2}\,\mathcal{C}_0$** defines a triangle △ABC whose circumcircle Γ has EA as its tangent at A, whose side quadrances satisfy $Q_3/Q_2 = AB^2/(AC^2) = 2$, and whose symmedian from A intersects BC at F = (4/3)$\mathcal{C}_0$.*
4. ***4D Hurwitz Lift & Resolution of the 3D Prime Gap ([ESTABLISHED / PROVEN]):*** *By Legendre's three-square theorem, the 3D slice w = 0 ($\mathbb{Z}^3$) omits every prime p ≡ 7 (mod 8) (7, 23, 31, 47, ..., comprising 1/4 of all primes). In the infinite 4D Hurwitz integer ring $\mathcal{H} = \mathbb{Z}^4 \cup (\mathbb{Z} + 1/2)^4$, every positive integer n has $24\sigma_{odd}(n)$ representations of norm N(q) = n, so every odd prime p has 24(p + 1) > 0 Hurwitz states (and 24 for p = 2), with every integer state on a prime shell automatically primitive (gcd(w,x,y,z) = 1). Within the finite bounded box $[-2,2]^4$, Hurwitz norms reach at most 16, and the primes realized are **exactly {2, 3, 5, 7, 13}**; the prime 11 is not a norm of any Hurwitz point in the box ([PROVEN]).* Corner-anchored comparison ([COMPUTED], after Hoard 2026e): *the integer box $\{0, ..., 3\}^4$ realizes every prime up to 23 and also 31, missing 29; the box $\{0, ..., 3\}^3$ misses the primes 7 and 23 (both ≡ 7 (mod 8)) among the primes up to its maximum norm 27.*
5. ***Non-Collinear Prime Triangles vs. Collinear Exclusion ([PROVEN]):*** *If three distinct collinear points of any Euclidean space have integer pairwise quadrances and two of them are prime, the third is not prime (Hoard 2026e Thm. 4.1). In particular, no three distinct collinear lattice points have all three pairwise quadrances (Q_1, Q_2, Q_3) prime. Non-collinear prime triangles exist in $[-2,2]^3$ (e.g., A = (0,0,0), B = (1,1,1), C = (1,0,2) with Q = (2,5,3) and $A_R = 24 > 0$).*
6. ***Overflow-Safe 128-Bit Gram-TQF Arithmetic & Real Sextic / K3 Correspondence ([PROVEN / CORRESPONDENCE]):***
   - ***(a) Two-Product 128-Bit Gram Evaluation:*** *For 4D difference vectors **u**, **v** ∈ $\mathbb{Z}^4$ with coordinate horizon $|u_k|, |v_k| \le 10^6 < 2^{20}$, the quadrances and dot product satisfy $Q_2, Q_3, |\langle \mathbf{u}, \mathbf{v}\rangle| \le 4 \times 10^{12} < 2^{42}$. Evaluating Lagrange's Gram identity*

     $$A_R = 4\left(\|\mathbf{u}\|^2\|\mathbf{v}\|^2 - \langle \mathbf{u}, \mathbf{v}\rangle^2\right) = 4\left(Q_2Q_3 - \langle \mathbf{u}, \mathbf{v}\rangle^2\right) \ge 0$$

*keeps $Q_2\,Q_3 < 2^{84}$ and $\langle \mathbf{u}, \mathbf{v}\rangle^2 < 2^{84}$, requiring **two widening 64 × 64 → 128-bit multiplications** (`__uint128_t`) with $A_R < 2^{86}$ (whereas naive evaluation of S^2 reaches $\sim 2^{91}$).*

- ***(b) Real Plane Sextics & Degree-2 K3 Surfaces:** Smooth plane sextic curves in $\mathbb{P}^{27}_{\mathbb{R}} \setminus \Delta$ (the space of real sextic forms minus the discriminant hypersurface Δ) have **64 rigid isotopy classes** grouped into 56 topological types, of which 48 types are unsplit and 8 types split into dividing and non-dividing pairs ($48 + 2 \cdot 8 = 64$, matching the cardinality of the 64-voxel lattice **[CORRESPONDENCE]**). A smooth sextic has **Degree 6** in $\mathbb{P}^2$, **Degree 12** when pulled back to a conic $\Gamma \cong \mathbb{P}^1$ or on its Hessian curve (yielding $6 \times 12 = 72$ complex flexes, at most 24 real), **Degree 30** on its dual curve $C^\vee$, and its double cover $w^2 = f(x,y,z)$ branched along the sextic is a **degree-2 algebraic K3 surface** (**[ESTABLISHED]**).*

Proof.

1. *Follows directly from Lemma 3 (Part 1) on $[0, 2\mathcal{C}_0]$.*
2. *The dilation $\mathbf{x} \mapsto 2\mathbf{x}$ bijects $K = [-1,1]^3$ onto $V = [-2,2]^3$, so $2\rho_g(K) = V \iff \rho_g(K) = K \iff \rho_g \in O$ (the 24 rotational symmetries of the centered cube). Under $\pi : S^3 \xrightarrow{2:1} SO(3)$, $\pi^{-1}(O)$ has order 48 and contains the 24-element binary tetrahedral group $2T = \mathcal{H}_1$.*
3. *Solving $(x_F - 0)/(2\mathcal{C}_0 - x_F) = (4\mathcal{C}_0 - 0)/(4\mathcal{C}_0 - 2\mathcal{C}_0) = 2$ gives $x_F = (4/3)\mathcal{C}_0 = \lambda/12$. For any point A off line BC on the circle centered at $E = 4\mathcal{C}_0$ with radius $R_{\mathcal{A}} = 2\sqrt{2}\,\mathcal{C}_0$, we have $EA^2 = 8\mathcal{C}_0^2 = (4\mathcal{C}_0)(2\mathcal{C}_0) = EB \cdot EC$. By the converse of the tangent-secant theorem, line EA is tangent to the circumcircle Γ of $\triangle ABC$ at A, and $\triangle EAB \sim \triangle ECA \implies AB^2/(AC^2) = EB/(EC) = 2$. By Lemma 3, the symmedian from A meets BC at the harmonic conjugate $F = (4/3)\mathcal{C}_0$.*
4. *The infinite-lattice statements are standard by Legendre's three-square theorem and Hurwitz's norm theorem (Conway & Smith, 2003). For the box: an integer coordinate in $[-2,2]$ has square 0, 1 or 4, so an integer point with k coordinates of absolute value 2 and j of absolute value 1 ($k + j \le 4$) has norm $4k + j$, which takes exactly the values 0–10, 12, 13, 16. A half-integer coordinate in $[-2,2]$ is $\pm 1/2$ or $\pm 3/2$, so a half-integer point with a coordinates equal to $\pm 3/2$ has norm $(9a + (4 - a))/4 = 2a + 1 \in \{1, 3, 5, 7, 9\}$. The union contains the primes 2, 3, 5, 7, 13 and omits 11 (which needs a coordinate of absolute value 3, e.g. $9 + 1 + 1$).*
5. Coordinate-free: *collinearity gives $\sqrt{Q_3} = |\sqrt{Q_1} \pm \sqrt{Q_2}|$, so $Q_3 = Q_1 + Q_2 \pm 2\sqrt{Q_1\,Q_2}$. Since $Q_3 \in \mathbb{Z}$, $Q_1\,Q_2$ is a perfect square, which for primes forces $Q_1 = Q_2 = p$; then $Q_3 \in \{0, 4p\}$, and 0 is excluded by distinctness (checked on all prime pairs below 2,000).* Lattice version: *if A, B, C are distinct and collinear on a lattice line, write $B - A = s\mathbf{v}$ and $C - A = t\mathbf{v}$ with $\mathbf{v}$ a primitive integer vector of norm $N = \|\mathbf{v}\|^2$ and $s, t \in \mathbb{Z} \setminus \{0\}$, $s \neq t$. The three pairwise quadrances are $s^2 N$, $t^2 N$, and $(s - t)^2 N$. For $s^2 N$ and $t^2 N$ to be prime, we must have $s^2 = t^2 = 1$ and $N = p$ prime; since $s \neq t$, $t = -s$, so $(s - t)^2 N = 4p$, which is composite.*
6. *In $\mathbb{Z}^4$ with $|u_k|, |v_k| \le 10^6$, $\|\mathbf{u}\|^2 \le 4 \times 10^{12} < 2^{42}$, so $Q_2\,Q_3 \le 1.6 \times 10^{25} < 2^{84}$, requiring two 64 × 64 → 128-bit products. For the double cover $\pi: X \to \mathbb{P}^2$ branched along a smooth sextic $B \sim 6H$, the ramification formula gives $K_X = \pi^*(K_{\mathbb{P}^2} + (1/2)B) = \pi^*(-3H + 3H) = 0$, and $(\pi^*H)^2 = 2H^2 = 2$. Together with $h^1(\mathcal{O}_X) = 0$ (standard), X is a K3 surface of degree 2 (**[ESTABLISHED]**).* ■

3.3 Lemma 10: Global Nesting and Spatial Separation Theorem

Lemma 10 (Global Nesting, Disjoint Spherical Shells and Cross-Manifold Separation Gap). **[PROVEN]** Tier data from Hoard (2026a).

In units of $\mathcal{C}_0$, let the 5 toroidal tiers T_n ($n = 0, \ldots, 4$) have major radii $R_n = 5n^2 + 15n + 15 \in \{15, 35, 65, 105, 155\}$ and minor radii $r_n = 2n + 4 \in \{4, 6, 8, 10, 12\}$, and let the centered Lower Manifold voxel cube be $V = [-2, 2]^3$. Then:

1. ***Embedded Tori:** Since $r_n < R_n$ for every $n \in \{0, \ldots, 4\}$, each T_n is a smoothly embedded torus in $\mathbb{R}^3$.*
2. ***Disjoint Spherical Shells Under Arbitrary Rotations:** Regardless of its 3D orientation $\mathrm{Rot}(T_n) \in SO(3)$, tier T_n is contained strictly within the closed spherical shell*

$$\mathcal{A}_n = \left\{\mathbf{x} \in \mathbb{R}^3 : R_n - r_n \le \|\mathbf{x}\| \le R_n + r_n\right\} \in \left\{[11,19],\,[29,41],\,[57,73],\,[95,115],\,[143,167]\right\}\mathcal{C}_0,$$

and consecutive shells $\mathcal{A}_n$ and $\mathcal{A}_{n+1}$ are strictly disjoint with radial wall clearances $\Delta_{n,n+1} = (R_{n+1} - r_{n+1}) - (R_n + r_n) = (6n + 10)\mathcal{C}_0 \in \{10, 16, 22, 28\}\mathcal{C}_0$.

3. ***Strict Separation Between Upper and Lower Manifolds:*** *The centered voxel cube $V = [-2, 2]^3$ is contained in the closed ball $\|\boldsymbol{x}\| \le \sqrt{2^2 + 2^2 + 2^2} = 2\sqrt{3}\,\mathcal{C}_0 \approx 3.4641\mathcal{C}_0$. Consequently, the Euclidean distance between the Lower Manifold V and the innermost Upper Manifold tier T_0 is at least:*

$$\operatorname{dist}(V, T_0) \ge (R_0 - r_0)\mathcal{C}_0 - 2\sqrt{3}\mathcal{C}_0 = \left(11 - 2\sqrt{3}\right)\mathcal{C}_0 \approx 7.5359\mathcal{C}_0 \approx 4.050\ \text{m}.$$

In particular, $\mathcal{M}_{upper} \cap \mathcal{M}_{lower} = \emptyset$, and the entire 5-tier geometric engine is contained in the closed ball of radius $(R_4 + r_4)\mathcal{C}_0 = 167\mathcal{C}_0 \approx 89.76$ m. (Cube 3 = $[-4, 4]^3$ of Lemma 8 Step 5 lies in $\|\boldsymbol{x}\| \le 4\sqrt{3}\,\mathcal{C}_0$, at distance at least $(11 - 4\sqrt{3})\,\mathcal{C}_0 \approx 4.07\,\mathcal{C}_0$ from T_0, by the same argument.)

Proof.

1. *Every point $\boldsymbol{x} \in T_n$ lies at distance r_n from some point $\boldsymbol{c}$ on the major core circle of T_n, which has $\|\boldsymbol{c}\| = R_n$. By the triangle inequality, $R_n - r_n \le \|\boldsymbol{c}\| - \|\boldsymbol{x} - \boldsymbol{c}\| \le \|\boldsymbol{x}\| \le \|\boldsymbol{c}\| + \|\boldsymbol{x} - \boldsymbol{c}\| = R_n + r_n$. Since $O \in SO(3)$ preserves Euclidean norms $\|\cdot\|$, this shell bound holds identically for every orientation $\mathrm{Rot}(T_n)$.*
2. *Evaluating $[R_n - r_n, R_n + r_n]$ for n = 0, 1, 2, 3, 4 gives $[15 - 4, 15 + 4] = [11, 19]$, $[35 - 6, 35 + 6] = [29, 41]$, $[65 - 8, 65 + 8] = [57, 73]$, $[105 - 10, 105 + 10] = [95, 115]$, and $[155 - 12, 155 + 12] = [143, 167]$, with gaps $29 - 19 = 10$, $57 - 41 = 16$, $95 - 73 = 22$, and $143 - 115 = 28$.*
3. *For $\boldsymbol{y} = (y_1, y_2, y_3) \in V = [-2, 2]^3$, $\|\boldsymbol{y}\| = \sqrt{y_1^2 + y_2^2 + y_3^2} \le \sqrt{4 + 4 + 4} = 2\sqrt{3} \approx 3.4641 < 11 = R_0 - r_0$. Thus $\operatorname{dist}(V, T_0) \ge (11 - 2\sqrt{3})\mathcal{C}_0 > 0$, proving $\mathcal{M}_{upper} \cap \mathcal{M}_{lower} = \emptyset$.* ■

3.4 Lemma 11: The Winding Dichotomy Theorem (Rational Closure vs. Golden Ergodicity)

Lemma 11 (Winding Dichotomy: Rational Stator Clock Closure, Golden Clifford Ergodicity and the Three-Distance Theorem). **[ESTABLISHED / PROVEN]** Part 3 is the three-distance theorem of Steinhaus and Sós (Sós 1958); Part 4 after Hoard (2026d), Proposition 4.1.

Let $\mathbb{T}^2 = \mathbb{R}^2 / (2\pi\mathbb{Z})^2$ be a flat 2-torus (including the Golden Clifford Torus $x^2 + y^2 = r_1^2 = \varphi^2/(\varphi + 2)$, $z^2 + w^2 = r_2^2 = 1/(\varphi + 2)$ in $S^3 \subset \mathbb{R}^4$, which has flat induced metric $ds^2 = r_1^2\, d\theta_1^2 + r_2^2\, d\theta_2^2$), and consider the linear phase flow $\gamma(t) = (\omega_1 t, \omega_2 t) \pmod{2\pi}$ with $\omega_2 \ne 0$. Then:

1. ***Rational Closure Regime (The Stator Clock):*** *If the frequency ratio $\omega_1/\omega_2 = a/b \in \mathbb{Q}$ is rational in lowest terms (gcd(a,b) = 1), then $\gamma(t)$ is periodic and closes to the identity after b turns in θ_2 and a turns in θ_1 ($T_{close} = 2\pi b/\omega_2$). The integer winding law $W_n = 16 - 2n \in \mathbb{Z}$ of the 5-tier stator operates strictly in this rational regime, so every stator orbit closes (***[PROVEN]***).* Remark (***[HEURISTIC]***): *the exact 30-pulse (0.5 s) caesium closure is pulse arithmetic of $W_\Sigma = 60$ (Proposition 11.4); it does not follow from torus winding, and any link between the two is heuristic.*
2. ***Irrational Ergodic Regime (The Golden Subdivision & Offload Layer):*** *If the frequency ratio $\omega_1/\omega_2 \notin \mathbb{Q}$ is irrational (in particular, $\omega_1/\omega_2 = \varphi = (1 + \sqrt{5})/2$), then $\gamma(t)$ **never closes** ($\gamma(t_1) \ne \gamma(t_2)$ for all $t_1 \ne t_2$), and by the Kronecker–Weyl theorem its trajectory is dense and equidistributed on $\mathbb{T}^2$. Note that the geometric radius ratio $r_1/r_2 = \varphi$ and the kinematic frequency ratio ω_1/ω_2 are independent parameters: ergodicity requires $\omega_1/\omega_2 \notin \mathbb{Q}$.*
3. ***Steinhaus-Sós Three-Distance Theorem & Fibonacci Two-Gap Perfection:*** *For any real number α and integer $N \ge 1$, the N points $\{k\alpha \bmod 1 : 0 \le k < N\}$ partition the unit circle $\mathbb{R}/\mathbb{Z}$ into N intervals having **at most three** distinct lengths. When $\alpha = 1/\varphi = \varphi - 1$ and $N = F_m$ is a **Fibonacci number** (e.g., $N \in \{5, 8, 13, 21, 34, 55, 89\}$), the partition has **at most two** distinct gap lengths, in the exact golden ratio φ: 1, with zero collisions (whereas binary splitting $\alpha = 1/2$ collapses all $N \ge 2$ points onto just 2 positions $\{0, 1/2\}$).*
4. ***Proposition 11.4 (Clock Closure)*** **[PROVEN]**, ***after Hoard 2026d Prop. 4.1:*** *$\nu_{Cs} = 9{,}192{,}631{,}770 = 2 \cdot 3^2 \cdot 5 \cdot 7^2 \cdot 47 \cdot 44{,}351 \equiv 30 \pmod{60}$. With pulses of 1/60 s, a window of k pulses holds whole periods of the caesium clock iff $2 \mid k$; of caesium and 40 Hz iff $6 \mid k$; of caesium and 432 Hz iff $10 \mid k$; and of all three iff $30 \mid k$. The shortest common window is 30 pulses (0.5 s). More generally, integer carriers $f_1, \ldots, f_r$ share a window of k pulses iff $\mathrm{lcm}_i\,(60/(\gcd(60, f_i)))$ divides k.*
5. ***Golden Radii Do Not Force Ergodicity (*****[PROVEN]*****):*** *The radius ratio and the frequency ratio are independent parameters. A winding with rational frequency ratio on the golden Clifford torus closes (for example the (2,3) trefoil of Lemma 8, Proposition 6.5); only an irrational frequency ratio gives an ergodic path.*

Proof.

1. ***Proof of Parts (1) and (2):*** *A trajectory closes at $t > 0$ iff $\omega_1 t = 2\pi a$ and $\omega_2 t = 2\pi b$ for non-zero integers a, b, which holds iff $\omega_1/\omega_2 = a/b \in \mathbb{Q}$. When $\omega_1/\omega_2 \notin \mathbb{Q}$, no such $(a, b) \neq (0, 0)$ exists, so γ is injective, and for every non-zero Fourier mode $(m_1, m_2) \in \mathbb{Z}^2 \setminus \{(0,0)\}$, $m_1\omega_1 + m_2\omega_2 \neq 0$, so $\lim_{T\to\infty} (1/T)\int_0^T e^{i(m_1\omega_1+m_2\omega_2)t}\,dt = 0$, proving equidistribution and density by Weyl's criterion. On the Golden Clifford Torus, $\varphi^2 + 1 = (\varphi + 1) + 1 = \varphi + 2$, so $r_1^2 + r_2^2 = (\varphi^2 + 1)/(\varphi + 2) = 1$.*
2. ***Proof of Part (3):*** *By the Three-Distance Theorem (Steinhaus; Sós, 1958), for any irrational α, the gaps between $\{k\alpha \bmod 1\}_{k=0}^{N-1}$ take at most 3 values: η_1, η_2, and $\eta_1 + \eta_2$, where η_1, η_2 are the distances of the closest left and right approximations connected to the continued fraction convergents p_m / q_m of α. For $\alpha = 1/\varphi = [0; 1, 1, 1, \ldots]$, the denominators of the convergents are the Fibonacci numbers $q_m = F_m$. When $N = F_m$ equals a convergent denominator, all compound gaps $\eta_1 + \eta_2$ have been subdivided into η_1 and η_2 with $\eta_1/\eta_2 = \varphi$, leaving **only two** distinct gap lengths. With $F_1 = F_2 = 1$ and $N = F_m$ ($m \geq 3$), there are **F_{m-1} gaps of length $\varphi^{-(m-2)}$ and F_{m-2} gaps of length $\varphi^{-(m-1)}$**.* Check: *using $\varphi^k = F_k\varphi + F_{k-1}$, the total length is $(F_{m-1}\varphi + F_{m-2})/\varphi^{m-1} = \varphi^{m-1}/\varphi^{m-1} = 1$, and the count is $F_{m-1} + F_{m-2} = F_m = N$. For $N = 8$: 5 gaps of $\varphi^{-4} \approx 0.1459$ and 3 of $\varphi^{-5} \approx 0.0902$.*
3. ***Proof of Parts (4) and (5):*** *k pulses of 1/60 s hold whole periods of f Hz iff $kf/60 \in \mathbb{Z}$, i.e. iff $60/(\gcd(60, f))$ divides k. Here $\nu_{Cs}/30 = 306{,}421{,}059$ is odd, so $\gcd(60, \nu_{Cs}) = 30$ and the condition is $2 \mid k$; $\gcd(60, 40) = 20$ gives $3 \mid k$; $\gcd(60, 432) = 12$ gives $5 \mid k$; and $\operatorname{lcm}(2, 3, 5) = 30$. Part (5) is Part (1) applied to the frequency ratio, which Part (2) shows is independent of r_1/r_2.* ■

4 Layer III — Cross-Manifold Dynamics, Holonomy, and the Fokker-Planck Sieve (Lemmas 6, 7, 8, 12, and 14)

4.1 Lemma 6: Cross-Gap 5-Moment Vandermonde Conservation

Lemma 6 (Cross-Gap 5-Moment Vandermonde Invertibility and Zero-Residual Exterior Locking). **[PROVEN]** Parts 3–4 relative to the profile definitions of OpenAI (2026), with its Lemma 4.3 and Corollary A.3 re-proved here.

*Across the radial separation zone bridging the Lower Manifold ($r \leq 2\sqrt{3}\,\mathcal{C}_0$) and Upper Manifold ($r \geq 11\mathcal{C}_0$), let $X = r^2/(2q) \in [X_a, X_b]$ denote the similarity radial coordinate at a fixed concentration scale $q > 0$, let (U, E) be the axial and azimuthal velocity profiles with angular momentum profile $H = \sqrt{(2X)}\,E$, and define the **five cumulative radial moments**:*

$$M(X) = \int_0^X U\,dx, \quad I(X) = \int_0^X H\,dx, \quad J(X) = \int_0^X UH\,dx, \quad S(X) = \int_0^X \left(U^2 - \frac{E^2}{2}\right) dx, \quad C_p(X) = \int_0^X \frac{E^2}{2x}\,dx.$$

Then:

1. ***Generalized Vandermonde Invertibility ([PROVEN]):*** *For any five ordered disjoint radial sub-intervals $I_0 < I_1 < I_2 < I_3 < I_4$ in $(0, \infty)$ (representing five radial coupling bands indexed by the 5 tiers), any five continuous non-negative, non-zero localization bumps β_j supported in I_j, and any five distinct real exponents $\alpha_0 < \alpha_1 < \ldots < \alpha_4$, the 5×5 moment matrix $B_{ij} = \int_0^\infty x^{\alpha_i}\beta_j(x)\,dx$ is strictly invertible ($\det B \neq 0$).*
2. ***Quadratic Moment Solvability ([PROVEN]):*** *If $\mathcal{Q}$ is a bounded symmetric bilinear map and $8\|B^{-1}\|^2\,\|\mathcal{Q}\|\,\|d\| \leq 1$, the nonlinear moment-matching equation $Bc + \mathcal{Q}(c,c) = d$ has a unique solution c in the ball $\|c\| \leq 2\|B^{-1}\|\|d\|$.*
3. ***Exterior Agreement ([PROVEN]):*** *If two profile pairs (U_1, E_1) and (U_2, E_2) agree pointwise for $X \geq X_b$ and their five cumulative moments agree at X_b ($\Delta m(X_b) = 0$), then $m_1(X) = m_2(X)$ for all $X \geq X_b$ ([PROVEN]). **Stress agreement ([PROVEN], relative to the profile definitions (4.7)–(4.11) of OpenAI 2026):** the exterior stress $\mathcal{T}_0(X)$ also agrees for all $X \geq X_b$. With the radial ODE integration constants set to zero, the stress scalars integrate in closed form (preprint Lemma 4.3, re-verified symbolically):*

$$Q_s = -W + \frac{(1-h)I - D\eta I_\eta - dJ_\eta + 2(h-D)\eta J}{XH}, \qquad N_s = -WU + \frac{D(M - \eta M_\eta) + 4h\eta S - dS_\eta}{X} + 4A\eta\Pi - d\Pi_\eta,$$

where $W = 1 - (2D\eta M + d\,\partial_\eta M)/X$, $H = \sqrt{(2X)}\,E$ and $\Pi = \Pi_{ax} + C_p$. Here η is the similarity time variable, h, D, d, A are the similarity parameters, S_q, S_n are the stress integrands, and Π_{ax}, F, p_s, s are the axial pressure and stress terms, all as defined in OpenAI (2026), eqs. (4.7)–(4.11); subscripts η denote ∂_η ([DEFINITION]). (The bands $I_0, \ldots, I_4$ in Part 1 are any five disjoint sub-intervals of the gap; indexing them by the tiers is a labelling, not a geometric identification, since by Lemma 10 no tier lies in the gap.)

4. ***The Five-Moment Jacobian: 2 + 3 Block Structure ([PROVEN]; after OpenAI 2026 Cor. A.3):*** *For two U-bumps and three E-bumps around $U_0 = u_c$ and $E_0 = e\ f\ x^\alpha$, the Jacobian of (M, I, J, S, C_p) is invertible if and only if $\alpha \notin \{-1/2, 1/2, 3/2\}$. After the row operations $J \to J - u_c I$ and $S \to S - 2u_c M$ it splits into a U-block with exponents $\{0, \alpha + 1/2\}$ and an E-block with exponents $\{1/2, \alpha, \alpha - 1\}$. This is the form in which Part 1 is actually used; Part 1 remains true as stated.*

Proof.

1. *By multilinearity of the determinant across columns, $\det B = \int_{I_0\times\dots\times I_4} \det[x_j^{\alpha_i}]_{0\le i,j\le 4} \prod_{j=0}^{4} \beta_j(x_j)\, dx_0\cdots dx_4$. A non-zero generalized polynomial $P(x) = \sum_{i=0}^{4} c_i\, x^{\alpha_i}$ with 5 distinct real exponents has at most 4 positive roots (divide by x^{α_0}, differentiate, and apply Rolle's theorem inductively 4 times). Thus $\det[x_j^{\alpha_i}] \neq 0$ with constant sign on the connected simplex $0 < x_0 < x_1 < x_2 < x_3 < x_4$. Since $\prod\beta_j \ge 0$ is positive on a set of positive measure, $\det B \neq 0$.*
2. *On the closed ball $\|c\| \le r = 2\|B^{-1}\|\|d\|$, the map $\Phi(c) = B^{-1}(d - \mathcal{Q}(c,c))$ satisfies $\|\Phi(c)\| \le r/2 + \|B^{-1}\|\|\mathcal{Q}\|r^2 \le r/2 + r/4 < r$, and $\|\Phi(c) - \Phi(c')\| = \|B^{-1}\mathcal{Q}(c - c', c + c')\| \le 2r\|B^{-1}\|\|\mathcal{Q}\|\|c - c'\| \le (1/2)\|c - c'\|$, so Banach's contraction mapping theorem yields a unique solution.*
3. *For $X \ge X_b$, $\Delta m(X) = \Delta m(X_b) + \int_{X_b}^{X} 0\, dx = 0$. For the stress: differentiating the displayed right-hand sides in X recovers the integrands $H\, S_q$ and S_n of the defining ODEs exactly, and both sides vanish at $X = 0$, so the formulas hold. They express Q_s and N_s algebraically in $(X, \eta, U, E, m, \partial_\eta m)$, and $\mathcal{T}_0 = F(p_s - s)$ adds only local radial derivatives of U and E. Every input agrees for $X \ge X_b$, hence so does $\mathcal{T}_0$. Nothing here uses the blowup construction.*
4. *After the row operations the U-rows are $\int \delta U\, x^0$ and $\int \delta U\, x^{\alpha+1/2}$, and the E-rows are $\int \delta E\, x^{1/2}$, $\int \delta E\, x^\alpha$ and $\int \delta E\, x^{\alpha-1}$, up to non-zero factors. Part 1 applies to each block exactly when its exponents are distinct: $0 \neq \alpha + 1/2$ iff $\alpha \neq -1/2$, and $1/2, \alpha, \alpha - 1$ are distinct iff $\alpha \notin \{1/2, 3/2\}$. Numerically, the singular-value ratio of the Jacobian drops to about 10^{-17} exactly at these three values and stays near 10^{-3} elsewhere ([COMPUTED]).* ■

4.2 Lemma 7: The 9 + 3 = 12 Affine Frame Bundle, Central Tier T_2, and Group Inverse H_4

Lemma 7 (The 9 + 3 = 12 Affine Frame Bundle $\mathfrak{aff}(3)$, Central Tier T_2 Equilibrium and S^3 Group Inverse H_4). [PROVEN / DEFINITION] Tier data from Hoard (2026a); gear chain from Hoard (2026b); valve kernel from Hoard (2026c); logarithm branch after Hoard (2026d).

1. ***The 9 + 3 = 12 Affine Frame Space $\mathfrak{aff}(3)$ ([PROVEN]):*** *Let the three 3D spatial groups $\mathcal{G}_1$, $\mathcal{G}_2$, $\mathcal{G}_3$ contribute three 3-vector micro-axis rows $\boldsymbol{e}_1$, $\boldsymbol{e}_2$, $\boldsymbol{e}_3 \in \mathbb{R}^3$ forming the 3×3 linear frame matrix $L = [\boldsymbol{e}_{a,b}] \in \mathbb{R}^{3\times3} \cong \mathfrak{gl}(3, \mathbb{R})$ (9 independent linear coordinates), and let $\boldsymbol{t} = (t_1, t_2, t_3)^T \in \mathbb{R}^3$ denote the system's combined 3D barycentric translation vector (3 independent translational coordinates). Then the space of affine transformations $\boldsymbol{x} \mapsto L\boldsymbol{x} + \boldsymbol{t}$ of $\mathbb{R}^3$ has exact dimension:*

$$\dim\left(\mathfrak{aff}(3)\right) = \dim\left(\mathfrak{gl}(3,\mathbb{R}) \ltimes \mathbb{R}^3\right) = \underbrace{9}_{\text{linear frame } L} + \underbrace{3}_{\text{barycentric translation } \mathbf{t}} = \mathbf{12},$$

equipped with the semidirect-product Lie bracket $[(A, \boldsymbol{a}), (B, \boldsymbol{b})] = (AB - BA, A\boldsymbol{b} - B\boldsymbol{a})$. (Defining $\boldsymbol{E}_a = \Sigma_b \boldsymbol{e}_{a,b}$ as sums of micro-axes would be linearly dependent on $\boldsymbol{e}_{a,b}$; the semidirect sum $\mathfrak{gl}(3,\mathbb{R}) \oplus \mathbb{R}^3$ makes all 12 coordinates linearly independent). The equality of cardinality 12 with the 12-tooth outer gear, the $6 \times 2 = 12$ voxel face-chirality channels, and the degree-12 Hessian is a [CORRESPONDENCE].

2. ***Central Tier T_2 as the Unique Arithmetic Mean & Median ([PROVEN]):*** *Across the 5 tiers $T_0, \dots, T_4$, the 3rd central tier T_2 ($n = 2$) is the unique arithmetic mean and median of both the minor radius and winding sequences:*

$$r_2 = \frac{1}{5}\sum_{n=0}^{4} r_n = \frac{r_0 + r_4}{2} = \mathbf{8}, \qquad W_2 = \frac{1}{5}\sum_{n=0}^{4} W_n = \frac{W_0 + W_4}{2} = \frac{W_\Sigma}{5} = \mathbf{12}, \qquad r_2 + W_2 = 20.$$

3. ***Markov Cycle Current & the S^3 Group Inverse H_4 ([ESTABLISHED / PROVEN / DEFINITION]):***
   - ***(a) Three-State Markov Cycle Current:*** *For a stationary irreducible Markov chain on the gear states $\{12, 10, 8\}$ with transition matrix K and stationary distribution π, probability conservation forces $J_{12,10} = J_{10,8} = J_{8,12} = J = (\Pi_+ - \Pi_-)/Z$ ($Z > 0$), so $J > 0 \iff \Pi_+ = K_{12,10}K_{10,8}K_{8,12} > K_{12,8}K_{8,10}K_{10,12} = \Pi_-$, with non-negative entropy production $\sigma = J \ln(\Pi_+ / \Pi_-) \ge 0$. (Note: for a general K, $\Pi_+ > \Pi_-$ is a design hypothesis; part (c) gives an explicit rational kernel for which it is proven.)*

- ***(b) Unique Group Inverse $H_4 \in S^3$ & BCH Cross-Product Expansion:*** *For any three holonomy quaternions H_1, H_2, $H_3 \in S^3$, there exists a unique compensating quaternion $H_4 \in S^3$ restoring the identity $H_1 H_2 H_3 H_4 = +1$, namely the group inverse:*

  $$H_4 = (H_1 H_2 H_3)^{-1} = \overline{H_3}\,\overline{H_2}\,\overline{H_1} \in S^3 \quad (\textbf{[PROVEN]}\ \text{group inverse};\ \textbf{[DEFINITION]}\ \text{as 4th manifold}\ \mathcal{M}^{(4)} \cong S^3).$$

  Writing $H_a = \exp(\epsilon_a)$ for small pure-quaternion drift vectors $\epsilon_a \in \mathfrak{su}(2) \cong \mathbb{R}^3$ (where $[\epsilon_a, \epsilon_b] = 2(\epsilon_a \times \epsilon_b)$), the Baker–Campbell–Hausdorff expansion yields:

  $$-\log H_4 = \epsilon_1 + \epsilon_2 + \epsilon_3 + \epsilon_1 \times \epsilon_2 + \epsilon_1 \times \epsilon_3 + \epsilon_2 \times \epsilon_3 + O\left(\|\epsilon\|^3\right).$$

  Logarithms use the principal branch $\log q = \phi\hat{u}$ with $\phi \in [0, \pi)$, defined on $S^3 \setminus \{-1\}$ (after Hoard 2026d).

- ***(c) A Pump That Provably Circulates ([PROVEN]; Hoard 2026c valve):*** *On the gear states $a = 12$, $b = 10$, $c = 8$, take symmetric positive weights W, the base walk $P^0_{ij} = W_{ij}/d_i$, the forward orientation $a \to b \to c \to a$, and multiply each reverse edge e by $\rho_e = 1/(1 + \lambda A_e)$ ($\lambda > 0$; $A_e \geq 0$, not all zero), returning the rejected mass to the diagonal. The resulting kernel $\hat{K}$ is rational, irreducible and row-stochastic, and its stationary current is*

  $$J = \frac{(1 - \rho_{ba}\rho_{cb}\rho_{ac})\,\Pi_+}{Z} > 0.$$

- ***(d) The Implied Fourth as a Parity Element ([PROVEN]):*** *Since $H_1 H_2 H_3 H_4 = 1$ is preserved by cyclic shifts, each of the four is determined by the other three: $H_1 = (H_2 H_3 H_4)^{-1}$, $H_2 = H_1^{-1} H_4^{-1} H_3^{-1}$, $H_3 = H_2^{-1} H_1^{-1} H_4^{-1}$. One lost element of a group of four is recoverable exactly; two are not. H_4 carries no information beyond its three parents, and offloading it returns only their product, never their individual histories. (One multiplication order must be fixed system-wide; see Lemma 12.)*

- ***(e) Positive Current on the 64-Cell Lattice ([PROVEN]):*** *On the torus $\Lambda = (\mathbb{Z}/4)^3$ take nearest-neighbour weights $w_1, w_2, w_3 > 0$ (the same at every cell), $W = 2(w_1 + w_2 + w_3)$, forward steps $+\boldsymbol{e}_k$ with probability w_k/W, reverse steps $-\boldsymbol{e}_k$ with probability $\rho_k\, w_k/W$ ($0 < \rho_k \leq 1$), and the remaining mass on the diagonal. The kernel is rational, irreducible and doubly stochastic, its unique stationary distribution is uniform, and every forward edge carries the stationary current*

  $$J_{x \to x+\mathbf{e}_k} = \frac{(1 - \rho_k)\, w_k}{64\, W} > 0 \quad \text{whenever } \rho_k < 1.$$

  For valves that vary from cell to cell (for example built from Q_h) the edge-wise statement fails: individual forward edges can carry negative current, while the net flux through each cross-section stays positive in all trials (companion paper, Problem P1).

Proof.

1. *The map $(L, \boldsymbol{t}) \mapsto (\boldsymbol{x} \mapsto L\boldsymbol{x} + \boldsymbol{t})$ is a vector space isomorphism from $\mathbb{R}^{3\times3} \times \mathbb{R}^3 \cong \mathbb{R}^{12}$ onto the affine transformations of $\mathbb{R}^3$, and composing two affine transformations gives $(I + \varepsilon_1 A, \varepsilon_1 \boldsymbol{a}) \circ (I + \varepsilon_2 B, \varepsilon_2 \boldsymbol{b}) - (I + \varepsilon_2 B, \varepsilon_2 \boldsymbol{b}) \circ (I + \varepsilon_1 A, \varepsilon_1 \boldsymbol{a}) = \varepsilon_1\varepsilon_2(AB - BA, A\boldsymbol{b} - B\boldsymbol{a})$.*
2. *For any 5-term arithmetic progression $a_n = a_0 + nd$ ($n = 0,1,2,3,4$), $(1/5)\Sigma_{n=0}^{4} a_n = a_0 + 2d = a_2$.*
3. *Stationarity $\Sigma_j \pi_i K_{ij} = \Sigma_j \pi_j K_{ji}$ at $i = 12$ and $i = 10$ gives $J_{12,10} = J_{8,12} = J_{10,8} = J$, and the Markov chain tree theorem gives $J = (\Pi_+ - \Pi_-)/Z$. In S^3, $(H_1 H_2 H_3)^{-1} = H_3^{-1} H_2^{-1} H_1^{-1} = \overline{H_3}\,\overline{H_2}\,\overline{H_1}$. Since $[\epsilon_a, \epsilon_b] = \epsilon_a \epsilon_b - \epsilon_b \epsilon_a = 2(\epsilon_a \times \epsilon_b)$ for pure quaternions, the $(1/2)[\epsilon_a, \epsilon_b]$ terms in the BCH formula equal $\epsilon_a \times \epsilon_b$. (c) P^0 is reversible, so by Kolmogorov's criterion $P^0_{ab}P^0_{bc}P^0_{ca} = W_{ab}W_{bc}W_{ca}/(d_a\, d_b\, d_c) = P^0_{ac}P^0_{cb}P^0_{ba}$. The valve leaves the forward entries unchanged and scales the three reverse entries, so $\Pi_- = \rho_{ba}\rho_{cb}\rho_{ac}\Pi_+ < \Pi_+$, and part (a) gives $J = (\Pi_+ - \Pi_-)/Z$ (checked on 5,000 exact rational cases). (d) Conjugating $H_1H_2H_3H_4 = 1$ by H_1 gives $H_2H_3H_4H_1 = 1$, and similarly for each rotation; solving each cyclic form for its last factor gives the formulas (checked on 500 groups of exact rational unit quaternions). If H_1 and H_2 are both unknown, every pair $(X, X^{-1}T)$ with $T = (H_3H_4)^{-1}$ is consistent with the record. (e) Each row sums to 1 by construction. Translation invariance gives column x the entries $P(x - \mathbf{e}_k, x) = w_k/W$, $P(x + \mathbf{e}_k, x) = \rho_k\, w_k/W$ and $P(x, x)$, the same multiset as row x, so columns also sum to 1 and the uniform distribution $\pi = 1/64$ is stationary. The steps $\pm\mathbf{e}_k$ generate Λ, so the chain is irreducible and π is unique. Then $J_{x\to x+\mathbf{e}_k} = (1/64)(P(x, x + \mathbf{e}_k) - P(x + \mathbf{e}_k, x)) = 1/64 \cdot (1 - \rho_k)w_k/W$ (checked on 200 random exact rational valves).* ■

4.3 Lemma 8: The 8-Step "π-Cycle" and C_9 Non-Degenerate K_5 4-Simplex Theorem

Lemma 8 (The 8-Step "π-Cycle" Zero-Point Sequence, C_9 Nine-Ray Construction and Non-Degenerate K_5 4-Simplex Lift). [PROVEN / DEFINITION] Rational Lorentz lift of Proposition 6.2 from Hoard

(2026c).

Parameterize the 8-step cycle by $\theta_k = k\pi/4$ ($k = 1, \ldots, 8$) on the centered cubes $K = [-1, 1]^3$ (Cube 1, edge $2\mathcal{C}_0$) and $V = [-2, 2]^3$ (Cube 2, edge $4\mathcal{C}_0$). Then:

- ***Step 1 ($\theta_1 = \pi/4$ — Circle Phase Lift)*** **[DEFINITION]:** *$s \in \mathbb{Z}/N\mathbb{Z} \mapsto (\cos(2\pi s/N), \sin(2\pi s/N)) \in S^1 \subset \mathbb{R}^2$ is well-defined and N-periodic, with origin anchor O = (0,0,0).* ***Step 1 → Step 2 Lift (*****[DEFINITION]*****):*** *Take N = 9 and send the circle phase onto the latitude circle of Lemma 8.2,*

$$\mathbf{x}(s) = -\tfrac{1}{3}(1,1,1) + \sqrt{\tfrac{8}{3}}\left(\cos\tfrac{2\pi s}{9}\,\mathbf{u} + \sin\tfrac{2\pi s}{9}\,\mathbf{v}\right), \qquad \mathbf{u} = \tfrac{(2,-1,-1)}{\sqrt{6}}, \quad \mathbf{v} = \tfrac{(0,1,-1)}{\sqrt{2}}.$$

  *Then $\|\boldsymbol{x}(s)\|^2 = 3$, consecutive phases are 40° apart, and $\boldsymbol{x}(0)$, $\boldsymbol{x}(3)$, $\boldsymbol{x}(6)$ are the cube corners $(1,-1,-1)$, $(-1,1,-1)$, $(-1,-1,1)$ (***[PROVEN]***). So the nine phases of Step 1 are exactly the nine ray endpoints of Step 2, and s mod 3 is the C_9 offset.*

- ***Step 2 ($\theta_2 = \pi/2$ — Conformal Null-Cone Lift, C_9 Nine-Ray Construction, & K_5 4-Simplex)*** **[PROVEN]:**

  1. ***Exact Conformal Null-Cone Embedding (*****[PROVEN]*****):*** *In Minkowski space $\mathbb{R}^{4,1}$ with null basis (e_o, e_∞) satisfying $e_o^2 = e_\infty^2 = 0$ and $e_o \cdot e_\infty = -1$, the conformal null-cone lift of $\boldsymbol{x} \in \mathbb{R}^3$ is:*

$$X(\mathbf{x}) = \mathbf{x} + \frac{1}{2}\|\mathbf{x}\|^2 e_\infty + e_o \cong \left(x,\, y,\, z,\, \frac{1}{2}Q(\mathbf{x}),\, 1\right) \in \mathbb{R}^5, \qquad -2\,X(\mathbf{x}) \cdot X(\mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|^2 = Q(\mathbf{x} - \mathbf{y}).$$

     (The factor 1/2 on $Q(\boldsymbol{x}) = \|\boldsymbol{x}\|^2$ is mandatory for the inner product to equal Euclidean quadrance). Each quaternion $q_a = w_a + x_a\,i + y_a\,j + z_a\,k$ acts via the left-multiplication matrix $Q_a = w_a\,I_4 + S(x_a, y_a, z_a) \in M_4(\mathbb{R})$ (scalar-plus-skew-symmetric) satisfying $Q_a^T Q_a = N(q_a) I_4$ and $\det Q_a = N(q_a)^2$.

  2. ***Lemma 8.2 (Nine Distinct Peripheral Rays from C_9 Offsets)*** **[PROVEN]:** *A regular tetrahedron Δ_0 inscribed in Cube 1 $K = [-1, 1]^3$ with polar vertex at (1,1,1) and peripheral vertices at $(1,-1,-1)$, $(-1,1,-1)$, $(-1,-1,1)$ is invariant under 120° rotations about its polar axis $\hat{\boldsymbol{p}} = (1/\sqrt{3})(1,1,1)^T$ (so rotating by {0°, 120°, 240°} yields only 3 distinct peripheral rays). Rotating Δ_0 about $\hat{\boldsymbol{p}}$ by the* ***C_9 phase offsets {0°, 40°, 80°}*** *(combining the rotor's C_3 with the tetrahedron's C_3) generates three distinct congruent regular tetrahedra $\Delta_1, \Delta_2, \Delta_3$ inscribed in the* ***circumsphere*** *$\|\boldsymbol{x}\| = \sqrt{3}\,\mathcal{C}_0$ of Cube 1, whose* ***9 peripheral vertices are all distinct*** *and lie equally spaced at 40° intervals ($2\pi j/9$, $j = 0, \ldots, 8$) around a common latitude circle of that sphere. Only Δ_1 (offset 0°) is inscribed in Cube 1 itself; the 40° and 80° copies have vertices outside K (largest coordinate ≈ 1.586). 2′.* ***Proposition 8.2′ (No Nine Cube-Vertex Rays)*** **[PROVEN]:** *No construction whatsoever can produce 9 distinct rays from O that all end at vertices of Cube 1, because Cube 1 has only 8 vertices. In particular, any requirement that the nine C_9 rays terminate on cube vertices is unsatisfiable; the rays must end on the circumsphere (as in Lemma 8.2) or on another polytope. 2″.* ***Proposition 8.2″ (Three of Nine)*** **[PROVEN]:** *Exactly three of the nine C_9 rays end at vertices of Cube 1, and they form one C_3 coset: the 0° tetrahedron Δ_1. (This is the exact reading of the corpus rule "3 of 9 streams pass to the cube".) 2‴.* ***Ray Endpoints (*****[DEFINITION]*****):*** *The nine C_9 rays terminate on the circumsphere $\|\boldsymbol{x}\| = \sqrt{3}\,\mathcal{C}_0$ of Cube 1, at the points $\boldsymbol{x}(s)$ of the Step 1 → Step 2 lift.*

  3. ***Lemma 8.3 (Paraboloid Quadrance Lift to a Genuine K_5 4-Simplex)*** **[PROVEN]:** *While the 4 vertices of Δ_a and their centroid O = (0,0,0) are affinely dependent in $\mathbb{R}^3$, lifting them to $\mathbb{R}^4$ via the quadrance coordinate $\boldsymbol{x} \mapsto (\boldsymbol{x}, \|\boldsymbol{x}\|^2) \in \mathbb{R}^4$ makes all* ***5 points affinely independent*** *(rank = 4), spanning a non-degenerate 4-simplex whose 1-skeleton is the complete pentagon graph K_5.*

- ***Step 3 ($\theta_3 = 3\pi/4$ — Four-Fold 1/16 Subdivision into 16 Congruent Half-Cubes)*** **[PROVEN]:** *Four successive volume/area halvings yield $A_4 = (1/2)^4 A_0 = (1/16)A_0$ and $L_4 = 2^4 = 16$ congruent layers. Because 16 is not a perfect cube m^3, a 3D cube cannot be partitioned into 16 congruent sub-cubes; instead, three halvings produce the $2 \times 2 \times 2 = 8$ core sub-cubes, and the fourth halving produces* ***16 congruent rectangular boxes (half-cubes)*** *(or the $4 \times 4 = 16$ square cells on a cube face).*

- ***Step 4 ($\theta_4 = \pi$ — 2 × Radial Dilation to Cube 2 & $E_{max} = 4$ Quadrance Scaling)*** **[PROVEN]:** *Dilating $K = [-1, 1]^3 \mapsto V = [-2, 2]^3$ ($\lambda = 2$) scales every quadratic quadrance energy functional $E = \Sigma k_{ij} Q_{ij}$ by $\lambda^2 = 2^2 = 4$ ($E_{max} = 4\,E_{comp}$). Note that the cycle phase at Step 4 is $\theta_4 = \pi$, whereas the acoustic plane-wave phase across the 4-cubit cube edge $4\mathcal{C}_0 = \lambda/4$ is $k(4\mathcal{C}_0) = \pi/2$.*

- ***Step 5 ($\theta_5 = 5\pi/4$ — (+,−,+) Sense-Reversing Fold & Golden Binomial Cube)*** **[PROVEN]:** *In the inward folding recursion $F_{r+1} = P_r(F_r)^{\sigma_r}$, where $F_r \in S^3$ is the frame at gear $r \in \{12, 10, 8\}$ and $P_r \in S^3$ is that gear's transfer quaternion, with $(\sigma_{12}, \sigma_{10}, \sigma_8) = (+1, -1, +1)$, quaternion inversion $q \mapsto q^{-1} = \overline{q} = \cos(\vartheta/2) - \hat{u}\sin(\vartheta/2)$ reverses the sense of rotation ($\vartheta \mapsto -\vartheta$, or axis $\hat{u} \mapsto -\hat{u}$). For any cube of side $n = m +$*

r with m = n/φ and r = n/φ² (1/φ + 1/φ² = 1), the volume expands identically as n³ = (m + r)³ = m³ + 3m²r + 3mr² + r³ (1 major cube, 3 slabs, 3 rods, 1 residual cube). ***Cube 3 ([DEFINITION]):*** *Since n/φ is irrational for rational n, an exact golden binomial cube cannot have rational sides. Cube 3 is the Fibonacci binomial cube W = [−4, 4]³ with n = 8 = F₆, m = 5, r = 3, so that 8³ = 125 + 225 + 135 + 27 exactly, and m/r = 5/3, n/m = 8/5 approximate φ. Its edge 8$\mathcal{C}_0$ = λ/2 continues the doubling 2 → 4 → 8 of Cube 1 → Cube 2 → Cube 3. Its corner radius 4√3 $\mathcal{C}_0$ ≈ 6.93 $\mathcal{C}_0$ is below $R_0 - r_0$ = 11 $\mathcal{C}_0$, so Cube 3 stays separated from tier T_0 by (11 − 4√3) $\mathcal{C}_0$ ≈ 4.07 $\mathcal{C}_0$ ≈ 2.19 m (Lemma 10). An exact golden version requires arithmetic in $\mathbb{Q}(\sqrt{5})$.*

- ***Step 6 (θ_6 = 3π/2 — D_3 Trefoil Symmetry) [PROVEN]:*** *Writing the (2,3)-trefoil as w(t) = x(t) + iy(t) = $i\,e^{-it} - 2i\,e^{2it}$ and z(t) = −sin(3t), we have $w(t + 2\pi/3) = e^{-2\pi i/3} w(t)$ and z(t + 2π/3) = z(t) (C_3 rotation), and (x, y, z)(−t) = (−x, y, −z)(t) (180° half-turn about the y-axis), generating the dihedral group D_3 of order 6. The C_3 subgroup alone has 3 fundamental arcs of normalized measure 1/3 (1/3 + 1/3 + 1/3 = 1); the full D_3 has 6 fundamental arcs of measure 1/6, so the arc t ∈ [0, π/3] and its six images rebuild the whole curve. With the C_3 axis along (1,1,1) and a half-turn axis along (1,−1,0), the six symmetries are the integer signed-permutation matrices generated by* $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ *and* $\begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$*, all in the cube group O of Lemma 5 Part 2, so every engine folds and unfolds with the same integer arithmetic.* ***Step 6 Projection ([PROVEN / DEFINITION]):***

  1. ***Proposition 6.1 (φ-Trefoil) [PROVEN]:*** *For every s > 0, $T_s(t)$ = (sin t + 2 sin 2t, cos t − 2 cos 2t, −s sin 3t) keeps both symmetries and is a trefoil. The corpus case is s = φ.*
  2. ***Proposition 6.2 (The Ghost Map) [PROVEN], adopted as the Step 6 projection:*** *For |**p**| < 1 define*

$$G(\mathbf{p}) = \Phi\big(\mathcal{C}_{\mathrm{rat}}(\mathbf{p})\big) = \left(\frac{1+|\mathbf{p}|^2}{1-|\mathbf{p}|^2},\ \frac{2\mathbf{p}}{1-|\mathbf{p}|^2}\right),$$

  *where $\mathcal{C}_{rat}$ is the rational Cayley map of Lemma 9 Part 3 and Φ(w, **m**) = (1/w, **m**/w) is the rational Lorentz lift of Hoard 2026c (applied to the representative of ±q with w > 0). Then: (i) G is a bijection from the open unit ball onto the upper sheet $\mathbb{H}^3$ = {$x_0^2 - |\mathbf{x}|^2 = 1, x_0 > 0$}; (ii) G is an isometry from the Poincaré ball model, $\cosh d_{\mathbb{H}}(G(\mathbf{p}), G(\mathbf{r})) = 1 + (2|\mathbf{p} - \mathbf{r}|^2)/((1 - |\mathbf{p}|^2)(1 - |\mathbf{r}|^2))$; (iii) G maps rational points to rational points; (iv) G(R**p**) = diag(1, R) G(**p**) for every R ∈ SO(3). Between two projected states, $Q_h = \langle G_i, G_j\rangle_L^2 - 1 = \sinh^2 d_{\mathbb{H}}(G_i, G_j)$ is a rational, symmetric, point-separating kernel. For any scale c < 1/√10 (e.g. c = 1/4), G(cT_1) is a D_3-symmetric trefoil in $\mathbb{H}^3$ whose fundamental arcs have equal hyperbolic length, and rational parameters u = tan(t/2) give exactly rational points.*
  3. ***Proposition 6.3 (Spin Lift) [PROVEN]:*** *The preimage of the trefoil's D_3 under $S^3 \to SO(3)$ is the binary dihedral group $2D_3$ of order 12. It lies in SU(2), which is the stabilizer in SL(2, $\mathbb{C}$) of the base point G(**0**) = (1, **0**). This is the precise content of "SU(2) ⊂ SL(2, $\mathbb{C}$)" in Step 6. (There is no homomorphism SO(3) → SL(2, $\mathbb{C}$) splitting the cover; the correct statement is the inclusion SU(2) ⊂ SL(2, $\mathbb{C}$).) The equality of the order 12 with the 12 channels is* **[CORRESPONDENCE]**.
  4. ***Proposition 6.4 (Ghost Projections from the {4,3,5} Honeycomb) [PROVEN]:*** *Place the cube cell $\mathcal{K}$ of the hyperbolic honeycomb {4,3,5} at the base point, with faces normal to the coordinate axes; in the Klein model it is the Euclidean cube $[-a, a]^3$ with $a^2 = \sqrt{5} - 2 = \varphi^{-3}$. Orient the trefoil with its C_3 axis along (1,1,1) and a half-turn axis along (1,−1,0) (Step 6), and take the scale c = 1/12. Let $B_{\pm k}$ be the boost along ±**e**$_k$ with*

$$\cosh\vartheta = \phi, \qquad \sinh\vartheta = \sqrt{\phi}, \qquad B_{+1} = \begin{pmatrix} \phi & \sqrt{\phi} & 0 & 0 \\ \sqrt{\phi} & \phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

  Then: (i) the dihedral angle of $\mathcal{K}$ is exactly 72°, its in-radius is r_{in} = artanh($\varphi^{-3/2}$) ≈ 0.5306, and ϑ = 2r_{in} = arccosh φ ≈ 1.0613; (ii) B_{+k} is the reflection in the face $x_k = a\,x_0$ composed with the reflection in the mid-plane x_k = 0, so it is a symmetry of the honeycomb carrying $\mathcal{K}$ onto its neighbour across that face; (iii) G(cT) lies inside $\mathcal{K}$ (maximum Klein radius 0.4706 < a ≈ 0.4859); (iv) the six ***ghosts*** *$B_{\pm k}$ G(cT) lie in the six face-neighbour cells, so they are pairwise disjoint and disjoint from G(cT), each congruent to it; (v) the 120° rotation about (1,1,1) cycles $B_{+1} \to B_{+2} \to B_{+3}$ (and the negative triple), while the half-turn exchanges the positive and negative triples. Every C_3-invariant energy therefore gives each ghost of a triple exactly 1/3 of the triple's total (the rigorous form of "1/3 + 1/3 + 1/3 =*

1"), and the full D_3 distributes it over all six. Ghost coordinates are exact in the number field $\mathbb{Q}(\sqrt{\varphi})$ of degree 4. The rapidity is fixed by the honeycomb, not chosen.

5. ***Proposition 6.5 (Equivalent Setting Inside SU(2))*** **[PROVEN]:** *With the golden Clifford radii $r_1 = \varphi/\sqrt{(\varphi+2)}$, $r_2 = 1/\sqrt{(\varphi+2)}$ of Lemma 11, the curve $K(s) = (r_1 e^{2is}, r_2 e^{3is}) \in \mathbb{C}^2 \cong \mathbb{H}$ consists of unit quaternions on the golden Clifford torus, closes at $s = 2\pi$, is the (2,3) torus knot (a trefoil), and is invariant under $s \mapsto s + 2\pi/3$ and $s \mapsto -s$ (both determinant +1 on $\mathbb{R}^4$). It closes because the winding ratio 2 : 3 is rational (Lemma 11 Part 1); the golden ratio sets only the radii. Setting A (Propositions 6.2–6.4) is adopted as the definition of Step 6, so the projection is the composite $\boldsymbol{p} \mapsto B_{\pm k}\, G(c\,\boldsymbol{p})$ with nothing left to choose; Setting B is recorded as an equivalent model of the same trefoil inside the spin group.*

- ***Steps 7 & 8 ($\theta_7 = 7\pi/4$, $\theta_8 = 2\pi$ — Apex Peaks & Holonomy Evaluation)*** **[PROVEN]:** *Across the 8 steps $\theta_k = k\pi/4$ ($k = 1, \ldots, 8$), the transverse amplitude $|\sin\theta_k|$ equals 1 at* ***exactly two steps****:* ***Step 2 ($k = 2$, $\theta_2 = \pi/2$)*** *and* ***Step 6 ($k = 6$, $\theta_6 = 3\pi/2$)*** *(whereas $|\sin(2\theta_k)| = |\sin(k\pi/2)| = 1$ at odd steps $k = 1, 3, 5, 7$ and 0 at even steps).* Step convention (**[DEFINITION]**): *one step is a 90° spatial rotation with spinor $(1+i)^k$, exact by Lemma 13 Part 4; 45° spatial steps are not exactly representable and are not used. At $k = 2$ and $k = 6$ the gcd-reduced spinors are i and $-i$: the same 180° half-turn about the x-axis with opposite spin sign. These are the handshake windows of §7.2.*

Proof (including Lemmas 8.2 and 8.3).

1. ***Conformal Inner Product:*** *Using $e_o^2 = e_\infty^2 = 0$ and $e_o \cdot e_\infty = -1$, we compute:*

$$X(\mathbf{x}) \cdot X(\mathbf{y}) = \mathbf{x}\cdot\mathbf{y} + \left(\frac{1}{2}\|\mathbf{x}\|^2\right)(1)(e_\infty \cdot e_o) + (1)\left(\frac{1}{2}\|\mathbf{y}\|^2\right)(e_o \cdot e_\infty) = \mathbf{x}\cdot\mathbf{y} - \frac{1}{2}\|\mathbf{x}\|^2 - \frac{1}{2}\|\mathbf{y}\|^2 = -\frac{1}{2}\|\mathbf{x}-\mathbf{y}\|^2,$$

   so $-2\, X(\boldsymbol{x}) \cdot X(\boldsymbol{y}) = \|\boldsymbol{x} - \boldsymbol{y}\|^2$.

2. ***Proof of Lemma 8.2 (C_9 Nine-Ray Construction):*** *The 3 peripheral vertices of Δ_0 have inner product $-1/\sqrt{3}$ with the polar unit axis $\hat{\boldsymbol{p}} = (1/\sqrt{3})(1,1,1)^T$ and norm $\sqrt{3}$, so they lie on the circle of radius $\sqrt{(3 - 1/3)} = \sqrt{(8/3)}$ in the plane $\hat{\boldsymbol{p}} \cdot \boldsymbol{x} = -1/\sqrt{3}$ at azimuthal angles 0°, 120°, 240°. Rotating Δ_0 around $\hat{\boldsymbol{p}}$ by {0°, 40°, 80°} shifts the azimuths to {0°, 120°, 240°} + {0°, 40°, 80°} = {40° · j : j = 0, 1, …, 8}, producing 9 distinct vertices equally spaced by 40°. Rotation preserves norms, so all nine vertices stay on $\|\boldsymbol{x}\| = \sqrt{3}$. A point of this sphere lies in $K = [-1,1]^3$ only if every coordinate has absolute value 1, i.e. only at a cube corner; exactly 3 corners lie on the latitude $\hat{\boldsymbol{p}} \cdot \boldsymbol{x} = -1/\sqrt{3}$ (azimuths 0°, 120°, 240°), so the 6 vertices at other azimuths lie outside K. The three corners on that latitude are exactly the rays with offset 0°, which proves Proposition 8.2″. Proposition 8.2′ is immediate: 9 distinct rays from O meet 9 distinct points, and K has 8 vertices.*

3. ***Proof of Lemma 8.3 (K_5 4-Simplex Affine Independence):*** *Suppose $a\|\boldsymbol{x}\|^2 + \boldsymbol{b} \cdot \boldsymbol{x} + c = 0$ holds at O = (0,0,0) and at all 4 vertices $\boldsymbol{v}_1$, $\boldsymbol{v}_2$, $\boldsymbol{v}_3$, $\boldsymbol{v}_4$ of Δ_a. Evaluating at O gives $c = 0$. At each vertex, $\|\boldsymbol{v}_i\|^2 = 3$ (true for Δ_1, whose vertices lie in $\{\pm1\}^3$, and preserved by the rotations producing Δ_2, Δ_3), so $3a + \boldsymbol{b} \cdot \boldsymbol{v}_i = 0$. Since $\Sigma_{i=1}^4 \boldsymbol{v}_i = \boldsymbol{0}$ (also preserved by rotation), summing over $i = 1,\ldots,4$ gives $12a = 0 \Longrightarrow a = 0$. Then $\boldsymbol{b} \cdot \boldsymbol{v}_i = 0$ for four vectors spanning $\mathbb{R}^3$, forcing $\boldsymbol{b} = \boldsymbol{0}$. Hence no hyperplane in $\mathbb{R}^4$ contains all 5 lifted points, proving they form a non-degenerate 4-simplex (K_5).*

4. ***Proof of D_3 Trefoil Symmetry:*** *Using $\sin t = (e^{it} - e^{-it})/(2i)$ and $\cos t = (e^{it} + e^{-it})/2$, we have $w(t) = x(t) + iy(t) = (\sin t + i\cos t) + 2(\sin 2t - i\cos 2t) = i\, e^{-it} - 2i\, e^{2it}$. Substituting $t \mapsto t + 2\pi/3$ gives $w(t + 2\pi/3) = i\, e^{-it}e^{-2\pi i/3} - 2i\, e^{2it}e^{4\pi i/3} = e^{-2\pi i/3}w(t)$ (since $e^{4\pi i/3} = e^{-2\pi i/3}$), and $z(t + 2\pi/3) = -\sin(3t + 2\pi) = z(t)$. Substituting $t \mapsto -t$ gives $(x(-t), y(-t), z(-t)) = (-x(t), y(t), -z(t))$, a 180° rotation in $\mathbb{R}^3$ about the y-axis. The group has order 6 and acts freely except at finitely many parameter values, so a fundamental domain is an arc of parameter length $2\pi/6$. Conjugating the generators by the orthonormal frame $(\boldsymbol{b} \times \boldsymbol{n}, \boldsymbol{b}, \boldsymbol{n})$ with $\boldsymbol{n} = (1,1,1)/\sqrt{3}$ and $\boldsymbol{b} = (1,-1,0)/\sqrt{2}$ gives the two integer matrices; the arc $[0, \pi/3]$ and its six integer images cover the curve (checked on 720 points).*

5. ***Proof of the Step 6 propositions.*** 6.1: *scaling z by s commutes with rotations about z and with $(x, y, z) \mapsto (-x, y, -z)$, and the maps diag(1, 1, λ), λ from 1 to s, are invertible linear maps forming an ambient isotopy, so the knot type is preserved.* 6.2: *$\mathcal{C}_{rat}$ sends the ball onto the hemisphere $w = (1 - |\boldsymbol{p}|^2)/(1 + |\boldsymbol{p}|^2) > 0$, and Φ sends that hemisphere bijectively onto $\mathbb{H}^3$: if $x_0^2 - |\boldsymbol{x}|^2 = 1$ and $x_0 > 0$, then $w = 1/x_0 \in (0, 1]$ and $w^2 + |\boldsymbol{m}|^2 = (1 + |\boldsymbol{x}|^2)/x_0^2 = 1$. Composing gives the displayed formula. With $a = |\boldsymbol{p}|^2$ and $b = |\boldsymbol{r}|^2$, $\langle G(\boldsymbol{p}), G(\boldsymbol{r})\rangle_L = ((1 + a)(1 + b) - 4\boldsymbol{p} \cdot \boldsymbol{r})/((1 - a)(1 - b))$, which expands to the right side of (ii). (iii) holds because only field operations occur, and (iv) because $|R\boldsymbol{p}| = |\boldsymbol{p}|$. Since $x^2 + y^2 = 5 - 4\cos 3t \le 9$ and $z^2 \le 1$, $|T_1(t)| \le \sqrt{10}$, so cT_1 lies in the ball for $c < 1/\sqrt{10}$, and (iv) carries the D_3 symmetry to $\mathbb{H}^3$.* 6.3: *the cover is 2-to-1 with kernel $\{\pm1\}$ contained in the preimage, so the preimage has order $2 \cdot 6 = 12$;*

SL(2, ℂ) acts on ℍ³ through PSL(2, ℂ) ≅ SO⁺(3,1), and point stabilizers are conjugates of SU(2). 6.4: *(i) In the Klein model the face $x_1 = a\,x_0$ has outward Lorentz normal $\boldsymbol{n} = (a, 1, 0, 0)$ with $\langle \boldsymbol{n}, \boldsymbol{n}\rangle = 1 - a^2$, and two adjacent faces meet at interior angle θ with $\cos\theta = a^2/(1 - a^2)$. Setting $\theta = 72°$, $\cos 72° = (\sqrt{5} - 1)/4$, gives $a^2 = (\sqrt{5} - 1)/(\sqrt{5} + 3) = \sqrt{5} - 2 = \varphi^{-3}$; the face centre is at Klein radius $a = \tanh r_{in}$, and $\cosh 2r_{in} = (1 + a^2)/(1 - a^2) = (\sqrt{5} - 1)/(3 - \sqrt{5}) = \varphi$, so $\sinh^2 2r_{in} = \varphi^2 - 1 = \varphi$. Since $a\sqrt{3} \approx 0.842 < 1$ the cell is compact. (ii) The product of reflections in two hyperplanes with a common perpendicular is the translation along that perpendicular by twice their distance, here $2r_{in}$; the mid-plane reflection is a symmetry of $\mathcal{K}$, the face reflection is a generator of the Coxeter group [4,3,5], so their product lies in the symmetry group of the honeycomb, and it maps the face $x_1 = -a\,x_0$ onto $x_1 = a\,x_0$. (iii) For $|\boldsymbol{p}| \le 3/12$ the Klein radius $2|\boldsymbol{p}|/(1 + |\boldsymbol{p}|^2) \le 0.5/1.0625 \approx 0.4706 < a$, and $|T| \le 3$ by the proof of 6.2. (iv) Distinct cells of a honeycomb have disjoint interiors, and boosts are isometries. (v) Conjugating a boost along $\boldsymbol{e}$ by a rotation R gives the boost along $R\boldsymbol{e}$; the cyclic permutation sends $\boldsymbol{e}_1 \to \boldsymbol{e}_2 \to \boldsymbol{e}_3$, and the half-turn about $(1,-1,0)$ sends $\boldsymbol{e}_1 \mapsto -\boldsymbol{e}_2$, $\boldsymbol{e}_3 \mapsto -\boldsymbol{e}_3$. With $R \cdot (B\,G(cT)) = (RBR^{-1})\,G(RcT) = (RBR^{-1})\,G(cT)$ (6.2(iv) and the C_3-invariance of T), the energy split follows. Checked in* `verify_honeycomb.py` *(exact symbolic cell geometry and reflection product; containment on 720 trefoil points; minimum trefoil-to-ghost distance 0.49).* 6.5: *$|K(s)|^2 = r_1^2 + r_2^2 = 1$ (Lemma 11); $K(s + 2\pi/3) = (e^{4\pi i/3} r_1 e^{2is}, r_2 e^{3is})$; conjugating both factors sends $s \mapsto -s$; and a (p, q) curve with $\gcd(p, q) = 1$ on a standard torus in S^3 is the (p, q) torus knot (***[ESTABLISHED]***). All checks are in* `verify_step6.py`*.* ■

4.4 Lemma 12: Holonomy Telescoping and Right-Acting Ordering Theorem

Lemma 12 (Exact Holonomy Telescoping, {7,8} Partition Sizes and Ordering Convention).
[PROVEN] Gear chain from Hoard (2026b); closing remark after Hoard (2026d), Remark 4.7.

*In the three-scale Torsional Markov Memory-Offload Pump across the 12-point ($n = 0,\ldots,719$), 10-point ($m = 0,\ldots,71$), and 8-point ($l = 0,\ldots,9$) gears, let $q_0^{(12)}, \ldots, q_{719}^{(12)} \in S^3$ be the outer-gear frame states, let the outer increments be $\Delta q_n^{(12)} = q_n^{(12)}(q_{n-1}^{(12)})^{-1}$ for $n = 0, \ldots, 719$, where the boundary value $q_{-1}^{(12)}$ is fixed by one of three conventions: **open** $q_{-1}^{(12)} := q_0^{(12)}$ (i.e. $\Delta q_0^{(12)} = 1$; **adopted in this monograph**), **anchored** $q_{-1}^{(12)} := 1$ (i.e. $\Delta q_0^{(12)} = q_0^{(12)}$), or **cyclic** $q_{-1}^{(12)} := q_{719}^{(12)}$, let the outer block products be $\Delta Q_m^{(12\to10)} = \Delta q_{10m+9}^{(12)} \cdots \Delta q_{10m}^{(12)}$ (later frames on the left), and let the middle inverse increments be $\delta_m = \Delta q_m^{(10)} = (\Delta Q_m^{(12\to10)})^{-1}$. Then:*

1. ***Exact {7,8} Block Partition:*** *The level sets $A_l = \{m \in \{0,\ldots,71\} : \lfloor 10m/72 \rfloor = l\}$ ($l = 0,\ldots,9$) partition $\{0,\ldots,71\}$ into 10 consecutive integer intervals of exact lengths:*

$$\big(|A_0|, |A_1|, \ldots, |A_9|\big) = (8,\,7,\,7,\,7,\,7,\,8,\,7,\,7,\,7,\,7), \qquad \sum_{l=0}^{9} |A_l| = 72.$$

2. ***Load-Bearing Right-Acting Middle Multiplication:*** *Define the net outer rotation left-to-right in reverse chronological order $Q^{(12)} = \Delta q_{719}^{(12)} \cdots \Delta q_0^{(12)}$, and define the middle and inner net rotations in **right-acting chronological order**:*

$$Q^{(10)} = \delta_0\delta_1\cdots\delta_{71}, \qquad \Delta q_l^{(8)} = \prod_{m\in A_l}^{\rightarrow} \delta_m, \qquad Q^{(8)} = \Delta q_0^{(8)}\Delta q_1^{(8)}\cdots\Delta q_9^{(8)}.$$

Then by associativity of quaternion multiplication and the reversal law $(AB)^{-1} = B^{-1}A^{-1}$:

$$Q^{(10)} = \Big(Q^{(12)}\Big)^{-1}, \qquad Q^{(8)} = Q^{(10)} = \Big(Q^{(12)}\Big)^{-1}, \qquad Q^{(10)}Q^{(12)} = +1.$$

(Conversely, if $Q^{(10)}$ is multiplied in left-acting order $\delta_{71}\cdots\delta_0$, then $Q^{(10)}Q^{(12)} \neq 1$ for generic non-commuting quaternions).

3. ***Exact Holonomy Identity & O(log k) Parallel Prefix Scan:*** *Under the right-acting convention of Part (2), the three-gear holonomy $H = Q^{(8)}\,Q^{(10)}\,Q^{(12)}$ satisfies:*

$$H = Q^{(8)}\underbrace{Q^{(10)}Q^{(12)}}_{=1} = Q^{(8)} = \Big(Q^{(12)}\Big)^{-1} = q_{-1}^{(12)}\Big(q_{719}^{(12)}\Big)^{-1}.$$

The three boundary conventions therefore give Table 3.

Table 3: Boundary conventions for the outer gear and the holonomy each gives.

Convention	$q_{-1}^{(12)}$	Holonomy H	Information carried
Open (adopted)	$q_0^{(12)}$	$q_0^{(12)}(q_{719}^{(12)})^{-1}$	Net rotation from start to end; H = 1 iff the chain returns to its start
Anchored	1	$(q_{719}^{(12)})^{-1}$	Net rotation from the identity
Cyclic	$q_{719}^{(12)}$	1 identically	None: closure is built in, so H = 1 tests nothing

*The open convention is adopted because it is the only one in which H is a genuine closure test. Furthermore, by associativity, any product of k consecutive quaternion deltas can be evaluated in **O(log k) parallel depth** via a binary prefix scan.*

Remark (after Hoard 2026d Remark 4.7): *when the increments are differences of stored states, H measures only the start against the end; a non-trivial holonomy requires increments generated independently, for example by integrating an angular velocity. Under no convention does the 720-frame cycle force the state back to the identity or remove floating-point drift; only the vacuous cyclic convention makes H ≡ 1.*

Proof.

1. *For $l \in \{0,\dots,9\}$, $m \in A_l \iff 7.2l \le m < 7.2(l+1) \iff \lceil 7.2l \rceil \le m \le \lceil 7.2(l+1) \rceil - 1$. Evaluating $\lceil 7.2l \rceil$ for $l = 0,\dots,10$ gives 0, 8, 15, 22, 29, 36, 44, 51, 58, 65, 72, whose first differences are (8, 7, 7, 7, 7, 8, 7, 7, 7, 7).*
2. *By associativity, $Q^{(12)} = \Delta Q_{71}^{(12\to10)}\,\Delta Q_{70}^{(12\to10)}\cdots\Delta Q_{0}^{(12\to10)}$. Taking the inverse of this product reverses the order of the factors:*

$$\left(Q^{(12)}\right)^{-1} = \left(\Delta Q_{71}^{(12\to10)}\cdots\Delta Q_{0}^{(12\to10)}\right)^{-1} = \left(\Delta Q_{0}^{(12\to10)}\right)^{-1}\cdots\left(\Delta Q_{71}^{(12\to10)}\right)^{-1} = \delta_0\delta_1\cdots\delta_{71} = Q^{(10)}.$$

   Since $A_0, \dots, A_9$ partition $\{0,\dots,71\}$ in increasing order, concatenating $\Delta q_l^{(8)} = \overrightarrow{\Pi}_{m\in A_l}\,\delta_m$ gives $Q^{(8)} = \delta_0\,\delta_1\cdots\delta_{71} = Q^{(10)}$.
3. *Telescoping gives $Q^{(12)} = (q_{719}q_{718}^{-1})(q_{718}q_{717}^{-1})\cdots(q_0\,q_{-1}^{-1}) = q_{719}^{(12)}(q_{-1}^{(12)})^{-1}$. Multiplying $H = Q^{(8)}Q^{(10)}Q^{(12)}$ gives $H = (Q^{(12)})^{-1}(Q^{(12)})^{-1}Q^{(12)} = (Q^{(12)})^{-1} = q_{-1}^{(12)}(q_{719}^{(12)})^{-1}$. Substituting the three values of $q_{-1}^{(12)}$ gives the table in Part 3.* ■

4.5 Lemma 14: Fokker-Planck Stationary Measure and the Circularity Boundary

Lemma 14 (Fokker-Planck Lattice Master Equation, Gibbs vs. Harmonic Trap Measure, Hurwitz Partition Function and the Circularity Boundary). [PROVEN] Numerics after Hoard (2026e).

Let Λ be a connected lattice graph equipped with a real potential $V: \Lambda \to \mathbb{R}$, inverse temperature / barrier parameter $\beta > 0$, and either the continuous-time master equation $d\rho_b/(dt) = \Sigma_{a\sim b}(\rho_a\,k_{ab} - \rho_b\,k_{ba})$ with $k_{ab} = D\,e^{-(\beta/2)(V(b)-V(a))}$ ($D > 0$) or the discrete-time Metropolis chain $K(a,b) = c_{ab}\min(1, e^{-\beta(V(b)-V(a))})$ ($c_{ab} = c_{ba} > 0$, with positive holding probability). Then:

1. ***Unique Gibbs Stationary Distribution ([PROVEN]):** On any finite connected lattice Λ with positive diffusion ($D > 0$), the unique stationary distribution is the **Gibbs distribution**:*

$$\pi(x) = \frac{e^{-\beta V(x)}}{Z(\beta)}, \qquad Z(\beta) = \sum_{y\in\Lambda} e^{-\beta V(y)},$$

   which is strictly positive on every node ($\pi(x) > 0\ \forall x \in \Lambda$) and concentrates on $\arg\min V$ only in the limit $\beta \to \infty$.
2. ***Absorbing Traps & Harmonic Hitting Measure ([PROVEN]):** If outgoing transition rates are set to zero on a non-empty subset $\mathcal{P} \subset \Lambda$ (the "$D_2 \to 0$ at prime nodes" absorbing trap rule), then from any initial state x_0, all probability mass is absorbed into $\mathcal{P}$ exponentially fast at rate $\lambda_{min} > 0$, converging to the **harmonic hitting measure** $H_{x_0}(p) = ((-L_{TT})^{-1}L_{T\mathcal{P}})_{x_0,p}$ (where L is the generator, $T = \Lambda \setminus \mathcal{P}$ the transient nodes, and L_{TT}, $L_{T\mathcal{P}}$ its blocks) supported exclusively on $\mathcal{P}$.*
3. ***Normalizability on the Infinite 4D Hurwitz Lattice ([PROVEN]):** On the infinite 4D Hurwitz lattice $\mathcal{H}$ with a radial potential $V(q) = V(N(q))$ depending only on the integer norm $n = N(q)$, the Gibbs partition function converges if and only if:*

$$Z_{\mathcal{H}}(\beta) = 1 + \sum_{n=1}^{\infty} 24\sigma_{\text{odd}}(n)e^{-\beta V(n)} < \infty,$$

which is satisfied by any confining potential growing at least logarithmically ($V(n) \geq \alpha \ln n$ with $\beta\alpha > 2$), and in particular by any linear/harmonic trap $V(n) = \kappa n$ ($\kappa > 0$).

4. ***The Circularity Boundary ([PROVEN]):*** *In both the $\beta \to \infty$ Gibbs limit (Part 1) and the absorbing trap limit (Part 2), the limiting support coincides identically with the set where V is minimized or outgoing rates vanish. Consequently, if $V(Q)$ or the trap rule is defined by testing whether $Q = N(q)$ is prime, evaluating a step of the dynamics already requires a primality test on Q, so the sieve **displays and samples** a prime set rather than computing primality from scratch. Accordingly, in the SHD-CCP Dual-Helix architecture, the Fokker-Planck dynamics operate in **Helix A (heuristic candidate sampler)**, paired with **Helix B (exact arithmetic certifier)**.*

Proof.

1. *Detailed balance $\pi(a)k_{ab} = (D/Z)e^{-(\beta/2)(V(a)+V(b))} = \pi(b)k_{ba}$ (and $\pi(a)K(a,b) = (c_{ab}/Z)\min(e^{-\beta V(a)}, e^{-\beta V(b)}) = \pi(b)K(b,a)$) holds on every edge. By irreducibility (and aperiodicity from holding probabilities) on a finite connected graph, Perron–Frobenius guarantees uniqueness and strict positivity $\pi(x) > 0$. As $\beta \to \infty$, $\pi(x)/\pi(y) = e^{-\beta(V(x)-V(y))} \to 0$ whenever $V(x) > \min V$.*
2. *Restricted to the transient non-trap nodes $T = \Lambda \setminus \mathcal{P}$, the sub-generator L_{TT} is strictly diagonally dominant along paths reaching $\mathcal{P}$, so all eigenvalues of L_{TT} have strictly negative real part ($\mathrm{Re}\,\lambda \leq -\lambda_{min} < 0$), $-L_{TT}$ is invertible, and $\lim_{t\to\infty} \rho_{\mathcal{P}}(t) = \rho_T(0)(-L_{TT})^{-1}L_{T\mathcal{P}} + \rho_{\mathcal{P}}(0)$.*
3. *By Hurwitz's norm theorem (Conway & Smith, 2003; Lemma 5 Part 4), the shell of norm $n \geq 1$ in $\mathcal{H}$ contains exactly $24\sigma_{odd}(n)$ Hurwitz integers, so $Z_{\mathcal{H}}(\beta)$ is exactly the stated series and converges iff it is finite. Since $1 \leq \sigma_{odd}(n) \leq \sigma_1(n) = n\Sigma_{d|n}(1/d) \leq n(1 + \ln n)$: (i) if $V(n) = \kappa n$ with $\beta\kappa > 0$, the terms are at most $24\, n^2\, e^{-\beta\kappa n}$, a convergent series; (ii) if $V(n) \geq \alpha \ln n$ with $\beta\alpha > 2$, the terms are at most $24(1 + \ln n)\, n^{1-\beta\alpha} = 24(1 + \ln n)\, n^{-1-\varepsilon}$ with $\varepsilon = \beta\alpha - 2 > 0$, which converges by comparison with $\Sigma n^{-1-\varepsilon/2}$ (as $\ln n = o(n^{\varepsilon/2})$).*
4. *Immediate from Parts (1) and (2), since the transition generator k_{ab} or L_{TT} takes $V(Q)$ or the indicator $\mathbf{1}_{\mathcal{P}}(Q)$ as an explicit input parameter.* ■

5 Layer IV — Modular-Cubic Arithmetic, Bit-Growth Limits, and Cryptographic Security (Lemmas 9, 13, and 15)

5.1 Lemma 9: Prime-Triplet Mod-30 CRT Cubic Indexing, Closed-Form c_{13}, and ECDLP Obstructions

Lemma 9 (Prime-Triplet Mod-30 Classification, Closed-Form Garner CRT Reconstruction, Maximal 10-Bit Triplet (881, 883, 887) and Cryptographic Obstructions). [PROVEN]

1. ***Mod-30 Two-Class Admissibility ([PROVEN]):*** *Every prime triplet $(p, p + 2, p + 6)$ with $p > 5$ lies in one of **only two** residue triples in $(\mathbb{Z}/30\mathbb{Z})^\times$:*

$$(p, p+2, p+6) \equiv (11, 13, 17) \pmod{30} \qquad \text{or} \qquad (p, p+2, p+6) \equiv (17, 19, 23) \pmod{30}.$$

2. ***Collision-Free O(1) Garner CRT Reconstruction with Closed-Form Inverses ([PROVEN]):*** *For **every** prime $p > 3$ (even if $p + 2$ or $p + 6$ is composite), the three moduli $(m_1, m_2, m_3) = (p, p + 2, p + 6)$ are strictly pairwise coprime ($\gcd(p, p + 2) = \gcd(p, p + 6) = \gcd(p + 2, p + 6) = 1$). By the Chinese Remainder Theorem, the map $\Phi_{CRT}(X) = (X \bmod p, X \bmod (p + 2), X \bmod (p + 6)) = (x_1, x_2, x_3)$ is a **bijective ring isomorphism** over the cubic range $M = p(p + 2)(p + 6) = p^3 + 8p^2 + 12p$. Any state $X \in [0, M)$ is reconstructed in **O(1) arithmetic operations with zero lookup tables** via Garner's mixed-radix algorithm:*

$$u_1 = x_1, \quad u_2 = (x_2 - u_1)c_{12} \bmod (p+2), \quad u_3 = \big((x_3 - u_1)c_{13} - u_2\big)c_{23} \bmod (p+6),$$
$$X = u_1 + u_2 p + u_3 p(p+2) \in [0, M),$$

where all three modular inverses are given by exact closed-form integer polynomials in p:

$$c_{12} = p^{-1} \bmod (p+2) = \frac{p+1}{2} \quad (\forall p > 3),$$

$$c_{23} = (p+2)^{-1} \bmod (p+6) = \begin{cases} \frac{3p+17}{4} & \text{if } p \equiv 1 \pmod 4, \\ \frac{p+5}{4} & \text{if } p \equiv 3 \pmod 4, \end{cases}$$

$$c_{13} = p^{-1} \bmod (p+6) = \begin{cases} \frac{p+5}{6} & \text{if } p \equiv 1 \pmod 6, \\ \frac{5p+29}{6} & \text{if } p \equiv 5 \pmod 6. \end{cases}$$

*For the 30-bit Form 2 payload of the 64-bit Einstein Tile (three 10-bit residues, requiring p + 6 ≤ 1024), the **largest prime triplet is (p, p + 2, p + 6) = (881, 883, 887)**, which bijectively indexes M = 690,017,701 ≈ $2^{29.36}$ cubic states (the next candidate, p = 1019, fails: p + 6 = 1025 = $5^2 \cdot 41$ is composite and needs 11 bits).*

3. ***Rational Cayley Parameterization ([PROVEN]):** The Cayley-stereographic map $\mathcal{C}_{rat}: \mathbb{Q}^3 \to S^3 \cap \mathbb{Q}^4$ given by $\mathbf{u} \mapsto (1 - \|\mathbf{u}\|^2, 2\mathbf{u})/(1 + \|\mathbf{u}\|^2)$ produces exact rational unit quaternions ($N(\mathcal{C}_{rat}(\mathbf{u})) = 1$) without square roots or transcendental functions.*
4. ***Unconditional Cryptographic Obstructions on O(1) ECDLP ([PROVEN]):***
   - ***(a) Pigeonhole Obstruction:** Any map from a 256-bit elliptic curve group ($|E(\mathbb{F}_q)| \approx 2^{256}$) into a 64-bit tile ($2^{64}$ states) has some fiber of size at least 2^{192}, so no such map is injective and no O(1) inversion from the tile alone exists; no amount of checkpointing changes this information-theoretic bound.*
   - ***(b) Torsion-Free Homomorphism Triviality:** Every group homomorphism $\varphi: E(\mathbb{F}_q) \to (\mathbb{Q}^d, +)$ from a finite group of order n into a torsion-free $\mathbb{Q}$-vector space satisfies $n\varphi(P) = \varphi(nP) = \mathbf{0} \implies \varphi \equiv \mathbf{0}$. Defining $\varphi(kP) := k\mathbf{v}_0$ non-homomorphically requires already knowing the discrete logarithm $k = \log_P(Q)$.*
   - ***(c) Torus-Valued ("Gauge") Characters ([PROVEN]):** Homomorphisms $\chi: \langle P \rangle \to S^1$ into a compact torus exist and can be faithful, unlike maps into $\mathbb{Q}^d$. But any algorithm that evaluates a faithful χ at $Q = kP$ solves the discrete logarithm, and a non-faithful χ of order m reveals only k mod m. So phase-winding ("gauge-covariant") projections do not bypass (b).*

Proof.

1. *Exhaustive verification over the 8 units {1, 7, 11, 13, 17, 19, 23, 29} (mod 30) leaves only $p \equiv 11 \implies (11, 13, 17)$ and $p \equiv 17 \implies (17, 19, 23)$ free of factors of 3 and 5.*
2. *Since p > 3 is prime, $2 \nmid p$ and $3 \nmid p$, so gcd(p, p + 2) = gcd(p, 2) = 1, gcd(p, p + 6) = gcd(p, 6) = 1, and gcd(p + 2, p + 6) = gcd(p + 2, 4) = 1. For the closed-form inverses:*
   - *Modulo p + 2, $p \equiv -2$, and $(-2)((p + 1)/2) = -(p + 1) = -(p + 2) + 1 \equiv 1 \pmod{p + 2}$.*
   - *Modulo p + 6, $p + 2 \equiv -4$. When $p \equiv 3 \pmod 4$, $(p + 5)/4 \in \mathbb{Z}$ and $(-4)((p + 5)/4) = -(p + 5) = -(p + 6) + 1 \equiv 1 \pmod{p + 6}$. When $p \equiv 1 \pmod 4$, $(3p + 17)/4 \in \mathbb{Z}$ and $(-4)((3p + 17)/4) = -3(p + 6) + 1 \equiv 1 \pmod{p + 6}$.*
   - *Modulo p + 6, $p \equiv -6$. When $p \equiv 1 \pmod 6$, $(p + 5)/6 \in \mathbb{Z}$ and $(-6)((p + 5)/6) = -(p + 5) = -(p + 6) + 1 \equiv 1 \pmod{p + 6}$. When $p \equiv 5 \pmod 6$, $(5p + 29)/6 \in \mathbb{Z}$ and $(-6)((5p + 29)/6) = -5(p + 6) + 1 \equiv 1 \pmod{p + 6}$. Testing primes $p \le 1018$ shows that (881, 883, 887) is the largest prime triplet with p + 6 ≤ 1024.*
3. *$N(\mathcal{C}_{rat}(\mathbf{u})) = ((1 - \|\mathbf{u}\|^2)^2 + 4\|\mathbf{u}\|^2)/((1 + \|\mathbf{u}\|^2)^2) = (1 + \|\mathbf{u}\|^2)^2/((1 + \|\mathbf{u}\|^2)^2) = 1$.*
4. *(a) If a map sends a set of size ≈ 2^{256} into 2^{64} states and every fiber had fewer than 2^{192} elements, the domain would have fewer than $2^{64} \cdot 2^{192} = 2^{256}$ elements, a contradiction (pigeonhole). (b) Let φ be a homomorphism and $P \in E(\mathbb{F}_q)$ of order n. Then $n\varphi(P) = \varphi(nP) = \varphi(\mathcal{O}) = \mathbf{0}$ in $\mathbb{Q}^d$; since $\mathbb{Q}^d$ is torsion-free and $n \ge 1$, $\varphi(P) = \mathbf{0}$. As P was arbitrary, $\varphi \equiv \mathbf{0}$. A map with $\varphi(kP) = k\mathbf{v}_0$ for $\mathbf{v}_0 \neq \mathbf{0}$ is therefore not a homomorphism, and evaluating it at $Q = kP$ requires k, i.e. the discrete logarithm. (c) Faithfulness gives $\chi(P) = e^{2\pi i a/n}$ with gcd(a, n) = 1, so $\chi(Q) = e^{2\pi i ak/n}$; reading the angle to $O(\log n)$ bits recovers ak mod n, hence $k = a^{-1}(ak) \bmod n$. If χ has order m, then $\chi(Q)$ depends only on k mod m. ■*

5.2 Lemma 13: Exact Quaternionic Arithmetic, Niven's Finite-Order Bound, and Bit-Growth Theorem

Lemma 13 (Unbounded Denominator Growth in $S^3 \cap \mathbb{Q}^4$, Niven's Finite-Order Subgroup Theorem and Projective Integer Bit-Growth). [PROVEN]

1. ***Unbounded Denominator Growth in $S^3 \cap \mathbb{Q}^4$ ([PROVEN]):** The Pythagorean rational unit quaternion $q = (3 + 4i)/5 \in S^3 \cap \mathbb{Q}^4$ ($N(q) = (9 + 16)/25 = 1$) has the property that for every integer $n \ge 1$, its n-th power $q^n = (a_n + b_n\, i)/5^n$ in lowest terms has **exact denominator 5^n** ($\gcd(a_n, 5) = \gcd(b_n, 5) = 1$).*

Consequently, q has infinite order, and no fixed-bit-width register can store arbitrary multi-step products in $S^3 \cap \mathbb{Q}^4$ without rounding or denominator overflow.

2. ***Complete Classification of Finite-Order Rational Unit Quaternions ([PROVEN]):*** *Every element $q \in S^3 \cap \mathbb{Q}^4$ of finite multiplicative order ($q^m = +1$) has order:*

$$\mathrm{ord}(q) \in \{1, 2, 3, 4, 6\}.$$

*Consequently, a rational unit quaternion of finite order can induce 3D spatial rotations $x \mapsto q\,x\,q^{-1}$ by angles $2\theta \in \{0°, 120°, 180°, 240°\}$ **only** (never 45°, 72°, or 90°). The 24 Hurwitz units $\mathcal{H}_1 = \{\pm 1, \pm i, \pm j, \pm k, (\pm 1 \pm i \pm j \pm k)/2\}$ form a finite subgroup of $S^3 \cap \mathbb{Q}^4$ of order 24 closed under exact fixed-width arithmetic.*

3. ***Projective Integer Quaternion Arithmetic & Linear Bit Growth ([PROVEN]):*** *For any non-zero integer (or Hurwitz half-integer) quaternion $Q \in \mathbb{Z}^4 \setminus \{0\}$ of norm $N(Q) = Q\overline{Q} \in \mathbb{Z}_{>0}$, the induced 3D rotation*

$$\rho_Q(\mathbf{x}) = \frac{Q\mathbf{x}\bar{Q}}{N(Q)} \in \mathbb{Q}^3$$

is exact and invariant under scalar integer multiples $Q \mapsto \lambda Q$. If $Q_1, \ldots, Q_k \in \mathbb{Z}^4$ have coordinates bounded by $|(Q_r)_j| < 2^b$, then their k-step product $P_k = Q_k \cdots Q_1 \in \mathbb{Z}^4$ has coordinates bounded by:

$$\left|(P_k)_j\right| < 4^{k-1}2^{kb} = 2^{k(b+2)-2},$$

requiring at most $k(b + 2)$ bits for zero-loss exact integer tracking before projective normalization.

4. ***Exact Projective Representation of the 8-Step Spinor Double-Cover Cycle ([PROVEN]):*** *While a 90° 3D rotation ($q = (1 + i)/\sqrt{2}$, spinor half-angle 45°) is irrational in S^3, in projective integer quaternion arithmetic the Gaussian integer $Q_{step} = 1 + i \in \mathbb{Z}[i] \subset \mathbb{Z}^4$ ($N(Q_{step}) = 2$) induces an exact 90° spatial rotation $\rho_{1+i}(\mathbf{x})$ per step and satisfies:*

$$(1+i)^2 = 2i, \qquad (1+i)^4 = -4 \;\; (360^\circ \text{ spinor inversion}), \qquad (1+i)^8 = +16 \;\; (720^\circ \text{ double-cover identity}),$$

*realizing the entire 8-step spinor double-cover cycle in exact integer arithmetic: the unreduced powers reach $|(1 + i)^8| = 16$ and need **6 signed bits**, while dividing each state by the gcd of its coordinates leaves the eight unit states $1 + i$, i, $-1 + i$, -1, $-1 - i$, $-i$, $1 - i$, 1 (up to positive scalars), which need **2 signed bits**. (By contrast, a 45° 3D spatial rotation requires $\tan(22.5°) = \sqrt{2} - 1 \notin \mathbb{Q}$ and cannot be represented even projectively in $\mathbb{Z}^4$).*

Proof.

1. *In the Unique Factorization Domain of Gaussian integers $\mathbb{Z}[i]$, $5 = (2 + i)(2 - i)$ factors into two non-associate Gaussian primes $\pi = 2 + i$ and $\overline{\pi} = 2 - i$ (since $(2 + i)/(2 - i) = (3 + 4i)/5 \notin \{\pm 1, \pm i\}$). Because $3 + 4i = (2 + i)^2 = \pi^2$, we have $(3 + 4i)^n = \pi^{2n}$. Since $\overline{\pi}$ is prime and $\overline{\pi} \nmid \pi$, unique factorization implies $\overline{\pi} \nmid \pi^{2n} = a_n + b_n\,i$. Therefore $5 = \pi\overline{\pi}$ cannot divide $\gcd(a_n, b_n)$, proving $(a_n + b_n\,i)/5^n$ is in lowest terms with exact denominator 5^n. For the stronger claim that 5 divides **neither** a_n nor b_n: $\mathbb{Z}[i]/(5) \cong \mathbb{F}_5 \times \mathbb{F}_5$ via $i \mapsto (2, 3)$, since $2^2 \equiv 3^2 \equiv -1 \pmod 5$. Under this map $3 + 4i \mapsto (11, 15) \equiv (1, 0)$, so $(3 + 4i)^n \mapsto (1, 0)$ for all $n \geq 1$, i.e. $a_n + 2b_n \equiv 1$ and $a_n + 3b_n \equiv 0 \pmod 5$. Subtracting gives $b_n \equiv -1 \equiv 4$, and then $a_n \equiv -3b_n \equiv 3 \pmod 5$. Hence $\gcd(a_n, 5) = \gcd(b_n, 5) = 1$.*

2. *Any $q \in S^3 \cap \mathbb{Q}^4$ of finite order m can be written as $q = \cos\theta + \hat{u}\sin\theta$ where $\theta = 2\pi k/m \in \mathbb{Q}\pi$ and $\mathrm{Re}(q) = \cos\theta \in \mathbb{Q}$. By **Niven's Theorem**, the only rational values of $\cos(\theta)$ for $\theta \in \mathbb{Q}\pi$ are:*

$$\cos\theta \in \left\{0, \pm\frac{1}{2}, \pm 1\right\},$$

which restricts $\theta \pmod{2\pi}$ to $\{0, \pi, \pm\pi/2, \pm\pi/3, \pm 2\pi/3\}$ and hence $\mathrm{ord}(q) \in \{1, 2, 3, 4, 6\}$. Moreover, for $\mathrm{ord}(q) = 4$, $\cos\theta = 0$ so $q = xi + yj + zk$ is a pure quaternion with $x^2 + y^2 + z^2 = 1$, which rotates 3D space by $2\theta = \pi = 180°$. Thus 3D rotations by 90° ($2\theta = \pi/2 \Longrightarrow \cos\theta = 1/\sqrt{2} \notin \mathbb{Q}$) do not exist in $S^3 \cap \mathbb{Q}^4$.

3. *Each component of a quaternion product AB is a signed sum of 4 pairwise products of components of A and B, so $\|AB\|_\infty \leq 4\|A\|_\infty\|B\|_\infty$. Induction over k steps gives $\|P_k\|_\infty < 4^{k-1}(2^b)^k = 2^{k(b+2)-2}$.*

4. *Direct expansion gives $(1 + i)^2 = 1 + 2i - 1 = 2i$, $(1 + i)^4 = (2i)^2 = -4$, and $(1 + i)^8 = (-4)^2 = +16$. For any projective quaternion $Q = w + \mathbf{v} \in \mathbb{Z}^4$ rotating by angle $\alpha = 2\theta$, $\|\mathbf{v}\|^2 / w^2 = \tan^2\theta \in \mathbb{Q}$; for $\alpha = 45°$ ($\theta = 22.5°$), $\tan^2(22.5°) = (\sqrt{2} - 1)^2 = 3 - 2\sqrt{2} \notin \mathbb{Q}$, proving impossibility.* ■

5.3 Lemma 15: Recursive Merkle Integrity and the South Halo Pre-Filter Bound

Lemma 15 (Pigeonhole Preimage Bounds, South Halo Cache Pre-Filtering and Recursive Merkle Verification across Nested Cubic Layers). [PROVEN]

Consider a 12-layer nested 64-voxel state $G = (G_1, G_2, \ldots, G_{12}) \in \mathcal{V}_{64}^{12}$ (representing $64^{12} = (2^6)^{12} = 2^{72}$ possible geometric configurations), and consider a single 64-bit Einstein Tile word $W \in \{0,1\}^{64}$ whose top 8 bits (Bits 56–63) form the South Halo header SH_ID $\in \{0,1\}^8$ and whose lower 56 bits (Bits 0–55) form the payload $P \in \{0,1\}^{56}$. Then:

1. ***12-Layer Pigeonhole Preimage Bound ([PROVEN]):*** *For any 64-bit summary label $\tau: \mathcal{V}_{64}^{12} \to \{0,1\}^{64}$, there exists at least one label value τ_0 with at least $2^{72}/2^{64} = 2^8 = 256$ distinct preimage states $G \in \mathcal{V}_{64}^{12}$. Furthermore, any verifier that inspects only layers $G_1, \ldots, G_m$ ($m < 12$) cannot detect arbitrary tampering in the deeper layers $G_{m+1}, \ldots, G_{12}$.*
2. ***South Halo Pre-Filter vs. Integrity Boundary ([PROVEN]):*** *Within a single 64-bit Einstein Tile, the 8-bit South Halo SH_ID partitions the 2^{56} possible 56-bit payloads into $2^8 = 256$ buckets of average size:*

$$\frac{2^{56}}{2^8} = 2^{48} \approx 2.81 \times 10^{14} \text{ payloads per South Halo ID.}$$

   *Consequently, the 8-bit South Halo check in the QRE middleware is strictly a **fast O(1) L1/L2 cache resonance pre-filter** (rejecting non-matching phase/wheel IDs in nanoseconds), and **never** a standalone cryptographic integrity check against adversarial payload modification.*
3. ***Recursive Bottom-Up Merkle Hash Chain ([PROVEN]):*** *Let $H: \{0,1\}^* \to \{0,1\}^\kappa$ be a collision-resistant cryptographic hash function (e.g., SHA-256 / BLAKE3 with $\kappa = 256$ bits stored in the packet attestation header), and define the bottom-up recursive tag chain across all 12 layers:*

$$\tau_{12}(G) = H(G_{12}), \qquad \tau_d(G) = H\big(G_d \,\|\, \tau_{d+1}(G)\big) \quad \text{for } d = 11, 10, \ldots, 1.$$

   Then for any two distinct 12-layer states $G \neq G'$, equality of the root tag $\tau_1(G) = \tau_1(G')$ explicitly yields a collision in H at the first layer index $d \in \{1, \ldots, 12\}$ where $(G_d, \tau_{d+1}(G)) \neq (G'_d, \tau_{d+1}(G'))$. Thus bottom-up recursive verification guarantees full 12-layer integrity with birthday security $2^{\kappa/2}$ (2^{128} for $\kappa = 256$, vs. 2^{32} if truncated to 64 bits). Remark: *linear checksums give no integrity. CRC-32 is affine over GF(2), $\mathrm{crc}(x \oplus y \oplus z) = \mathrm{crc}(x) \oplus \mathrm{crc}(y) \oplus \mathrm{crc}(z)$ for equal-length inputs, so an adversary can reach any target checksum.*

Proof.

1. *By the Pigeonhole Principle, partitioning 2^{72} states into 2^{64} buckets yields at least one bucket of size $\geq \lceil 2^{72}/2^{64} \rceil = 2^8 = 256$. If a verifier depends only on $(G_1, \ldots, G_m)$, modifying G_{m+1} leaves the verifier's input unchanged.*
2. *Partitioning 2^{56} payloads into 2^8 Halo IDs gives an average fiber size of $2^{56} / 2^8 = 2^{48}$.*
3. *Suppose $G \neq G'$ but $\tau_1(G) = \tau_1(G')$. Let $d_0 = \max\{d \in \{1,\ldots,12\} : G_d \neq G'_d\}$ be the deepest layer where G and G' differ. Then for all $d > d_0$, $G_d = G'_d$, so $\tau_{d_0+1}(G) = \tau_{d_0+1}(G')$, which implies $(G_{d_0}, \tau_{d_0+1}(G)) \neq (G'_{d_0}, \tau_{d_0+1}(G'))$. Since $\tau_1(G) = \tau_1(G')$, the sequence of pairs $(\tau_d(G), \tau_d(G'))$ starts unequal at some $d \leq d_0$ and is equal at $d = 1$; at the largest index $d^* \leq d_0$ where $\tau_{d^*}(G) = \tau_{d^*}(G')$, the inputs $(G_{d^*}, \tau_{d^*+1}(G))$ and $(G'_{d^*}, \tau_{d^*+1}(G'))$ are distinct while their hash outputs under H are equal, exhibiting an explicit collision of H.* ■

6 Unified Protocol Specification: The 64-Bit Einstein Tile and Metrological Reference Tables

6.1 Bit-Level Specification of the 64-Bit Einstein Tile

Definition (64-Bit Einstein Tile). [DEFINITION] Form ID adopted from the SHD-CCP Tensor Transfer Protocol (Hoard 2026g).

A tile is one 64-bit word made of the fixed fields of Table 4, present in every packet, and the form-dependent core word (Bits 0–31) and gate byte (Bits 48–55) of Table 5, whose reading the 4-bit Form ID selects.

Design rule. A receiver must know how to decode a packet before decoding it. The fields below are therefore at fixed positions in every packet, and the 4-bit **Form ID** (adopted from the SHD-CCP Tensor Transfer Protocol) selects the reading of the remaining 40 bits. The header byte (Bits 56–63) is the South Halo pre-filter of Lemma 15, so its bound of 2^{48} payloads per header value holds.

Table 4: Fixed fields of the Einstein Tile, present in every form.

Bits	Width	Field	Definition	Governing lemmas
61–63	3	Cycle step k	$k \in \{0, \ldots, 7\} \leftrightarrow \theta_{k+1} = (k+1)\pi/4$ and the exact spinor state $(1+i)^{k+1}$	8, 13 (Part 4)
58–60	3	Wheel unit	$u \in \{0, \ldots, 7\} \leftrightarrow \{1, 7, 11, 13, 17, 19, 23, 29\}$ (mod 30), split by $((\cdot)/5)$ into $\mathcal{B}_{\pm}$	4 (Part 5)
56–57	2	Structure selector	In one engine: 00, 01, 10 = $\mathcal{M}^{(1)}$, $\mathcal{M}^{(2)}$, $\mathcal{M}^{(3)}$ and 11 = $\mathcal{M}^{(4)}$, the group inverse $H_4 = (H_1H_2H_3)^{-1}$. In the four-structure system (§7.6): 00 input I, 01 balancer A, 10 balancer $\overline{A}$, 11 emergent E	7 (Part 3), §7.6
44–47	4	Step-multiplexed index	Steps 1–2: circle phase $s \in \mathbb{Z}/9$ (values 0–8); Step 3: half-cube index $l \in \{0, \ldots, 15\}$; Step 6: ghost index (0 = central cell, 1–6 = B_{+1}, B_{+2}, B_{+3}, B_{-1}, B_{-2}, B_{-3}); other steps: one of the 12 coordinates of $\alpha\hat{f}\hat{f}(3)$. The step is read from Bits 61–63	7 (Part 1), 8
40–43	4	Form ID	Selects the reading of Bits 0–31 and 48–55 (table below)	§6.1
36–39	4	Stream slot	0–11: engine position in the twelve-engine ring; 12–15: the projections w, x, y, z. With Bits 56–57 it forms the 6-bit stream ID (§7.7). In a stand-alone engine the slot is 0	§7.4, §7.7
32–35	4	Scale register	Dyadic level l (scale 2^{-l}), Fibonacci index m ($N = F_m \le F_{15} = 610$), or subdivision frequency ν (Form 8)	8 (Steps 3–4), 11

Table 5: Form-dependent fields: Bits 0–31 (core word) and Bits 48–55 (gate byte).

Form ID	Name	Core word (Bits 0–31)	Gate byte (Bits 48–55)	Exactness
0	Smallest-three	Unit quaternion: 2-bit index of the largest component + three 10-bit quantized components, reconstruction derivative $\lvert g'(v)\rvert \le 1$	Geometric gate (below)	Lossy, bounded error (Lemma 13 Part 1)
1	Hurwitz	The doubled coordinates 2Q = (2w, 2x, 2y, 2z) of an integer or Hurwitz quaternion, gcd-reduced and sign-normalized (first non-zero coordinate positive, which leaves the rotation $Q\mathbf{x}\overline{Q}/N(Q)$ unchanged; the spinor sign is carried by the cycle step, Bits 61–63). The w byte holds $2w \in [0, 127]$ in its low 7 bits and the **loss flag** in its top bit; the x, y, z bytes hold values in [−127, 127]. On overflow the state is renormalized to the nearest representable direction and the loss flag is set. The cycle $(1+i)^k$ never overflows	Geometric gate	Exact while the flag is clear (Lemma 13 Parts 2–4)
2	Prime-triplet CRT	Three 10-bit residues (x_1, x_2, x_3) mod (881, 883, 887), decoded by Garner's algorithm with the closed-form (c_{12}, c_{23}, c_{13}); the top 2 bits are reserved (zero)	Geometric gate	Exact on [0, 690,017,701) (Lemma 9 Part 2)
3	Exact scalar	One IEEE-754 binary32 value, stored bit for bit	Bits 54–55: component w, x, y, z of a quad-slice v = [w, x, y, z]; Bits 48–53: slice counter	Exact
4–15	Codex shape	Bits 0–15: point index in the shape; Bits 16–31: two 8-bit refinement coordinates within the point's cell or face	Sub-region index	Exact about the codex point named; lossy about the original vector

Geometric gate (Forms 0–2). Bits 53–55: locus (0, …, 4 = T_0, …, T_4; 5 = core K; 6 = cube V; 7 = Cube 3 W = $[-4, 4]^3$). Bits 50–52: 2-bit C_9 offset or D_3 arc sector plus the 1-bit sense flag σ_r of Lemma 8 Step 5. Bits 48–49: harmonic flag (00 midpoint $\mathcal{C}_0$; 01 Klein gate $(4/3)\mathcal{C}_0$ / radial $(2/3)\mathcal{C}_0$; 10 Apollonius circle; 11 holonomy H).

Codex shapes (Forms 4–15). Each shape is an object whose coordinates the lemmas fix exactly, so every party can rebuild the same codex.

Table 6: Codex shapes of Forms 4–15.

Form	Shape	Points	Source
4	Voxel cube V	64 cells	Lemma 5
5	Core K	8 cells	Lemma 5
6	Stella octangula ($\Delta_1 \cup -\Delta_1$)	8 vertices	§7.5
7	Cuboctahedron (nearest neighbours in D_3)	12	§7.1
8	Geodesic icosahedron of frequency ν (Bits 32–35); the icosahedron is the vertex figure of {4,3,5}	$10\nu^2 + 2$	Lemma 8 Prop. 6.4
9	Hurwitz units	24	Lemmas 5, 13

Form	Shape	Points	Source
10	Binary dihedral group $2D_3$	12	Lemma 8 Prop. 6.3
11	Nine-ray circle $\mathbf{x}(s)$	9	Lemma 8 Step 1
12	Central cell and the six ghost cells of {4,3,5}	7	Lemma 8 Prop. 6.4
13	Golden circle points $k/\varphi \bmod 1$, $N = F_m$ (m in Bits 32–35)	F_m	Lemma 11
14	Wheel units mod 30	8	Lemma 4 Part 5
15	Escape: a user-defined codex, named by its hash in the Merkle attestation	—	Lemma 15

Exactness of the forms. Form 3 stores the value itself and is exact. A split into an 8-bit coarse value and a 16-bit residual would use 24 bits and could reproduce at most 2^{24} of the 2^{32} binary32 patterns (380 of 20,000 random values in a test), so it is not used. A codex packet is deterministic (every receiver reconstructs the same codex point) and lossy (it carries at most $\log_2$ of the number of points plus the refinement and sub-region bits).

6.2 Unified Metrological Standardization Table

Exact formulas are **[PROVEN]** and physical lengths **[COMPUTED]**. Evaluated at f_{cal} = 40 Hz, 20°C, 50% RH, 101.325 kPa (v_{local} = 343.98 m/s, λ = 8.5995 m, $\mathcal{C}_0$ = 53.748 cm ± 0.022 cm). The expanded uncertainty U = 0.22 mm (k = 2; combined relative standard uncertainty 2.09×10^{-4} from the six-term GUM budget of Hoard 2026d) and the temperature coefficient $d\mathcal{C}_0/dT \approx 0.98$ mm/K are **[COMPUTED]**.

Table 7: Engine lengths in cubits, wavelength fractions, acoustic phases and physical lengths.

Manifold Domain	Structural Feature / Coordinate	Exact Formula in Cubits ($\mathcal{C}_0$)	Wavelength Fraction (λ)	Acoustic Phase ($k\Delta x$)	Physical Length (20°C, 50% RH)	Governing Lemma & Status
Lower Manifold ($\mathcal{M}_{lower}$)	Four-Fold 1/16 Footprint	$(1/16)\mathcal{C}_0$	$\lambda/256$	$\pi/128$	3.359 cm	**Lemma 8 (Step 3)** **[PROVEN]**
Lower Manifold ($\mathcal{M}_{lower}$)	Harmonic Offset MF = $x_F - x_M$	$(1/3)\mathcal{C}_0$	$\lambda/48$	$\pi/24$	17.916 cm	**Lemma 5 (Part 3)** **[PROVEN]**
Lower Manifold ($\mathcal{M}_{lower}$)	Centered Radial Harmonic Conjugate F_{rad}	$(2/3)\mathcal{C}_0$	$\lambda/24$	$\pi/12$	35.832 cm	**Lemma 5 (Part 3)** **[PROVEN]**
Lower Manifold ($\mathcal{M}_{lower}$)	**Standardized Single Voxel (x_M, Core Half-Edge R_1)**	$1\mathcal{C}_0$	$\lambda/16$	$\pi/8$	53.748 cm	**Lemma 5 (Part 1)** **[DEFINITION / PROVEN]**
Lower Manifold ($\mathcal{M}_{lower}$)	**Klein-Wave Merge Gate (x_F, Edge Harmonic Conjugate)**	$(4/3)\mathcal{C}_0$	$\lambda/12$	$\pi/6$	71.664 cm	**Lemma 5 (Part 3)** **[PROVEN]**
Lower Manifold ($\mathcal{M}_{lower}$)	**Cube 1 Core Edge (L_{core}) = Cube 2 Half-Edge (R_2)**	$2\mathcal{C}_0$	$\lambda/8$	$\pi/4$	1.0750 m	**Lemma 5 (2–3) & Lemma 8** **[PROVEN]**
Lower Manifold ($\mathcal{M}_{lower}$)	Apollonius Circle Radius $R_{\mathcal{A}}$ = √(EB · EC)	$2\sqrt{2}\,\mathcal{C}_0 \approx 2.828\mathcal{C}_0$	$(\sqrt{2}/8)\lambda$	$\sqrt{2}\,\pi/4$	1.5202 m	**Lemma 5 (Part 3b)** **[PROVEN]**
Lower Manifold ($\mathcal{M}_{lower}$)	Centered Voxel Cube Corner Radius $\|\mathbf{x}\|_{max}$	$2\sqrt{3}\,\mathcal{C}_0 \approx 3.464\mathcal{C}_0$	$(\sqrt{3}/8)\lambda$	$\sqrt{3}\,\pi/4$	1.8619 m	**Lemma 10 (Part 3)** **[PROVEN]**
Lower / Upper Match	**Cube 2 Full Edge (L_{cube}) = Tier T_0 Minor Radius (ϱ_0)**	$4\mathcal{C}_0$	$\lambda/4$	$\pi/2$	2.1499 m	**Lemmas 4 (1), 5 (3), 8 (Step 4)** **[PROVEN]**
Cross-Manifold Gap	**Radial Separation dist(V, T_0) = 11 − 2√3**	$(11 - 2\sqrt{3})\mathcal{C}_0 \approx 7.536\mathcal{C}_0$	0.471λ	0.942π	4.0504 m	**Lemma 10 (Part 3)** **[PROVEN]**
Upper Manifold ($\mathcal{M}_{upper}$)	Tier Minor Radii $\varrho_n = (2n + 4)\mathcal{C}_0$ (n = 0…4)	4, 6, **8**, 10, $12\,\mathcal{C}_0$	(1/4)…(3/4)λ	(π/2)…(3π/2)	2.15, 3.22, **4.30**, 5.37, 6.45 m	**Lemmas 4 (1) & 7 (2)** **[PROVEN]**
Upper Manifold ($\mathcal{M}_{upper}$)	Tier Winding Spans $\mathcal{W}_n = (16 - 2n)\mathcal{C}_0$ (n = 0…4)	16, 14, **12**, 10, $8\,\mathcal{C}_0$	1…(1/2)λ	2π…π	8.60, 7.52, **6.45**, 5.37, 4.30 m	**Lemmas 4 (1) & 7 (2)** **[PROVEN]**

Manifold Domain	Structural Feature / Coordinate	Exact Formula in Cubits ($\mathcal{C}_0$)	Wavelength Fraction (λ)	Acoustic Phase ($k\Delta x$)	Physical Length (20°C, 50% RH)	Governing Lemma & Status
Upper Manifold ($\mathcal{M}_{upper}$)	**Tier-Winding Invariant ($\varrho_n + \mathcal{W}_n$)**	$20\mathcal{C}_0$ ($\forall n$)	$(5/4)\lambda$	$5\pi/2$	10.7495 m	**Lemma 4 (Part 1)** **[PROVEN]**
Upper Manifold ($\mathcal{M}_{upper}$)	Disjoint Spherical Shells $[R_n - r_n, R_n + r_n]$	[11,19]... [143,167]$\mathcal{C}_0$	(11/16)... (167/16)λ	(11π/8)... (167π/8)	[5.91, 10.21]... [76.86, 89.76] m	**Lemma 10 (Part 2)** **[PROVEN]**
Upper Manifold ($\mathcal{M}_{upper}$)	Inter-Shell Clearances $\Delta_{n,n+1} = (6n + 10)\mathcal{C}_0$	10, 16, 22, 28 $\mathcal{C}_0$	5/8, 1, 11/8, (7/4)λ	(5π/4)... (7π/2)	5.37, 8.60, 11.82, 15.05 m	**Lemmas 4 (2) & 10 (2)** **[PROVEN]**

7 Layer V — System Level: Projection, Tessellation, Handshake, Memory, and the Four-Structure System

One engine is one cube: the voxel cube V with its core K and five separated tiers. This section derives, from Lemmas 1–15, how engines stack, how they exchange data at the two orthogonal steps of the cycle, how they interact without sharing points, and how a hierarchy of twelve engines behaves as memory. It introduces no new lemma. Every statement is tagged, and the computational checks are in `verify_system.py` and `verify_memory.py`.

7.1 Tessellation and the 3:1 Interleave

Theorem 7.1 (Stacking). **[PROVEN]**

Translates of $V = [-2, 2]^3$ by $4\mathcal{C}_0\,\mathbb{Z}^3$ tile space with disjoint interiors. Identifying opposite faces turns one cube's 64 voxels into the torus $(\mathbb{Z}/4)^3$, the lattice on which the Hoard 2026c kernel runs. Each engine is one cell of an unbounded periodic stack, and neighbouring cubes meet only along faces.

Theorem 7.2 (3:1 Interleave and the Implied Fourth). **[PROVEN]**

Let $L = 2\mathbb{Z}^3$ and $o_0 = (0,0,0)$, $o_1 = (1,1,0)$, $o_2 = (1,0,1)$, $o_3 = (0,1,1)$. Then:

1. *The four translates $L + o_i$ are pairwise disjoint, and their union is the face-centred cubic lattice $D_3 = \{\boldsymbol{x} \in \mathbb{Z}^3 : x_1 + x_2 + x_3$ even$\}$.*
2. *The 120° rotation about (1,1,1) (the cyclic permutation matrix) fixes L and cycles the three translates $L + o_1$, $L + o_2$, $L + o_3$. This is the body-diagonal axis of Lemmas 4 and 8.2.*
3. *$o_1 + o_2 + o_3 \equiv 0 \pmod 2$: the offsets form the Klein four-group, so any three of the four translates determine the fourth.*
4. *The points $2o_i - (1,1,1)$ are the vertices of a regular tetrahedron inscribed in the cube (the 0° tetrahedron Δ_1 of Lemma 8.2).*
5. *Each point of D_3 has 12 nearest neighbours, the cuboctahedron vectors $(\pm1, \pm1, 0)$ and their permutations (kissing number 12). Along (1,1,1) they split 3 + 6 + 3, the middle six forming a regular hexagon.*

Proof. Each point of D_3 reduces mod 2 to exactly one even-weight vector, and the even-weight vectors are exactly $o_0, \ldots, o_3$; odd-weight points lie in no translate. The cyclic permutation maps (1,1,0) → (0,1,1) → (1,0,1) → (1,1,0). The even-weight vectors of $(\mathbb{Z}/2)^3$ form a subgroup of order 4 in which every element is its own inverse, so the three non-zero elements sum to zero. The four points $2o_i - (1,1,1)$ are pairwise at distance $2\sqrt{2}$. The norm-2 points of D_3 are exactly the 12 vectors listed, and their inner products with (1,1,1) take the values 2, 0, −2 three, six and three times. ■

The implied fourth has three exact twins in the monograph: the lattice (three translates force the fourth), the group ($H_4 = (H_1H_2H_3)^{-1}$, Lemma 7), and the quaternion algebra (ijk = −1, so the three imaginary axes generate the scalar axis). In each, "3:1" means three free parts and one part fixed by closure.

Theorem 7.3 (Growth at Every Scale). **[PROVEN]**

The chain $2\mathbb{Z}^3 \subset D_3 \subset \mathbb{Z}^3 \subset (1/2)D_3 \subset (1/2)\mathbb{Z}^3 \subset \cdots$ has indices 4, 2, 4, 2, ... Each 3:1 interleave (index 4) followed by a centring step (index 2) is one 2 × 2 × 2 subdivision (index 8), the three halvings of Lemma 8 Step 3.

Proof. The densities per unit cell are 1/8, 1/2, 1, and scaling by 1/2 multiplies density by 8. ■

The construction repeats inward without limit and outward by scaling by 2, so the same 3:1 step appears at every level.

Theorem 7.4 (Hyperbolic Stacking). **[ESTABLISHED]**; criterion **[COMPUTED]**.

A regular honeycomb {p, q, r} is spherical, Euclidean or hyperbolic as $\sin(\pi/p)\sin(\pi/r)$ is greater than, equal to, or less than $\cos(\pi/q)$ (Coxeter). For cubes: {4,3,3} gives 0.612 > 0.5 (spherical), {4,3,4} gives 0.5 = 0.5 (the cubic tiling), and {4,3,5} gives 0.416 < 0.5, the compact hyperbolic honeycomb in which five cubes with 72° dihedral angles meet at each edge. The number of cubes within distance R grows exponentially in R, and under the ghost map G (a Poincaré-ball isometry) the whole honeycomb lies inside the unit ball.

Seam-free labels. **[ESTABLISHED]**; checked. Label the 16 cells of a cube face with a 4 × 4 panmagic square, for example rows (1, 12, 7, 14), (8, 13, 2, 11), (10, 3, 16, 5), (15, 6, 9, 4). In the face tiling of Theorem 7.1 every 4 × 4 window at any offset is again panmagic and every 2 × 2 block sums to 34; all 384 panmagic squares on 1, ..., 16 have this property. The 3D analogue is Problem P4 of the companion paper.

7.2 The Handshake Windows

Theorem 7.5 (Two Handshake Windows). **[PROVEN]** From Lemmas 8 and 13.

In the gcd-reduced cycle $q_k = (1 + i)^k$, the transverse amplitude $|\sin(k\pi/4)|$ equals 1 only at k = 2 and k = 6, where $q_2 = i$ and $q_6 = -i$. Both give the same rotation, the 180° half-turn about the x-axis, with opposite spin sign.

Proof. $\sin(k\pi/4) = \pm 1$ iff $k \equiv 2 \pmod 4$. The gcd-reduced powers are $1 + i, i, -1 + i, -1, -1 - i, -i, 1 - i, 1$. A unit quaternion q and −q induce the same rotation $\boldsymbol{x} \mapsto q\boldsymbol{x}\bar{q}$, and a pure unit quaternion induces a half-turn. ■

Call k = 2 the **onload window** (expansion meets the orthogonal boundary) and k = 6 the **offload window** (collapse meets it; symmetry folding happens here). At both windows a partner sees the same spatial orientation, so two engines need not agree on their position in the double cover; the spin sign is the one bit the geometry never exposes. What crosses the handshake is finite and exactly specified (Table 8).

Table 8: What crosses a handshake window.

Item	Size	Why it suffices	Basis
Five cumulative moments M, I, J, S, C_p	5 numbers per band	Exterior agreement; any small target is reachable since the Jacobian is invertible for $\alpha \notin \{-1/2, 1/2, 3/2\}$	Lemma 6 Parts 1–4
One fundamental arc of the trefoil	1/6 of the curve	The six integer matrices of D_3 rebuild the rest exactly	Lemma 8 Step 6
Address residues	Three 10-bit residues	CRT with closed-form inverses reconstructs the address in O(1)	Lemma 9
Root hash	256 bits	Any change in the 12 nested layers changes it unless H has a collision	Lemma 15
Net rotation	One unit quaternion	Open-convention holonomy $H = q_0\, q_{719}^{-1}$	Lemma 12

What cannot cross. A product forgets its factors: the $24^3 = 13{,}824$ three-step chains of Hurwitz units give only 24 products, 576 chains each (**[COMPUTED]**). No scheme can rebuild a chain's history from its current quaternion.

Timing between different engines. Two engines on different integer carriers can exchange only at instants when both are at whole periods. By Lemma 11, Proposition 11.4, carriers f_1, f_2 share a window of k pulses of 1/60 s iff $\mathrm{lcm}(60/(\gcd(60, f_1)), 60/(\gcd(60, f_2)))$ divides k.

7.3 Interaction Without Contact

Theorem 7.6 (Separation Inside the Stack). **[PROVEN]** From Lemma 10.

Each engine lies in the closed ball of radius $167\,\mathcal{C}_0$ about its centre. If one engine is centred in each cell of a cubic tiling with edge greater than $334\,\mathcal{C}_0$, distinct engines are disjoint. Cubes touch along faces; engines never do.

Proof. Each closed ball lies in the interior of its own cell when the edge exceeds twice the radius, and distinct cells have disjoint interiors. ■

Theorem 7.7 (The Five-Moment Channel). **[PROVEN]** Relative to the profile definitions of OpenAI (2026);

from Lemma 6.

Beyond the joining radius X_b, the exterior stress of an engine is determined by its exterior profile and the five numbers M, I, J, S, C_p at X_b. Two engines with different interiors but equal moments are indistinguishable from outside (offload). Any small target set of moments can be produced by localized bumps, since the moment Jacobian is invertible and the quadratic terms are handled by Banach's fixed point (onload).

Theorem 7.8 (The Projected Channel). **[PROVEN]** From Lemma 8, Proposition 6.2, and Hoard (2026c).

Each engine maps its spinor state q (representative with w > 0) into $\mathbb{H}^3$ by the Lorentz lift Φ(q); in Cayley coordinates $q = \mathcal{C}_{rat}(\mathbf{p})$ this is the ghost map $G(\mathbf{p})$. Two engines i, j are related by

$$Q_h(i,j) = \langle G_i, G_j \rangle_L^2 - 1 = \sinh^2 d_{\mathbb{H}}(G_i, G_j),$$

which is rational, symmetric, zero only when the states coincide, and computed from q_i and q_j alone. A boost $B \in SO^+(3,1)$ moves an engine's image as an isometry without changing its internal geometry; this is the ghost of Proposition 6.4, whose boosts are the face-neighbour translations of the {4,3,5} honeycomb.

Theorem 7.9 (Directed Exchange). **[PROVEN]** for three states. From Hoard (2026c) with Lemma 7 Part 3(c).

Logits built from Q_h and projected by sparsemax give a rational, exactly sparse, row-stochastic kernel. With the rational valve on reverse edges, the three-state gear cycle carries the stationary current $J = (1 - \rho_{ba}\rho_{cb}\rho_{ac})\Pi_+/Z > 0$. Onload and offload are the forward and reverse directions of this valve. On the 64-cell lattice the current is positive for translation-invariant valves (Lemma 7 Part 3(e)); valves that vary from cell to cell are Problem P1 of the companion paper.

Theorem 7.10 (What Any Exchange Preserves and Loses). **[PROVEN]**

Exchanges preserve the shared geometry at the handshake windows (Theorem 7.5), the five moments (Theorem 7.7), the hyperbolic distance (Theorem 7.8), the address (Lemma 9) and the root-hash certificate (Lemma 15). They lose the chain history (products are not injective) and anything beyond 64 bits per packet (pigeonhole). No geometric map turns elliptic-curve multiples into an O(1) lookup (Lemma 9 Part 4).

7.4 Memory in the Twelve-Engine Hierarchy

Definition (Six-Tier Engine and the Twelve-Engine Hierarchy). **[DEFINITION]**

A six-tier engine *is the core plus the five tiers of Lemma 4. Its memory state is its holonomy $H \in S^3$ (Lemmas 7, 12) together with its 64-bit packet. Three data engines A_1, A_2, A_3 determine an implied fourth $F_A = (A_1A_2A_3)^{-1}$, itself a full six-tier engine. Doing this three times gives nine data engines and three implied fourths, twelve engines in all. An optional thirteenth engine $F_\bullet = (F_AF_BF_C)^{-1}$ is the fourth of the fourths. The memory cycle has three 120° phases, of which onload and offload use two.*

Proposition 7.11 (Implied Fourths Are Parity Engines). **[PROVEN]** From Lemma 7 Part 3(d).

Each group of four tolerates the loss of any one engine, so the hierarchy survives up to three losses, one per group. Two losses in the same group are unrecoverable. Only 9 of the 12 engines carry data (75%). The thirteenth engine adds exactly one recovery case: the loss of a data engine together with its own fourth (for example A_1 and F_A), since $F_\bullet$, F_B, F_C rebuild F_A, which then rebuilds A_1.

Proposition 7.12 (Offload Onto the Inverse Tier System Is Exact). **[PROVEN]** From Lemmas 4, 7, 13.

Onload carries data from the core out through the tiers by the word $W = U_3U_2U_1U_0$ with $U_{even} = U$, $U_{odd} = U^{-1}$. Then:

1. *The inverse word $W^{-1} = U_0^{-1}U_1^{-1}U_2^{-1}U_3^{-1}$ is again U, U^{-1}, U, U^{-1}, so the exact inverse tier system is the same tier system traversed from tier 4 to tier 0.*
2. *$W = I$, and the spinor word is $h\,\overline{h}\,h\,\overline{h} = 1$ exactly: no residual rotation and no sign ambiguity.*
3. *The radial scale is $R_4/R_0 = 155/15 = 31/3$ out and 3/31 back, both exact.*
4. *Left multiplication $q \mapsto hq$ is a norm-preserving bijection of the Hurwitz integers with inverse $q \mapsto \overline{h}q$, so data passed through any number of tier rotations keeps its norm and bit width. Products of data with data grow (Lemma 13 Part 3); rotations by unit spinors do not.*
5. *Offloading an implied fourth returns exactly the product of its three parents, never their individual histories.*

Proof. Items 1–3 are direct computation with integer matrices and exact quaternions. Item 4: N(h) = 1, multiplication by a unit maps the Hurwitz order onto itself, and N(hq) = N(h)N(q). Item 5 is Proposition 7.11 with the history-loss count of §7.2. Checked in `verify_memory.py` *(5,000 Hurwitz data quaternions).* ■

Proposition 7.13 (Twelve Engines as the Cuboctahedral Shell). [PROVEN]

The 12 nearest neighbours of Theorem 7.2(5) split by their zero coordinate into three planar groups of four. Each group sums to zero, so any three of its vectors determine the fourth, and the 120° rotation about (1,1,1) carries each group onto another. The twelve-engine hierarchy is therefore the cuboctahedral shell of one engine in the stack, one group of four per coordinate plane.

Counts. [CORRESPONDENCE] 12 × 6 = 72 tier slots match the 72 middle-gear frames (720/10) of Lemma 12, split 8, 7, 7, 7, 7, 8, 7, 7, 7, 7 by the blocks A_i; 12 × W_Σ = 720 matches the outer frames. Two different 72s occur in the corpus: the count 72 = 12 × 6 and the 72° dihedral angle of {4,3,5}. They are unrelated beyond 360°/5 = 72° versus 360°/72 = 5°.

Proposition 7.14 (The Third 120° Phase). [PROVEN]

Let three real channels (a, b, c) occupy the three phases, and set $s = a + b + c$ and $z = a + \omega b + \omega^2 c$ with $\omega = e^{2\pi i/3}$.

1. *(a, b, c) is recovered exactly from s and z, and a 120° cyclic shift of the channels fixes s and multiplies z by ω. So three phases are one invariant line plus one rotating plane.*
2. *If the phases are disjoint time slots, the third phase is a free channel: 3/2 the throughput of two.*
3. *If the phases are phasors on one complex carrier, $1 + \omega + \omega^2 = 0$ and the carrier holds only two real numbers, so the third phase is not free.*
4. *Under the same rotation the coset $L + o_0$ of Theorem 7.2 is fixed and the three face cosets are cycled. The rotating plane matches the three data engines and the invariant line matches the implied fourth.*

Proof. $a = (1/3)(s + 2\,\mathrm{Re}\, z)$, $b = (1/3)(s + 2\,\mathrm{Re}(\omega^2 z))$, $c = (1/3)(s + 2\,\mathrm{Re}(\omega z))$, and the shift (a, b, c) ↦ (c, a, b) gives $c + \omega a + \omega^2 b = \omega z$. Items 2–3 count real degrees of freedom. Item 4 is Theorem 7.2(2). ■

Default use. [DEFINITION]; the phase model is Problem P3 of the companion paper. Carry the implied fourths' traffic, or the reference clock of Proposition 11.4, in the third phase. Both data phases then carry only data, a 4/3 data-rate gain over sharing them with parity (Table 9).

Table 9: Data, parity and erasure tolerance of the memory configurations.

Configuration	Data bits	Parity bits	Losses survived
12 engines, parity sharing two phases	576	192	one per group
12 engines, parity in the third phase	576 per two data phases (×4/3 data rate)	192 in phase 3	one per group
13 engines	576	256	one per group, or a data engine with its own fourth
Third phase used for data (time slots only)	864	192	one per group

No arrangement stores more than the bits written (pigeonhole). The hierarchy adds erasure tolerance and phase throughput, not capacity.

7.5 The Second Lift: Three Ghost Tetrahedra Collapse Into the Central Cube

The second lift is the chain of Lemma 8: the Step 1 circle phase goes to the point **x**(s) of the nine-ray circle and then to the null cone of $\mathbb{R}^{4,1}$ by the conformal lift. In $\mathbb{R}^{4,1}$ every sphere is a single vector and every isometry of the hyperbolic cells acts linearly. The **ghost triple** is the set of three positive ghosts B_{+1}, B_{+2}, B_{+3} of Proposition 6.4, carried into the cells across the faces x = a, y = a, z = a. (It is distinct from the triple of Lemma 8.2, rotated by 0°, 40°, 80° about a shared axis.)

Proposition 7.15 (Orthogonal Directions Are 120° Apart Along the Diagonal). [PROVEN]

Projected onto the plane perpendicular to (1,1,1), the axis directions $\mathbf{e}_1$, $\mathbf{e}_2$, $\mathbf{e}_3$ are exactly 120° apart, each of length √(2/3); so are the face offsets (1,1,0), (1,0,1), (0,1,1). The three ghosts are therefore mutually orthogonal in space and 120° apart in phase.

Proof. Remove the component along (1,1,1). The cyclic permutation, a 120° rotation fixing (1,1,1), carries each projected vector to the next and preserves length. ■

Proposition 7.16 (A Tetrahedron and Its Inverse Make the Cube). [PROVEN]

$\Delta_1 = \{(1,1,1), (1,-1,-1), (-1,1,-1), (-1,-1,1)\}$ and its point reflection $-\Delta_1$ are disjoint, and $\Delta_1 \cup -\Delta_1 = \{\pm 1\}^3$, the 8 vertices of the cube. The point reflection commutes with every rotation and reverses handedness, so the point-reflected engine runs the identical cycle with the mirror trefoil.

Proof. The vertices of Δ_1 have an even number of minus signs and those of $-\Delta_1$ an odd number; together they exhaust $\{\pm 1\}^3$. $-I$ is central in O(3) with determinant -1. ■

Proposition 7.17 (Collapse at Step 6). [PROVEN]

(1) The inverse boosts B_{+k}^{-1} carry the three ghosts back into the central cell, where they coincide with the original tetrahedron; with $-\Delta_1$ they fill the 8 vertices of the central cube. (2) Step 6 is the offload window (spinor $-i$); its half-turn maps Δ_1 to itself and exchanges the positive and negative ghost triples, and the six ghost cells are exactly the six face-neighbours of the central cube. (3) In $\mathbb{R}^{4,1}$, each tetrahedron's circumsphere is one vector, and "the three coincide" is the linear condition that each pulled-back sphere vector equals the central one, $e_o - (3/2)e_\infty$ in the cube's coordinates.

Proof. (1) Proposition 6.4(ii) and Proposition 7.16. (2) The half-turn about x maps $(1,1,1) \leftrightarrow (1,-1,-1)$ and $(-1,1,-1) \leftrightarrow (-1,-1,1)$; Proposition 6.4(v) gives the exchange. (3) Isometries of the ball model are Möbius maps preserving the ball, which act linearly on $\mathbb{R}^{4,1}$ and send sphere vectors to sphere vectors ([ESTABLISHED]); all ten vertices of $\Delta_1, \Delta_2, \Delta_3$ are orthogonal to $e_o - (3/2)e_\infty$ (checked). ■

Definition (Collapsed Distribution and Disagreement). [DEFINITION]; properties **[PROVEN]**.

*Each ghost k carries a probability distribution μ_k on the 8 vertices of its cell's cube. The **collapsed distribution** is $\mu = (1/3)\Sigma_k (B_{+k}^{-1})_* \mu_k$ on the central cube, and the **disagreement** is $D = \max_{k \neq l} d_{TV}((B_{+k}^{-1})_* \mu_k, (B_{+l}^{-1})_* \mu_l)$. Then μ is a probability distribution; $D = 0$ exactly when the three agree, and then μ equals each of them; and any one ghost is recovered from the other two and μ: $(B_{+3^{-1}})_* \mu_3 = 3\mu - (B_{+1^{-1}})_* \mu_1 - (B_{+2^{-1}})_* \mu_2$. The central cube is the implied fourth of the three ghosts.*

The ten-node core. Δ_1, Δ_2, Δ_3 of Lemma 8.2 share the pole (1,1,1) and have 9 distinct outer vertices: 10 nodes. With $-(1,1,1)$ and the centre they form a 12-node set that splits 1 + 9 + 1 along the axis (a 9-sided bipyramid). The 12-point cuboctahedron of §7.1 splits 3 + 6 + 3; the two are different objects.

7.6 The Four-Structure System

Definition (Four-Structure System). [DEFINITION]

*Let A be a twelve-engine structure (§7.4) with holonomies $A_1, \ldots, A_{12}$, and $\overline{A}$ its **dynamical inverse**, $\overline{A}_j = A_j^{-1}$. A third structure I carries the input. The **balancers** are A and $\overline{A}$, placed on either side of the input (the* sandwich*), and the **emergent structure** E is fixed by the slot-wise closure*

$$A_j \, I_j \, A_j^{-1} \, E_j = 1, \qquad j = 1, \ldots, 12.$$

In the default configuration the balancer frame is global, $A_j = A$ for all j, and carries no data. (The point reflection of Proposition 7.16 is the spatial *inverse: same holonomy, opposite handedness. A dual-helix pair may carry both inverses.)*

Theorem 7.18 (Balancers Conserve Rotation Angles). [PROVEN]

$E_j = A \, I_j^{-1} \, A^{-1}$, so $\mathrm{Re}(E_j) = \mathrm{Re}(I_j)$: the emergent structure has exactly the input's rotation angles, about axes turned by A. Had the balancers been placed next to each other ($I_j \, A \, A^{-1} \, E_j = 1$), they would cancel and give $E_j = I_j^{-1}$.

Proof. Solve the closure for E_j; conjugation preserves the real part of a unit quaternion, which fixes the rotation angle. ■

Theorem 7.19 (Forward and Inverse Streams Cancel Only When Nested). [PROVEN]

A product in which every factor also appears inverted equals 1 for all values of the factors if and only if it reduces to the empty word by cancelling adjacent inverse pairs. In particular $q_1 \cdots q_9 \, q_9^{-1} \cdots q_1^{-1} = 1$, while $q_1 \cdots q_9 \, q_1^{-1} \cdots q_9^{-1} \neq 1$ for generic values.

Proof. "If" is associativity. "Only if": generic tuples of unit quaternions generate a free group (Epstein 1971, **[ESTABLISHED]***), so a word that does not reduce to nothing is not the identity for generic values. Checked: 300 of 300 nested words gave 1, 0 of 300 same-order words did.* ■

Proposition 7.20 (The Emergent Structure Closes Its Own Groups). **[PROVEN]**

If $I_1I_2I_3I_4 = 1$ in a group of four, then $E_4E_3E_2E_1 = A\,(I_1I_2I_3I_4)^{-1}A^{-1} = 1$. The emergent structure is the input inverted, read in reverse and seen in the balancer's frame; its groups close with no extra parity engines. With the slot relation and both group parities, every loss of up to 3 of the 8 engines (4 input, 4 emergent) in a group is recoverable, as are 64 of the 70 losses of 4.

Proof. Multiply out; the inner $A^{-1}A$ pairs cancel. The erasure counts are by exhaustive peeling over all patterns ([COMPUTED]*).* ■

The default configuration stores 9 data engines among 48. The gain at this scale is erasure tolerance and a conserved invariant, not capacity.

7.7 The 64 Streams

Definition (The 64 Streams). **[DEFINITION]**; counts **[PROVEN]**.

The four structures I, A, $\overline{A}$, E each log 12 engine streams and 4 projection streams: 4 × 16 = 64 streams. The stream ID is the 6-bit pair (structure selector, stream slot) of the Einstein Tile. On the voxel cube V, structure z is the layer z; its 12 engines occupy the 12 boundary cells of the 4 × 4 layer in ring order and its 4 projections the 4 interior cells. On the ring of 12, a shift by 4 cells (the 120° step) has three orbits of four, the groups of §7.4, and a shift by 3 cells (the square's 90° rotation) has four orbits of three.

The four orthogonal projections of a structure are the quaternion components w, x, y, z of its net holonomy, the ordered product of its 12 engine holonomies (E read in reverse order). They are exactly orthogonal in $\mathbb{R}^4$ and are sent as Form 3 or Form 1 packets. They provide checks that exist only at this scale (**[PROVEN]**; each checked on 200–500 exact cases):

- **Norm check:** $w^2 + x^2 + y^2 + z^2 = 1$ for every structure.
- **Equilibrium monitor:** the w stream of E equals the w stream of I (Theorem 7.18 with a global frame).
- **Balancer check:** the net of A times the net of $\overline{A}$ read in reverse is exactly 1.
- **Collapse check:** the disagreement D of §7.5 travels in the codex sub-region field.

At one packet per stream per step, the system logs 64 × 64 = 4,096 bits per cycle step. The equality of 64 streams, 64 voxels and 64 tile bits is a count (**[CORRESPONDENCE]**).

8 Status of Claims

Table 10 lists every result with its status and the method by which it was checked. Every proof was re-read line by line, and no result in this monograph is conditional. The scripts are described in Appendix A.

Table 10: Consolidated status of claims.

Result	Basis and verification	Status
Lemmas 1, 2, 3 (Homothety, Miquel center z_0, Symmedian & Apollonius circle)	Symbolic algebra & synthetic angle/sine chase	**[ESTABLISHED / PROVEN]**
Lemma 4 (1–4) ($r_n + W_n = 20$, $\mathbf{U}_{even}$ axis $(-1,1,1)$, $\chi = 1/\sqrt{3}$, $h^2 = -\underline{h}$, $h^4 = -h$, $h^6 = 1$)	Exact matrix & rational quaternion arithmetic	**[PROVEN]**
Lemma 4 (5–6) ($\mathcal{B}_+ = \ker((\cdot)/5) \pmod{30}$, $\det J_g = 14$, $(\Delta + 1)$-coloring)	Exhaustive mod-30 check & graph induction; explicit torus-separation instance	**[PROVEN]**
Lemma 5 (1–5) (Symmedian foot, 2O cube group, $(0,2;4,4/3) = -1$, $24\sigma_{odd}(n)$, box primes {2,3,5,7,13}, no collinear prime triple)	Exhaustive lattice scans ($n < 30$; $[-2,2]^4$; $[-4,4]^3$; $\{0..3\}^3$, $\{0..3\}^4$); coordinate-free collinear-prime check on all prime pairs < 2,000	**[PROVEN]**
Lemma 5 (6) (Two-product 128-bit Gram $A_R < 2^{86}$; $48 + 2 \cdot 8 = 64$ sextics & degree-2 K3)	Bit-width bound & Rokhlin-Nikulin census	**[PROVEN / CORRESPONDENCE]**
Lemma 6 (5×5 generalized Vandermonde $\det B \neq 0$ & Banach contraction)	Rolle's theorem induction & contraction proof; symbolic check of the integrated stress formulas; Jacobian singular values	**[PROVEN]**

Result	Basis and verification	Status
Lemma 7 ($\dim \mathfrak{aff}(3) = 9 + 3 = 12$, $T_2 = (8,12)$, Markov tree current J, BCH cross-products)	Lie bracket algebra; tree theorem on 20,000 random 3-state chains (pass); BCH error decay ~ $\|\epsilon\|^3$; valve kernel on 5,000 exact cases; parity recovery on 500 exact groups	**[PROVEN]**
Lemma 8 (8.2, 8.3) ($(1/2)\|\mathbf{x}\|^2$ conformal lift, C_9 {0°,40°,80°} 9 rays, rank-4 K_5 lift, D_3 trefoil, rotated vertices leave Cube 1)	Exact rank/azimuth check, vertex-coordinate check & random pair test; 3 of 9 rays; integer D_3 matrices and 1/6 arc (720 points); ghost-map identities on ~3,000 exact pairs; $2D_3$ of order 12; golden torus knot	**[PROVEN]**
Lemma 9 (Mod-30 triplets, closed-form c_{12}, c_{23}, c_{13} for all primes 5...20000, (881,883,887))	Verified on all 2,260 primes < 20,000 & 2,000 random X	**[PROVEN]**
Lemma 10 (Disjoint shells [11,19]...[143,167]$\mathcal{C}_0$, gap $11 - 2\sqrt{3} \approx 7.536\mathcal{C}_0$, $\mathcal{M}_{\text{upper}} \cap \mathcal{M}_{\text{lower}} = \emptyset$)	Triangle inequality & norm maximization	**[PROVEN]**
Lemma 11 (Rational closure vs. φ ergodicity; 2 gap lengths with exact counts and lengths at Fibonacci N = 5, 8, ..., 89)	Weyl's criterion & exact circle gap computation; factorization of ν_{Cs} and shortest windows (Prop. 11.4)	**[ESTABLISHED / PROVEN]**
Lemma 12 (A_l sizes (8,7,7,7,7,8,7,7,7,7), $Q^{(10)} = (Q^{(12)})^{-1}$, $H = (Q^{(12)})^{-1} = q_{-1}q_{719}^{-1}$ under right-acting order, all three boundary conventions)	Associative telescoping & 720-frame random test	**[PROVEN]**
Lemma 13 ($((3 + 4i)/5)^n$ denominator 5^n with $a_n \equiv 3$, $b_n \equiv 4 \pmod 5$, Niven order $\in \{1,2,3,4,6\}$, $(1 + i)^8 = 16$ in 6 signed bits, $O(k(b + 2))$ bits)	Gaussian prime factorization, $\mathbb{Z}[i]/(5)$ splitting & Niven's theorem; 2,000 random chains	**[PROVEN]**
Lemma 14 (Gibbs $\pi(x) \propto e^{-\beta V(x)}$, absorbing harmonic measure, Hurwitz $Z_{\mathcal{H}}(\beta)$, circularity)	Detailed balance & 30-node Metropolis verification; Hoard 2026e numerics reproduced (prime mass 0.39 → 0.9995, $\lambda_{\min} = 1.402$)	**[PROVEN]**
Lemma 15 ($2^{72}/2^{64} = 256$ preimages, 2^{48} South Halo preimages, recursive Merkle collision reduction)	Pigeonhole principle & backward induction on d; CRC-32 affinity on 2,000 triples	**[PROVEN]**
§7 System propositions (FCC interleave, scale chain, {4,3,5}, handshake windows, panmagic tiling, history loss, twelve-engine parity, inverse tiers, third phase; collapse, balancers, streams)	Exact lattice, matrix and rational-quaternion checks	**[PROVEN / ESTABLISHED / CORRESPONDENCE]** as tagged in §7
§6.1 Einstein Tile (fixed fields, Forms 0–15, codex shapes 4–15, Form 1 normalization)	Bit partition 0–63 and form coverage checked against this text; sign normalization on 500 cases	**[DEFINITION]**
§6.2 Metrological table	Exact formulas from Lemmas 4, 5, 7, 8 and 10; lengths evaluated at $v_{\text{local}} = 343.98$ m/s	**[PROVEN / COMPUTED]**

9 Conclusion

The fifteen lemmas and the propositions of Section 7 describe one enclosed engine, its projection and the four-structure system completely: every object in Appendix B has a definition, a governing result and a tile field or a stated reason for having none, and every claim is proven or tagged. Lemma 8 carries both the cycle and the full Step 6 projection; splitting the projection into a lemma of its own would make sixteen, a matter of presentation, so the count is kept at fifteen.

The system level adds no new lemma. Stacking, the handshake windows, contact-free exchange, parity memory and the four-structure sandwich all follow from Lemmas 1–15, and what they add is erasure tolerance, exact inverse offload and a conserved invariant, not capacity. Everything still open is in the companion paper *Open Problems of the Geometric Engine*.

A Computational Verification

Every computational claim in this monograph is checked by ten reproducible scripts (Table A1). All pass. Lines printing `False` in the first two scripts report that a rejected alternative fails, for example the cyclic-permutation reading of $\mathbf{U}_{\text{even}}$ or the prime p = 1019 for Form 2.

Running them. Each script runs on its own with `python3` and prints its checks; `verify_publication.py` and `verify_supplementary.py` take the path of this monograph as an optional argument.

Table A1: The verification scripts.

Script	What it checks
`verify_lemmas_part1.py`, `verify_lemmas_part2.py`	Lemmas 1–15
`verify_reinforcements.py`	Results drawn from Hoard 2026c–e and OpenAI 2026
`verify_step6.py`	The Step 6 projection (Lemma 8, Propositions 6.1–6.5)
`verify_system.py`	§7.1–7.3
`verify_memory.py`	§7.4
`verify_supplementary.py`	Clock windows, trefoil bounds, cuboctahedron, separation, Jacobian exponents, markdown lint
`verify_honeycomb.py`	Honeycomb boosts, lattice current, Step 1 lift, Cube 3
`verify_tile_system.py`	Einstein Tile, §7.5–7.7
`verify_publication.py`	Definitions, cross-references and tile layout of this text

B Object Inventory

A single engine is completely described when every object it uses has a definition, a lemma that governs it, and a place in the 64-bit tile or a stated reason for having none. Table B1 covers this text, including the Einstein Tile fields and the system-level objects of §7.5–7.7.

Table B1: Object inventory of one engine and the four-structure system.

#	Object	Definition	Governing lemma	Tile field
1	Cubit $\mathcal{C}_0$	Conventions, item 1	4 (Part 3), §6.2	Implicit unit
2	Core K and voxel cube V	Conventions, items 2–3	5, 10	Bits 53–55 (values 5, 6)
3	Cube 3 W = $[-4,4]^3$	Lemma 8 Step 5	8, 10	Bits 53–55 (value 7)
4	Tiers T_0, …, T_4	Lemma 4	4, 10	Bits 53–55 (values 0–4)
5	Tier rotations $\mathbf{U}_n$, spinor h	Lemma 4 Parts 2, 4	4, 13	Form 1 core word
6	Harmonic lengths 1/3, 2/3, (4/3) $\mathcal{C}_0$	Lemma 5 Part 3	3, 5	Bits 48–49
7	30-node wheel	Lemma 4 Part 5	4	Bits 58–60
8	Eight-step phase k	Lemma 8	8, 13 (Part 4)	Bits 61–63
9	Circle phase $s \in \mathbb{Z}/9$ and its lift	Lemma 8 Step 1	8	Bits 44–47 at Steps 1–2
10	Nine rays, C_9 offset	Lemma 8.2, 8.2‴	8	Bits 50–51
11	K_5 simplex lift	Lemma 8.3	8	None (derived from the tetrahedron)
12	Sixteen half-cubes	Lemma 8 Step 3	8	Bits 44–47 at Step 3
13	Sense reversal σ	Lemma 8 Step 5	8	Bit 52
14	Trefoil fundamental arc (1/6)	Lemma 8 Step 6	8	Bits 50–52
15	Ghost map G	Lemma 8 Prop. 6.2	8, 9 (Part 3)	None (computed from the payload)
16	Ghost boosts $B_{\pm k}$	Lemma 8 Prop. 6.4	8	Bits 44–47 at Step 6
17	Affine frame (L, $\mathbf{t}$)	Lemma 7 Part 1	7	Bits 44–47 (other steps)
18	Gear chain, holonomy H	Lemmas 7, 12	7, 12	Bits 48–49 = 11; value in payload
19	Implied fourth H_4	Lemma 7 Part 3(b), (d)	7	Bits 56–57 (value 11)
20	Pump current J	Lemma 7 Part 3(a), (c), (e)	7	None (property of the kernel)
21	Golden subdivision index F_m	Lemma 11	11	Bits 32–35
22	Clock windows	Lemma 11 Prop. 11.4	11	None (arithmetic of $W_\Sigma = 60$)
23	CRT address	Lemma 9	9	Form 2 core word
24	Five moments M, I, J, S, C_p	Lemma 6	6	None by design: exterior channel, 5 reals per band

#	Object	Definition	Governing lemma	Tile field
25	Gibbs sampler	Lemma 14	14	None by design: runs in Helix A
26	South Halo	Lemma 15 Part 2	15	Bits 56–63
27	Merkle root τ_1	Lemma 15 Part 3	15	None by design: external 256-bit attestation
28	Separation gaps	Lemma 10	10	None (geometric constants)
29	Form ID and payload forms 0–15	§6.1	9, 13, 15	Bits 40–43
30	Hurwitz loss flag	§6.1 (Form 1)	13	Top bit of the w byte
31	Codex shapes	§6.1 (Forms 4–15)	4, 5, 8, 11, 13, 15	Core word and gate byte
32	Ghost triple and collapsed distribution μ, disagreement D	§7.5	8	Form 6 codex packet; D in the sub-region field
33	Balancers A, $\overline{A}$ and emergent structure E	§7.6	7, 12	Bits 56–57 (values 01, 10, 11)
34	64 streams and projections w, x, y, z	§7.7	7, 13	Stream ID = Bits 56–57 and 36–39

C Sources

All five Hoard (2026a–e) reports were consulted (2026b in a related version titled *Mathematical Model for the Torsional Markov Memory-Offload Pump*):

Table C1: Series sources and where this monograph relies on them.

Ref.	Title	Where this monograph relies on it
Hoard 2026a	*The Geometric Engine of Time: Mathematical Architecture of the 6-Channel, 5-Tier Toroidal Orbital Calculator*	Tier data of Lemmas 4, 7, 10
Hoard 2026b	*The Torsional Markov Memory-Offload Pump: A Three-Scale Quaternionic Delta-Chain with Inverse Compensation and Non-Equilibrium Circulation*	Lemma 7 Part 3, Lemma 12
Hoard 2026c	*The Hyperbolic Sparsemax Pump: A Rational, Parity-Safe Executable Kernel for the SHD-CCP Lattice*	Rational Lorentz lift and Q_h (Lemma 8, Prop. 6.2; §7.3); valve kernel (Lemma 7 Part 3(c))
Hoard 2026d	*The Geometric Engine of Time: A Proposal for a Standardization of the Cubit*	Nine-tag system; $\mathcal{C}_0$ metrology and GUM budget; Prop. 11.4; room modes (Lemma 4 Part 3); Remark 4.7 (Lemma 12); logarithm branch (Lemma 7)
Hoard 2026e	*The Geometric Engine of Time: The Fokker-Planck Prime Sieve in Quaternionic Lattices*	Coordinate-free collinear-prime theorem (Lemma 5 Part 5); corner-box comparison (Lemma 5 Part 4); Lemma 14 numerics
Hoard 2026g	SHD-CCP protocol notes (Tensor Transfer, Torsional Twistor, Geometric Parity)	Form ID, codex and quad-slice (§6.1); four-structure system (§7.6–7.7)

No proof depends on a paper that is not restated here: every result taken from these sources is given with its proof and checked by script. Of the BioChain AI papers (references 1–10), reference 9 (*Topological Symmetry Folding*) and reference 10 (*Dual-Helix Framework*) were used; the others are cited for context only.

References

1. BioChain AI — SHD-CCP Architecture Research (2026). *Architectural Design Document: Quaternionic Runtime Environment (QRE), Technical Specification v1.5*.
2. BioChain AI — SHD-CCP Architecture Research (2026). *From Sequences to Spheres: How Quaternion Chain Compression Turns Time into Space (QCC)*.
3. BioChain AI — SHD-CCP Architecture Research (2026). *Geometric Data Layouts and Resonance Stability via Golden Subdivision for Panmagic Square Chains*.
4. BioChain AI — SHD-CCP Architecture Research (2026). *Quaternion Data Compression: The Math*.
5. BioChain AI — SHD-CCP Architecture Research (2026). *Quaternion Equilibrium & Voxel Alignment Architecture: Resolving Topological Singularities in the SHD-CCP Protocol via Higher-Dimensional Manifolds*.
6. BioChain AI — SHD-CCP Architecture Research (2026). *Quaternionic Delta Encoding on Golden Strassen-Clifford Toroids: Wave Collision and Sub-Cubic Matrix Optimization in the SHD-CCP 64-Bit Protocol*.

7. BioChain AI — SHD-CCP Architecture Research (2026). *SHD-CCP Extension Analysis: Per-Node Quaternion Scaling & Optimization*.
8. BioChain AI — SHD-CCP Architecture Research (2026). *The Hyperbolic Oracle: Collapsing ECDLP via SHD-CCP Quadrance Lattice Reduction and Prime Triplet Indexing*.
9. BioChain AI — SHD-CCP Architecture Research (2026). *Topological Symmetry Folding and Hyperbolic Projection in the SHD-CCP Framework: A 720:1 Quaternionic Cyclic System*.
10. BioChain AI — SHD-CCP Architecture Research (2026). *A Unified Dual-Helix Framework for Geometrically-Guided Pollard's Rho: Cyclic Feature Embeddings, Quadrance Oracles, and Exact Certification*.
11. Conway, J. H., & Smith, D. A. (2003). *On Quaternions and Octonions: Their Geometry, Arithmetic, and Symmetry*. A K Peters.
12. Coxeter, H. S. M. (1973). *Regular Polytopes* (3rd ed.). Dover.
13. Cramer, O. (1993). The variation of the specific heat ratio and the speed of sound in air with temperature, pressure, humidity, and CO_2 concentration. *Journal of the Acoustical Society of America*, 93(5), 2510–2516.
14. Epstein, D. B. A. (1971). Almost all subgroups of a Lie group are free. *Journal of Algebra*, 19, 261–262.
15. Garvan, F. G. (2019). New fifth and seventh order mock theta function identities. *Annals of Combinatorics*, 23(3–4), 765–783. arXiv:1907.04803.
16. Hoard, G. (2026). *Audit and Revision Record of the Fifteen Lemmas of the Geometric Engine*. Companion paper.
17. Hoard, G. (2026). *Open Problems of the Geometric Engine: A Research Program for the SHD-CCP Architecture*. Companion paper.
18. Hoard, G. (2026a). *The Geometric Engine of Time: Mathematical Architecture of the 6-Channel, 5-Tier Toroidal Orbital Calculator*. BioChain AI — SHD-CCP Architecture Research.
19. Hoard, G. (2026b). *The Torsional Markov Memory-Offload Pump: A Three-Scale Quaternionic Delta-Chain with Inverse Compensation and Non-Equilibrium Circulation*. Owl Academy / BioChain AI Technical Report.
20. Hoard, G. (2026c, June 21). *The Hyperbolic Sparsemax Pump: A Rational, Parity-Safe Executable Kernel for the SHD-CCP Lattice*. BioChain AI / Owl Academy Companion Paper.
21. Hoard, G. (2026d). *The Geometric Engine of Time: A Proposal for a Standardization of the Cubit*. BioChain AI — SHD-CCP Architecture Research.
22. Hoard, G. (2026e). *The Geometric Engine of Time: The Fokker-Planck Prime Sieve in Quaternionic Lattices*. BioChain AI — SHD-CCP Architecture Research.
23. Hoard, G. (2026f). *The Geometric Engine of Time: A Rigorous Audit of the Nine Lemmas and the Case for Fifteen*. Technical Audit Report, October 15, 2026.
24. Hoard, G. (2026g). *SHD-CCP Tensor Transfer Protocol (TTP)*; *The Torsional Twistor Protocol*; *Geometric Adjustment Protocol: 10-to-12 Core Strassen Fold*. BioChain AI design notes.
25. Kaihnsa, N., Kummer, M., Plaumann, D., Namin, M. S., & Sturmfels, B. (2017). Sixty-four curves of degree six. *arXiv:1703.01660v2 [math.AG]*.
26. Kapouleas, N., & McGrath, P. (2024). New minimal surface doublings of the Clifford torus. *arXiv:2411.00613*.
27. Niven, I. (1956). *Irrational Numbers*. Carus Mathematical Monographs, No. 11, Mathematical Association of America.
28. OpenAI (2026). *Finite Time Blowup for Navier–Stokes* (preprint, 166 pp.; announced September 8, 2026 as "On the Navier–Stokes Millennium Prize Problem", https://openai.com/index/navier-stokes-solution/). Overall correctness publicly disputed; only Lemmas 4.3, 4.4(i), 4.7, 6.1 and A.1–A.3 were re-checked here, and no other result of the preprint is used.
29. Schütte, K., & van der Waerden, B. L. (1953). Das Problem der dreizehn Kugeln. *Mathematische Annalen*, 125, 325–334.
30. Sós, V. T. (1958). On the distribution mod 1 of the sequence $n\alpha$. *Annales Universitatis Scientiarum Budapestinensis de Rolando Eötvös Nominatae, Sectio Mathematica*, 1, 127–134.
31. Wildberger, N. J. (2005). *Divine Proportions: Rational Trigonometry to Universal Geometry*. Wild Egg Books.
32. Williams, R. (2025). Simulating time with square-root space. *Proceedings of the ACM Symposium on Theory of Computing (STOC)*.
33. Zhao, Y. (2010). *Three Lemmas in Geometry*. MIT Winter Camp Lecture Notes.
34. Zwegers, S. P. (2002). *Mock Theta Functions*. Ph.D. Thesis, Universiteit Utrecht.

Companion papers. Two companion papers accompany this monograph. *Open Problems of the Geometric Engine* collects the questions that lie outside the defined system, with what is known and how to approach each; the problem numbers P1–P16 cited above refer to it. *Audit and Revision Record of the Fifteen Lemmas* records how the present text was reached: the audit of the nine-lemma source, the registers of corrections, and the changes made at each stage.